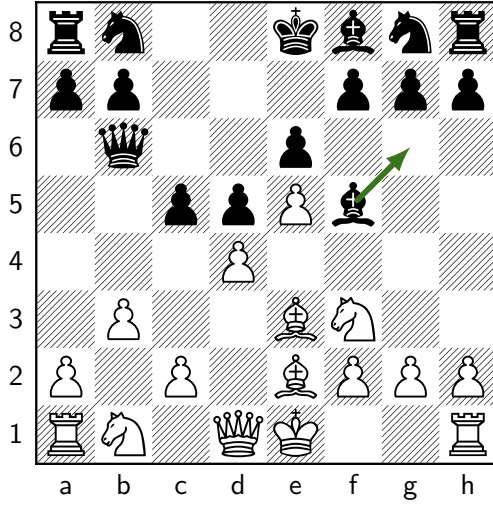


# Melanie-ANALOGY — Caro-Kann, Advance

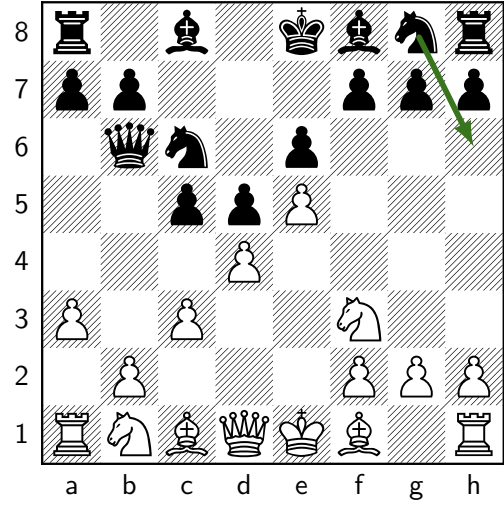
Melanie (Deep\_Vision)

## Abstract

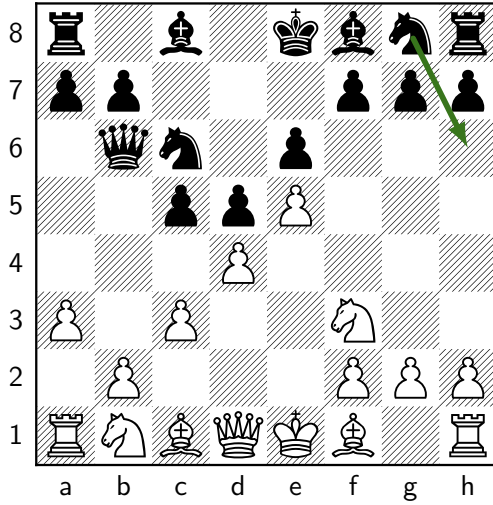
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



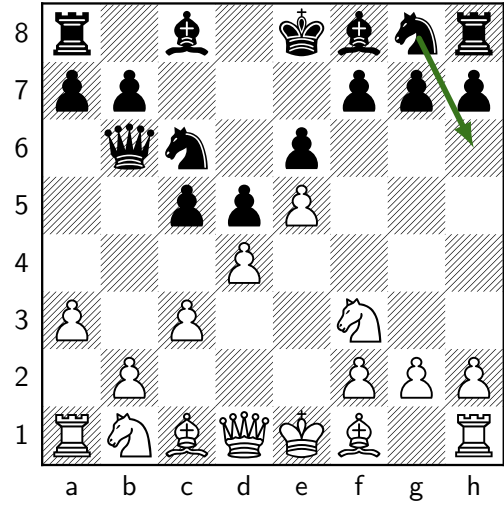
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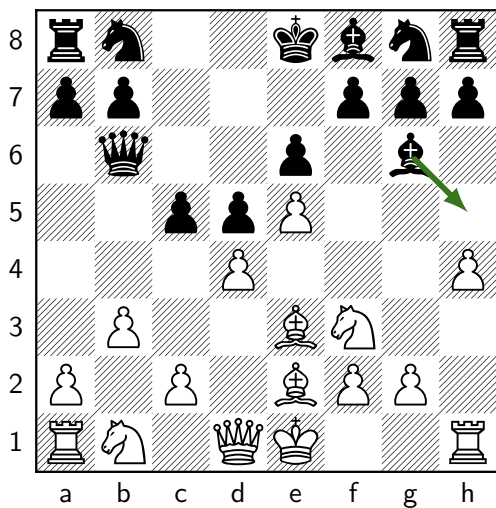


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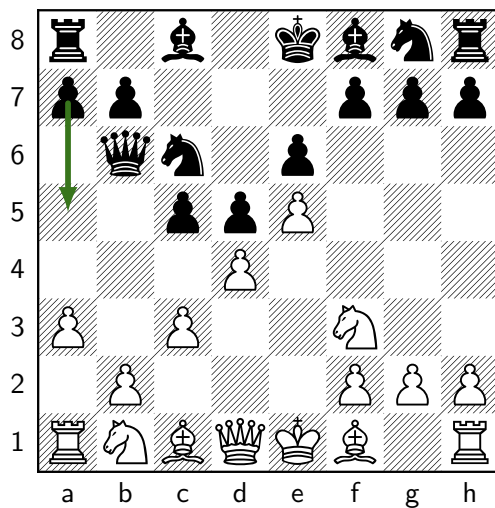


(d)

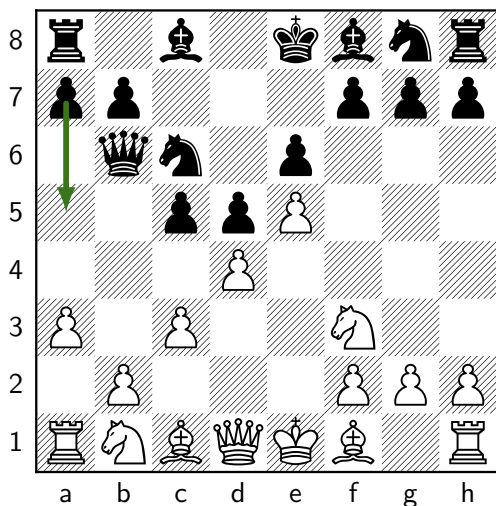
Figure 1: **Move 7, Black: Bg6**. (a) Our position; Melanie plays Bg6 (f5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2550` (ply 11) [structural distance  $d=64$ , QAP obj=93]: the master played Nh6; under  $\mu$  our Bf4 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Ra8, our Pg7 as their Bc8; transferred move f5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2551` (ply 11) [structural distance  $d=64$ , QAP obj=93]: the master played Nh6; under  $\mu$  our Bf4 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Ra8, our Pg7 as their Bc8; transferred move f5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2954` (ply 11) [structural distance  $d=64$ , QAP obj=93]: the master played Nh6; under  $\mu$  our Bf4 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Ra8, our Pg7 as their Bc8; transferred move f5g6 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Bg6 (summed QAP objective 1584, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



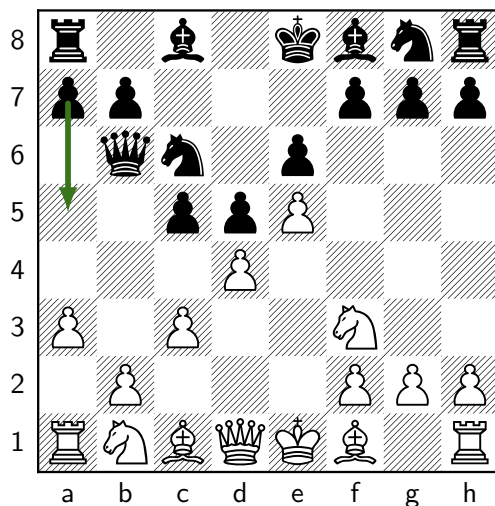
(a)



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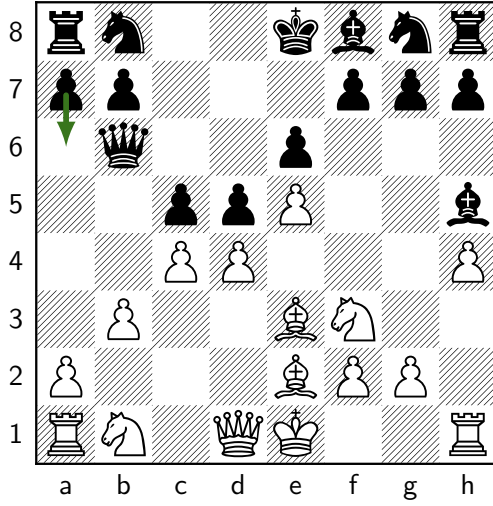


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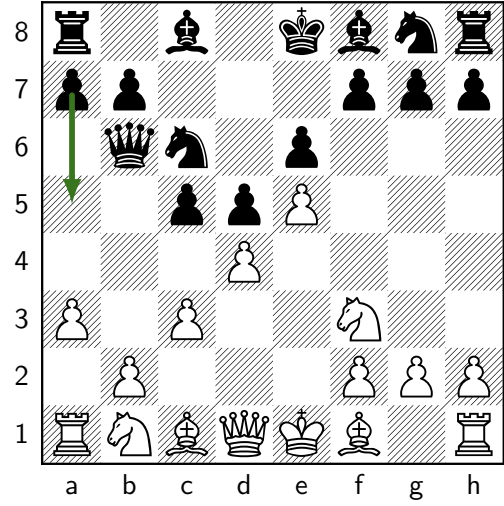


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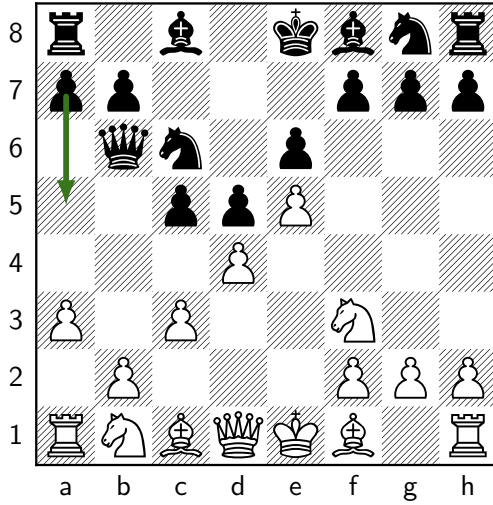
Figure 2: **Move 8, Black: Bh5.** (a) Our position; Melanie plays Bh5 (g6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2482` (ply 11) [structural distance  $d=68$ , QAP obj=92]: the master played a5; under  $\mu$  our  $Bg3$  plays the role of that game’s  $Pa2$  (a **cross-type** correspondence); cross-type roles: our  $Pd5$  as their  $Ra8$ , our  $pe5$  as their  $Bc8$ ; transferred move g6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2483` (ply 11) [structural distance  $d=68$ , QAP obj=92]: the master played a5; under  $\mu$  our  $Bg3$  plays the role of that game’s  $Pa2$  (a **cross-type** correspondence); cross-type roles: our  $Pd5$  as their  $Ra8$ , our  $pe5$  as their  $Bc8$ ; transferred move g6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2952` (ply 11) [structural distance  $d=68$ , QAP obj=92]: the master played a5; under  $\mu$  our  $Bg3$  plays the role of that game’s  $Pa2$  (a **cross-type** correspondence); cross-type roles: our  $Pd5$  as their  $Ra8$ , our  $pe5$  as their  $Bc8$ ; transferred move g6h5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bh5 (summed QAP objective 732, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



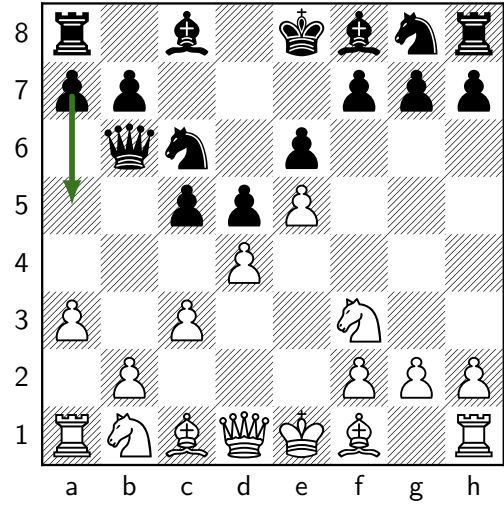
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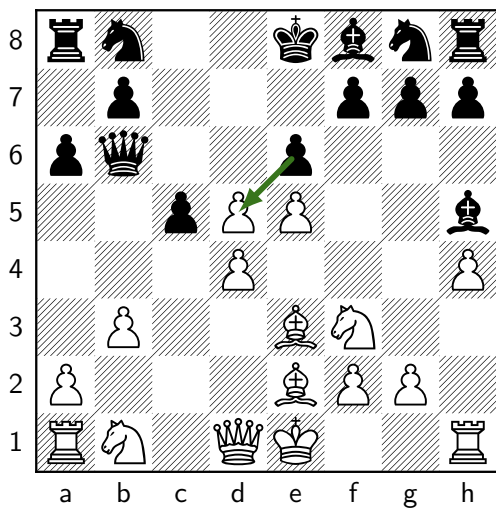


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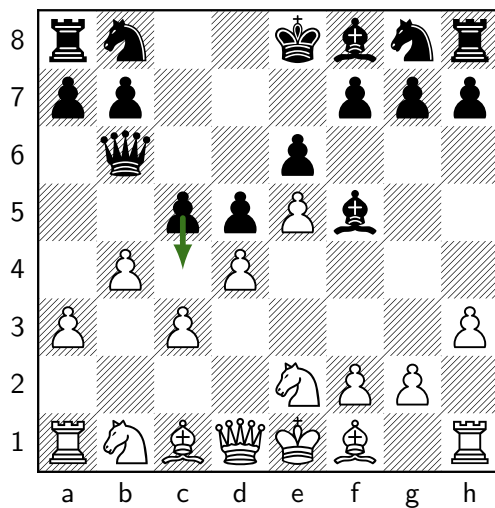


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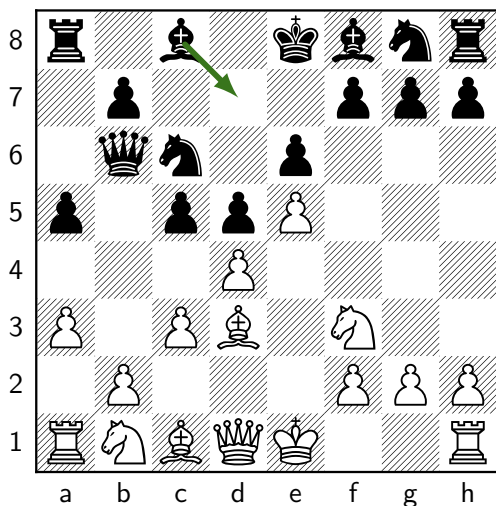
Figure 3: **Move 9, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2482` (ply 11) [structural distance  $d=74$ , QAP obj=69]: the master played a5; under  $\mu$  our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Ng8* as their *Ra8*, our *pe5* as their *Bc8*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2483` (ply 11) [structural distance  $d=74$ , QAP obj=69]: the master played a5; under  $\mu$  our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Ng8* as their *Ra8*, our *pe5* as their *Bc8*; transferred move a7a6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2952` (ply 11) [structural distance  $d=74$ , QAP obj=69]: the master played a5; under  $\mu$  our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *Ng8* as their *Ra8*, our *pe5* as their *Bc8*; transferred move a7a6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to a6 (summed QAP objective 552, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



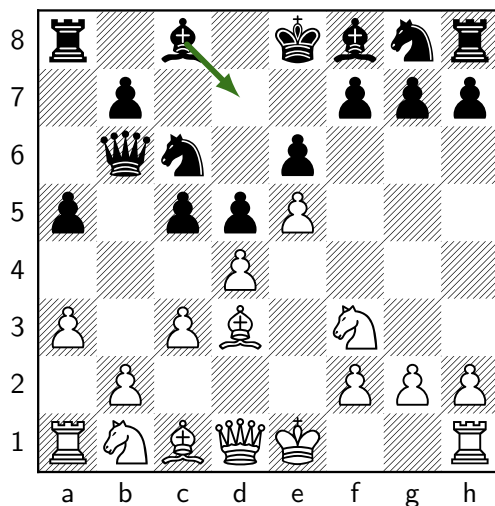
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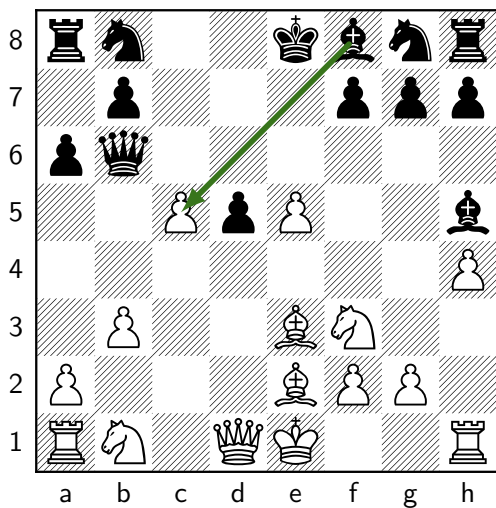


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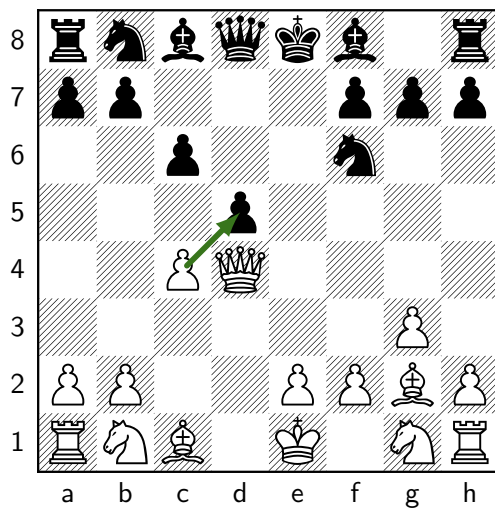


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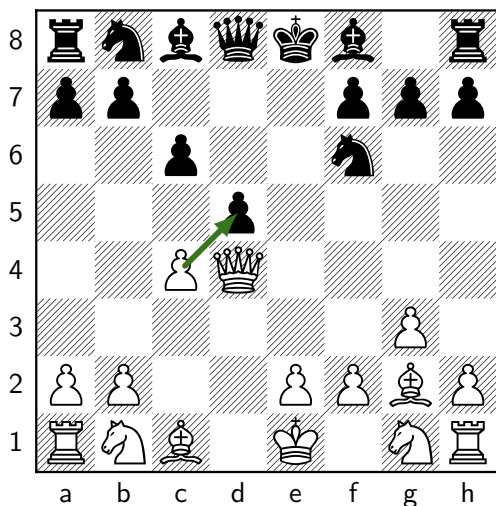
Figure 4: **Move 10, Black: exd5.** (a) Our position; Melanie plays **exd5** (**e6d5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1332` (ply 15) [structural distance  $d=69$ , QAP obj=65]: the master played **c4**; under  $\mu$  our *Pe3* plays the role of that game’s *Pc4*; cross-type roles: our *Pg7* as their *Ra8*, our *Rh8* as their *Nb8*; transferred move **e6d5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2482` (ply 13) [structural distance  $d=84$ , QAP obj=62]: the master played **Bd7**; under  $\mu$  our *Pe3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Ng8*, our *Pc5* as their *Nc6*; transferred move **e6d5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2483` (ply 13) [structural distance  $d=84$ , QAP obj=62]: the master played **Bd7**; under  $\mu$  our *Pe3* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Ng8*, our *Pc5* as their *Nc6*; transferred move **e6d5** (**endorses this move**). Synthesis: 19 retrieved analogs transferred to **exd5** (summed QAP objective 1073, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



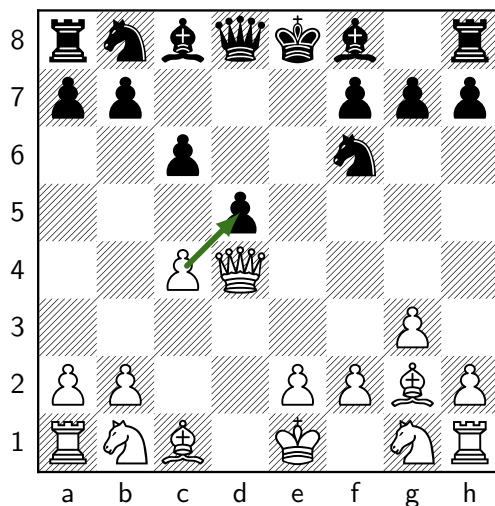
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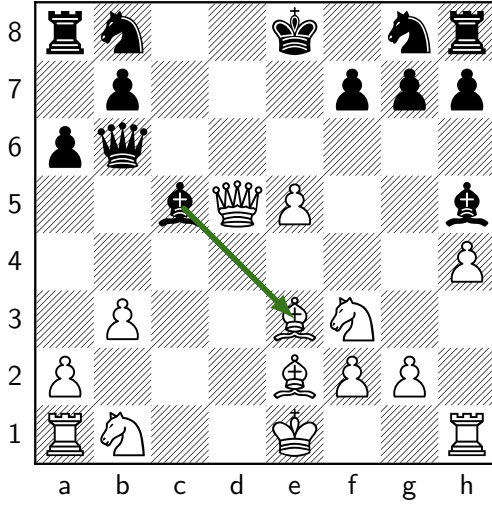


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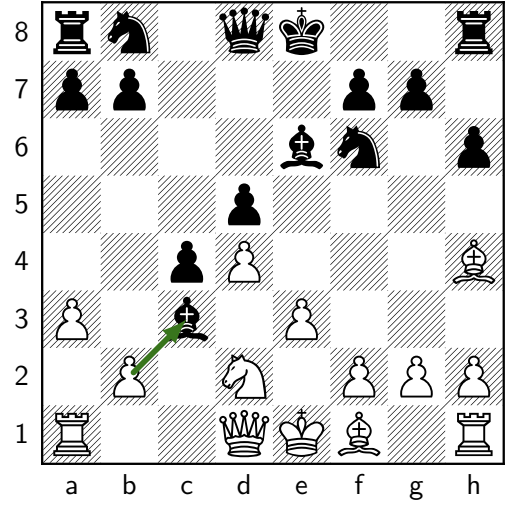


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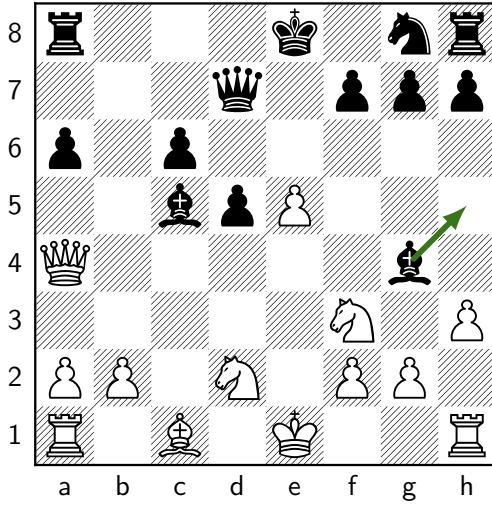
Figure 5: **Move 11, Black: Bxc5.** (a) Our position; Melanie plays Bxc5 (f8c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#232` (ply 10) [structural distance  $d=61$ , QAP obj=38]: the master played cxd5; under  $\mu$  our *Bf1* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *nb1* as their *Ra1*, our *Pb7* as their *Nb1*; transferred move f8c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#233` (ply 10) [structural distance  $d=61$ , QAP obj=38]: the master played cxd5; under  $\mu$  our *Bf1* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *nb1* as their *Ra1*, our *Pb7* as their *Nb1*; transferred move f8c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1989` (ply 10) [structural distance  $d=61$ , QAP obj=38]: the master played cxd5; under  $\mu$  our *Bf1* plays the role of that game’s *Pc4* (a **cross-type** correspondence); cross-type roles: our *nb1* as their *Ra1*, our *Pb7* as their *Nb1*; transferred move f8c5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Bxc5 (summed QAP objective 304, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



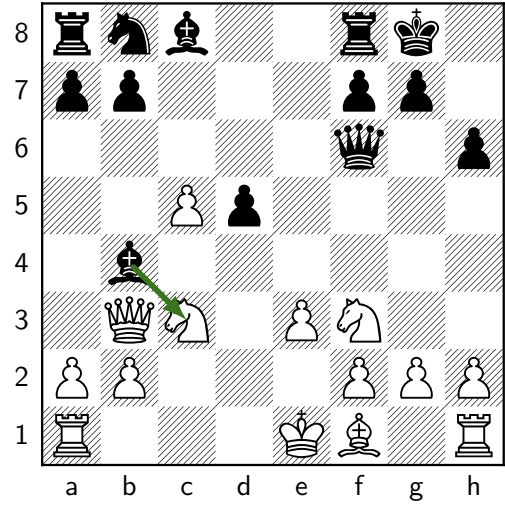
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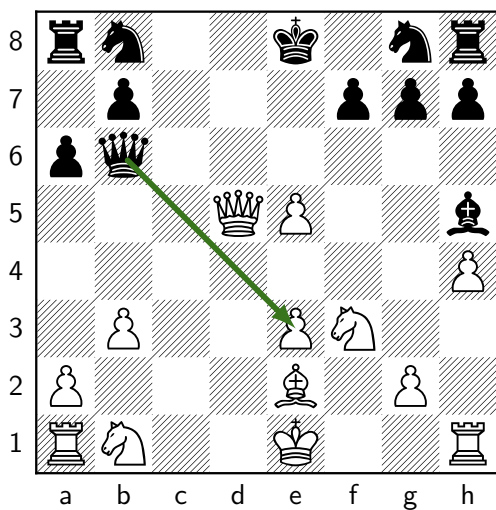


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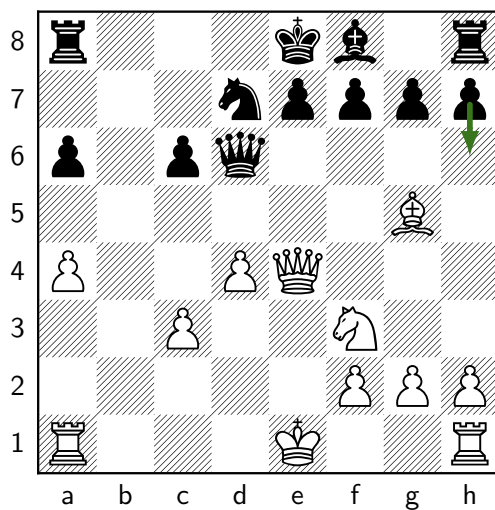


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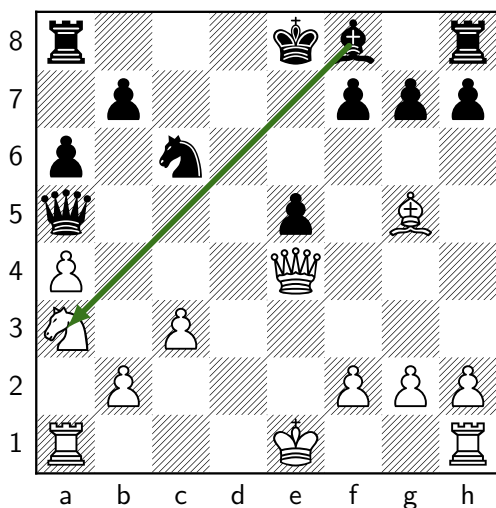
Figure 6: **Move 12, Black: Bxe3**. (a) Our position; Melanie plays Bxe3 (c5e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3229` (ply 20) [structural distance  $d=107$ , QAP obj=60]: the master played `bxc3`; under  $\mu$  our `Bc4` plays the role of that game’s `Pb2` (a **cross-type** correspondence); cross-type roles: our `be2` as their `Ra1`, our `qd5` as their `Bf1`; transferred move `c5e3` (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#436` (ply 23) [structural distance  $d=106$ , QAP obj=59]: the master played `Bh5`; under  $\mu$  our `Bc4` plays the role of that game’s `Bg5`; cross-type roles: our `Bh5` as their `Ra8`, our `Rh8` as their `Ph7`; transferred move `c5e3` (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3218` (ply 19) [structural distance  $d=99$ , QAP obj=57]: the master played `Bxc3+`; under  $\mu$  our `Bc4` plays the role of that game’s `Bb5`; cross-type roles: our `Pa6` as their `Nb8`, our `pe5` as their `Bc8`; transferred move `c5e3` (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Bxe3 (summed QAP objective 489, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



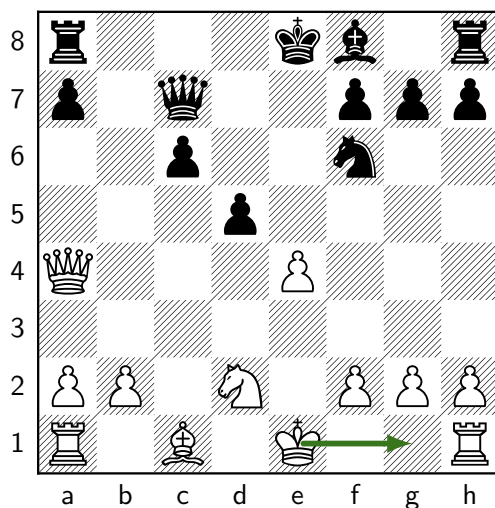
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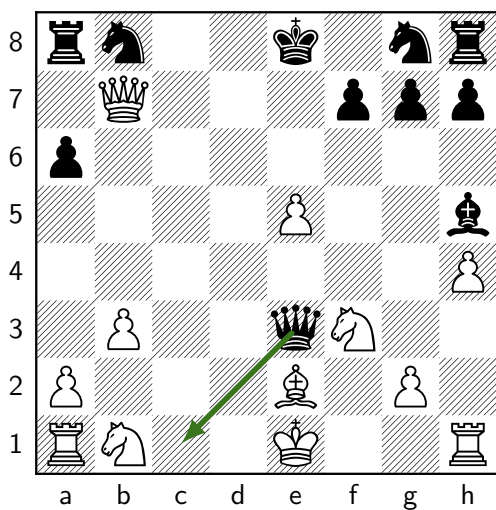


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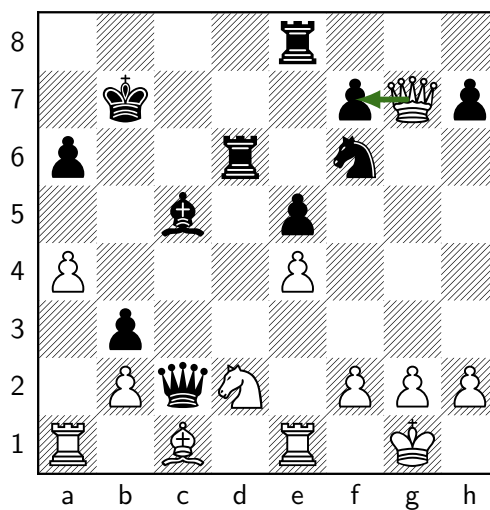


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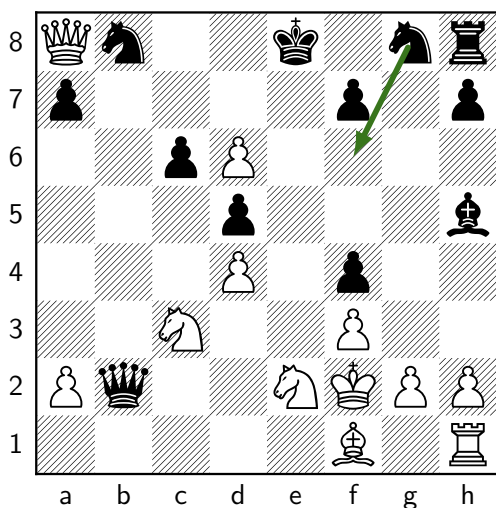
Figure 7: **Move 13, Black: Qxe3.** (a) Our position; Melanie plays Qxe3 (b6e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1562` (ply 27) [structural distance  $d=82$ , QAP obj=86]: the master played h6; under  $\mu$  our *Pf2* plays the role of that game’s *Ph2*; cross-type roles: our *ph4* as their *Ra8*, our *Pa6* as their *Nd7*; transferred move f7f6. (c) Retrieved from `ccr1_top_p0001#3481` (ply 27) [structural distance  $d=89$ , QAP obj=82]: the master played Bxa3; under  $\mu$  our *Bh4* plays the role of that game’s *Bf1*; cross-type roles: our *ph4* as their *Ra8*, our *Ph7* as their *Rh8*; transferred move h5f3. (d) Retrieved from `ccr1_top_p0000#141` (ply 22) [structural distance  $d=82$ , QAP obj=79]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pg7* as their *Bc1*, our *pa2* as their *Rh1*; transferred move e8f8. Synthesis: no retrieved analog transferred to Qxe3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



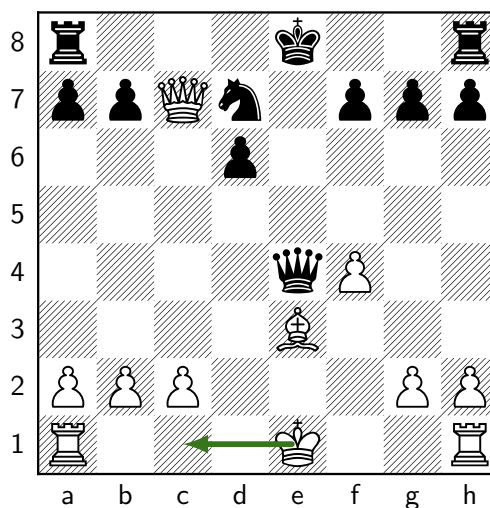
(a)



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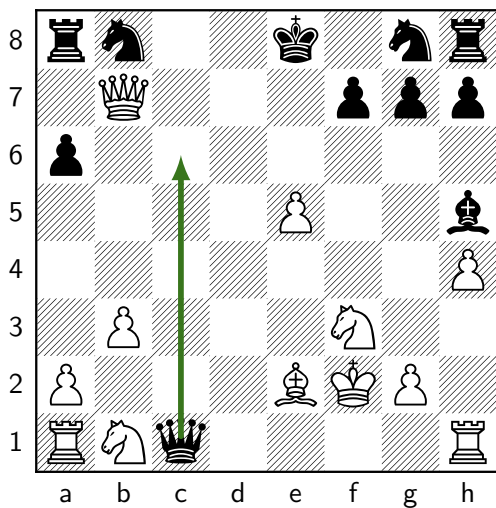


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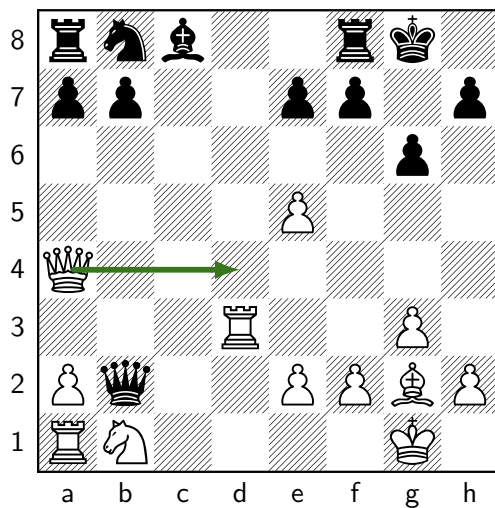


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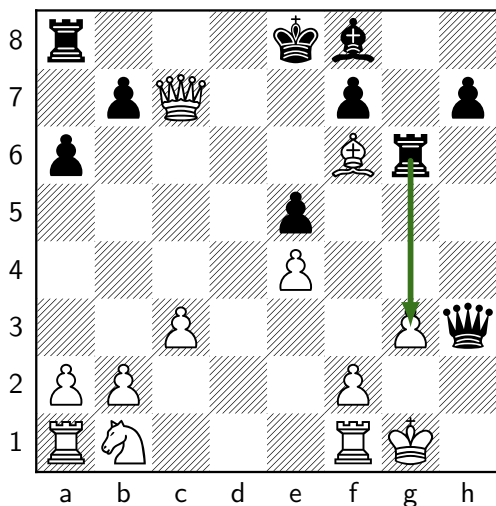
Figure 8: **Move 14, Black: Qc1+.** (a) Our position; Melanie plays Qc1+ (e3c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2252` (ply 42) [structural distance  $d=115$ , QAP obj=85]: the master played Qxf7+; under  $\mu$  our Qe6 plays the role of that game’s Qg7; cross-type roles: our Bh5 as their Ra1, our Ra8 as their Bc1; transferred move e3c1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3543` (ply 27) [structural distance  $d=132$ , QAP obj=118]: the master played Nf6; under  $\mu$  our Pf2 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our be2 as their Nb8, our nf3 as their Pa7; transferred move f7f6. (d) Retrieved from `ccr1_top_p0002#3992` (ply 26) [structural distance  $d=126$ , QAP obj=109]: the master played 0-0-0; under  $\mu$  our Ke1 plays the role of that game’s Ke1; cross-type roles: our pg2 as their Ra1, our Pf7 as their Rh1; transferred move e8f8. Synthesis: 1 retrieved analog transferred to Qc1+ (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



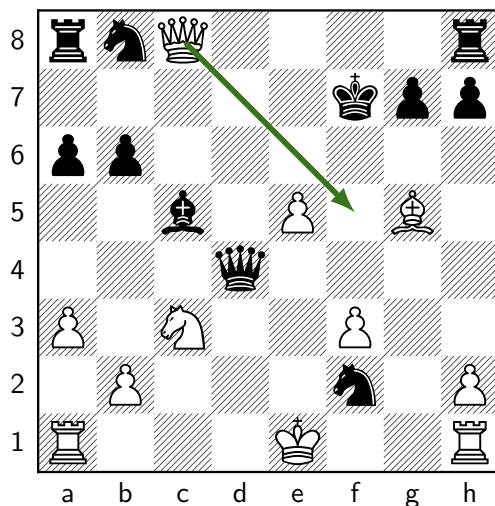
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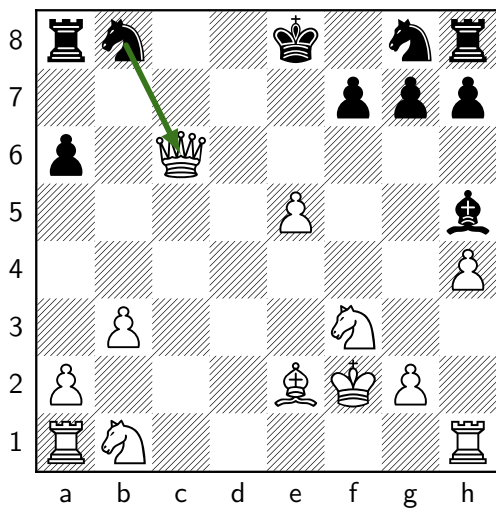


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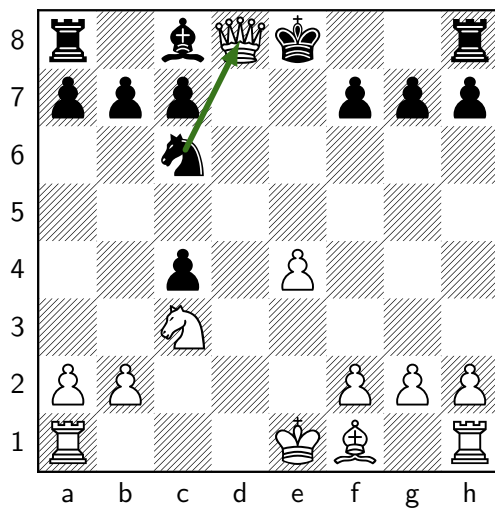


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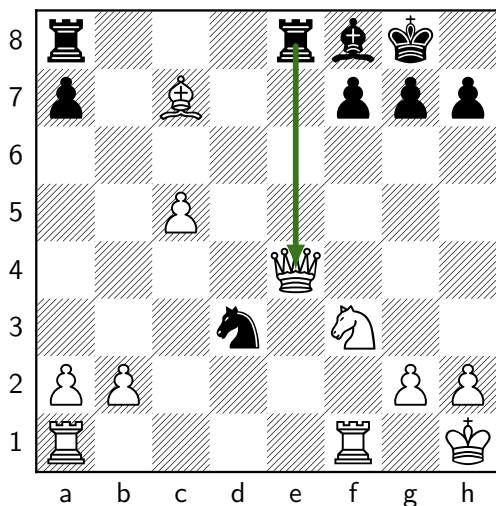
Figure 9: **Move 15, Black: Qc6**. (a) Our position; Melanie plays Qc6 (c1c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3849` (ply 28) [structural distance  $d=133$ , QAP obj=46]: the master played Qd4; under  $\mu$  our Qc8 plays the role of that game’s Qa4; cross-type roles: our Nb8 as their Pa2, our Ng8 as their Pf2; transferred move c1c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#737` (ply 31) [structural distance  $d=136$ , QAP obj=25]: the master played Rxg3+; under  $\mu$  our Qc8 plays the role of that game’s Rg3 (a **cross-type** correspondence); cross-type roles: our ph4 as their Ra8, our pg2 as their Bf8; transferred move c1c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2506` (ply 34) [structural distance  $d=117$ , QAP obj=80]: the master played Qf5+; under  $\mu$  our Qc8 plays the role of that game’s Qc8; cross-type roles: our nf3 as their Rh1, our Bh5 as their nf2; transferred move c1f4. Synthesis: 2 retrieved analogs transferred to Qc6 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



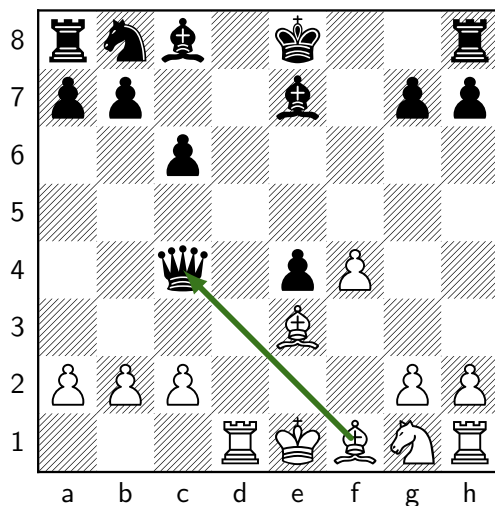
(a)



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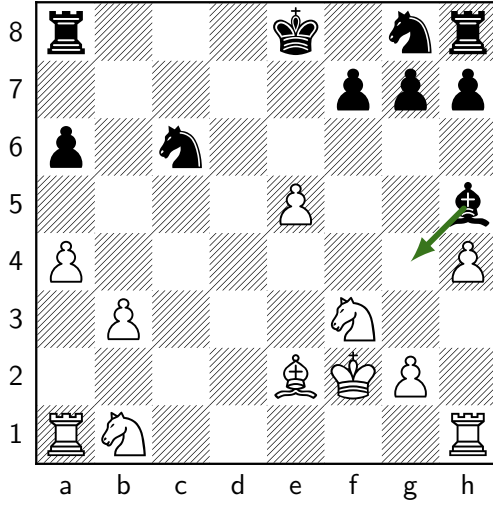


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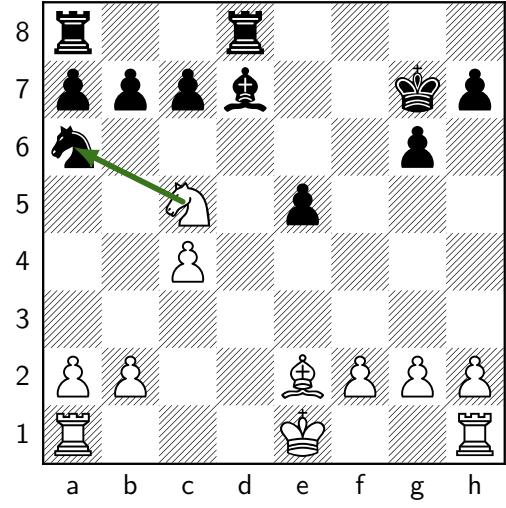


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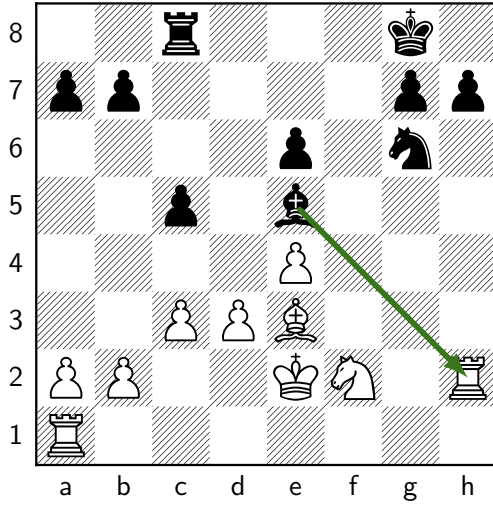
Figure 10: **Move 16, Black: Nxc6.** (a) Our position; Melanie plays Nxc6 (b8c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3355` (ply 19) [structural distance  $d=102$ , QAP obj=37]: the master played Nxd8; under  $\mu$  our Nb1 plays the role of that game’s Nc3; cross-type roles: our Ng8 as their Bc8, our be2 as their Pc4; transferred move b8c6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#724` (ply 37) [structural distance  $d=119$ , QAP obj=7]: the master played Rxe4; under  $\mu$  our Nb1 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Pa6 as their Bf8, our Ng8 as their Pa7; transferred move b8c6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1963` (ply 22) [structural distance  $d=125$ , QAP obj=2]: the master played Bxc4; under  $\mu$  our Nb1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rd1, our Pg7 as their Ng1; transferred move b8c6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nxc6 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



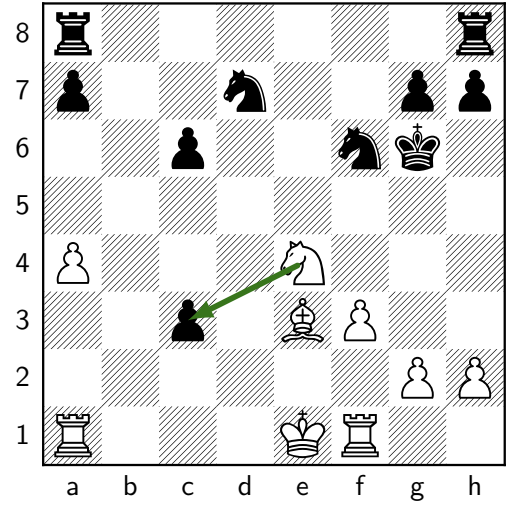
(a)



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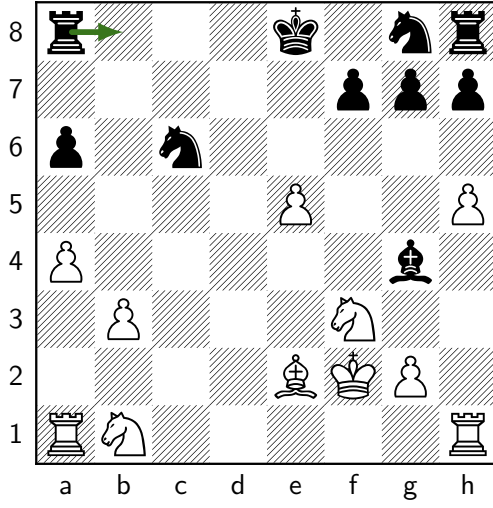


(c)

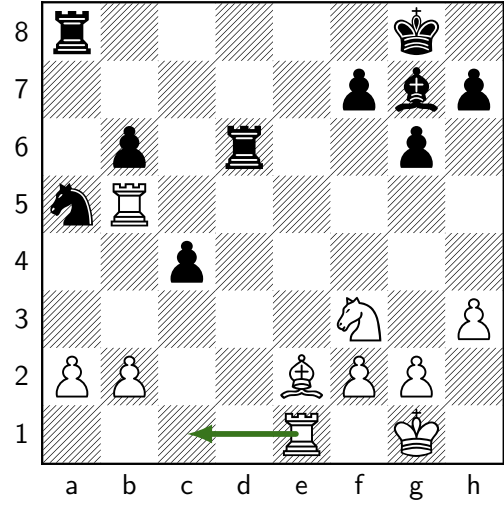


(d)

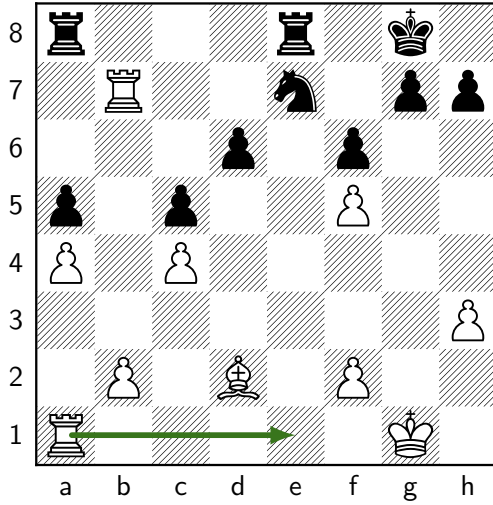
Figure 11: **Move 17, Black: Bg4**. (a) Our position; Melanie plays Bg4 (h5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1761` (ply 32) [structural distance  $d=67$ , QAP obj=39]: the master played Nxa6; under  $\mu$  our Bh4 plays the role of that game’s Nc5 (a **cross-type** correspondence); cross-type roles: our nb1 as their Rh1, our Ng8 as their Pa2; transferred move h5g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3470` (ply 43) [structural distance  $d=61$ , QAP obj=28]: the master played Bxh2; under  $\mu$  our Bh4 plays the role of that game’s Be4; cross-type roles: our Ng8 as their Pb7, our pa4 as their Ng6; transferred move h5g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3768` (ply 42) [structural distance  $d=61$ , QAP obj=26]: the master played Nxc3; under  $\mu$  our Bh4 plays the role of that game’s Ne4 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rf1, our Rh8 as their Ph2; transferred move h5g4 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bg4 (summed QAP objective 170, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



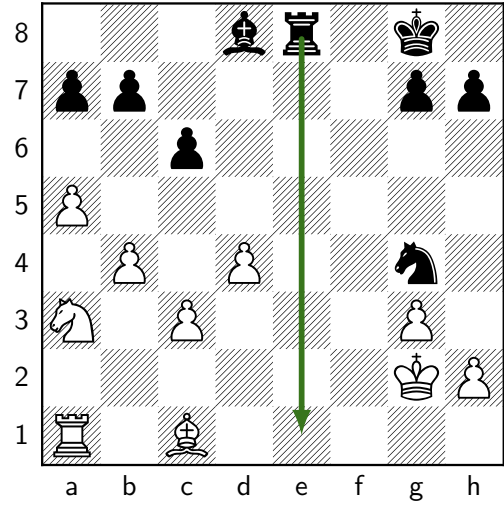
(a)



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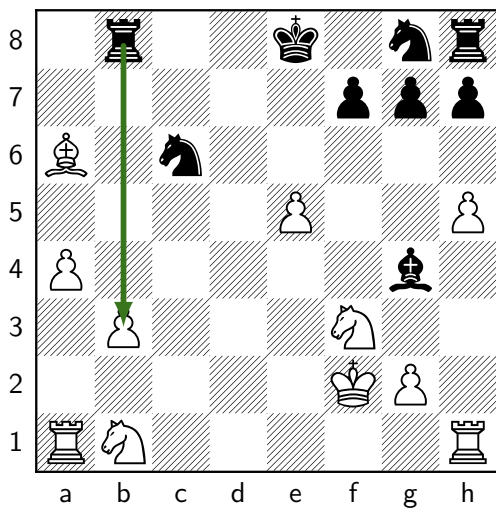


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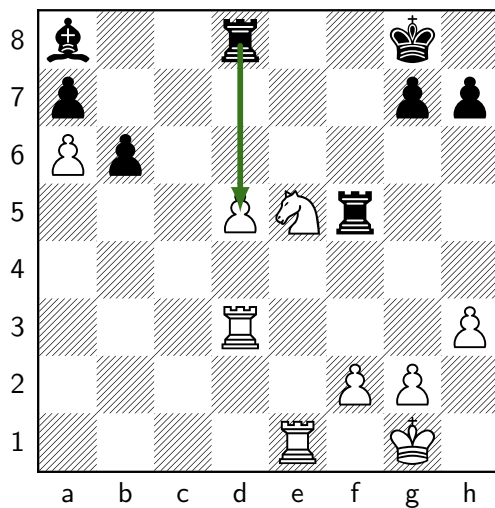


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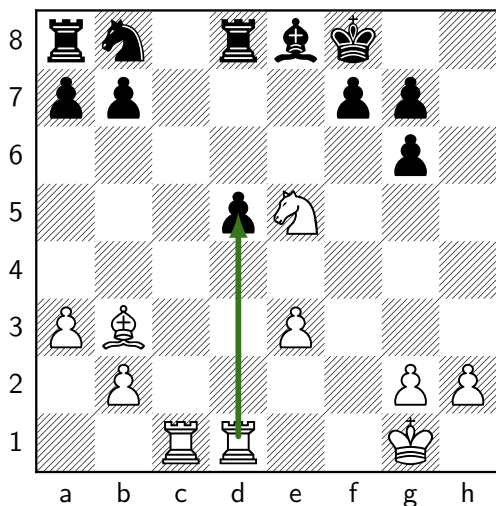
Figure 12: **Move 18, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3742` (ply 40) [structural distance  $d=77$ , QAP obj=32]: the master played Rc1; under  $\mu$  our Ra1 plays the role of that game’s Re1; cross-type roles: our Nc6 as their Be2, our Bg4 as their Rb5; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2236` (ply 50) [structural distance  $d=78$ , QAP obj=11]: the master played Re1; under  $\mu$  our Ra1 plays the role of that game’s Ra1; cross-type roles: our Rh8 as their Pb2, our pa4 as their Bd2; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2312` (ply 51) [structural distance  $d=77$ , QAP obj=7]: the master played Re1; under  $\mu$  our Ra1 plays the role of that game’s Re1; cross-type roles: our Pf7 as their Bd8, our Ng8 as their Pb7; transferred move a8b8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb8 (summed QAP objective 56, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



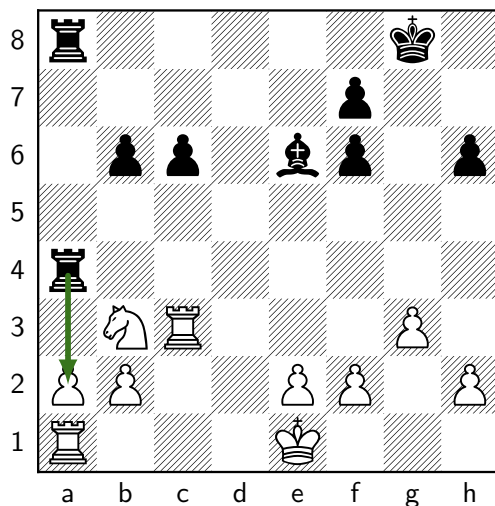
(a)



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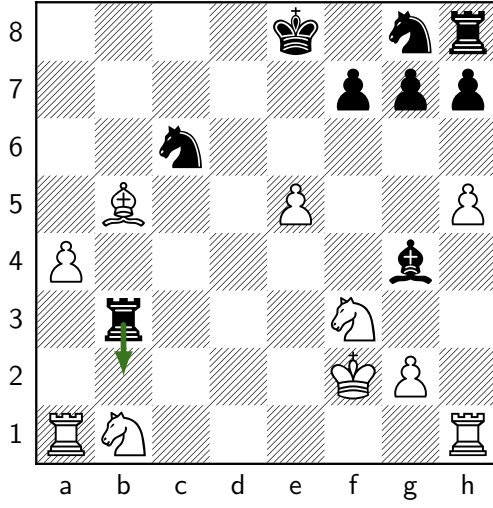


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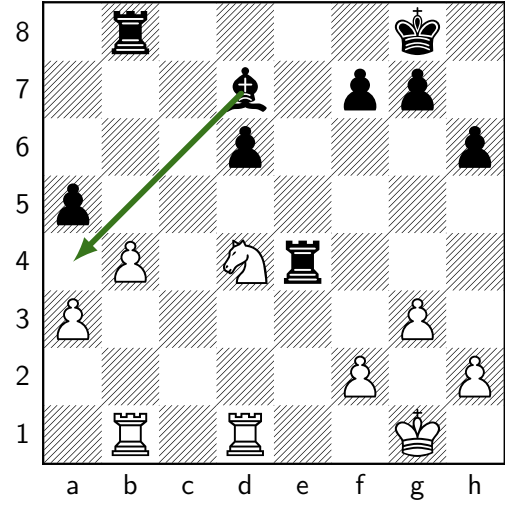


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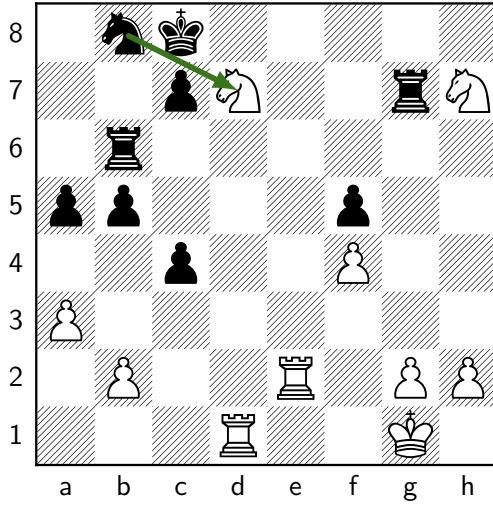
Figure 13: **Move 19, Black: Rxb3.** (a) Our position; Melanie plays Rxb3 (b8b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1484` (ply 53) [structural distance  $d=76$ , QAP obj=40]: the master played Rxd5; under  $\mu$  our Rb1 plays the role of that game’s Rd1; cross-type roles: our Nc6 as their Ba8, our Rh8 as their Pa7; transferred move b8b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1909` (ply 44) [structural distance  $d=70$ , QAP obj=27]: the master played Rxd5; under  $\mu$  our Rb1 plays the role of that game’s Rd1; cross-type roles: our Ng8 as their Pb2, our Rh8 as their Ph2; transferred move b8b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3955` (ply 41) [structural distance  $d=71$ , QAP obj=19]: the master played Rxa2; under  $\mu$  our Rb1 plays the role of that game’s Ra5; cross-type roles: our Nc6 as their Ra8, our Ng8 as their Pf7; transferred move b8b3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rxb3 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



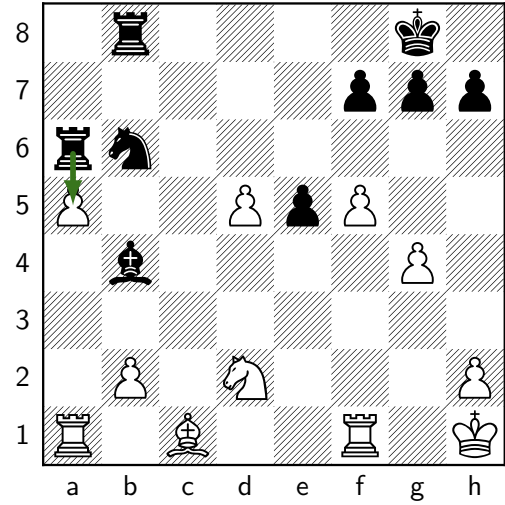
(a)



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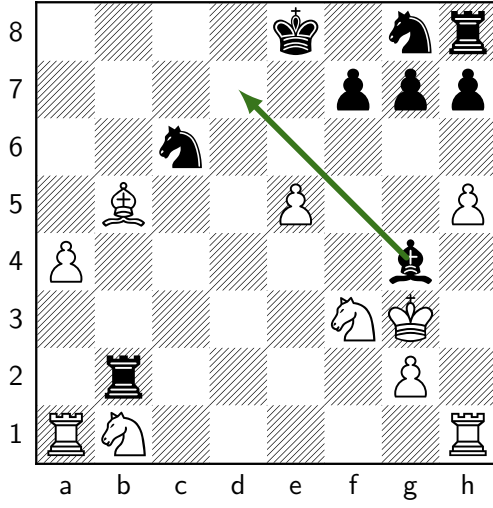


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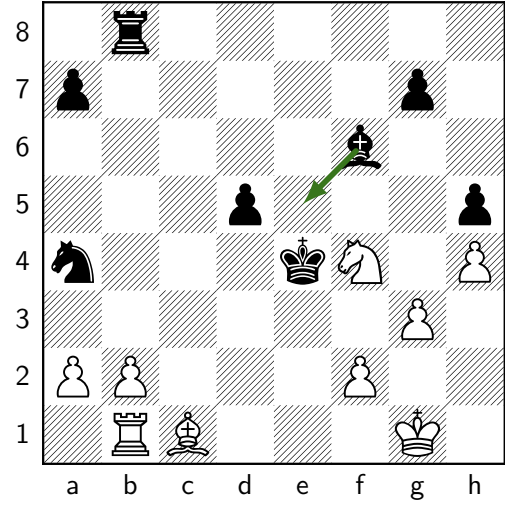


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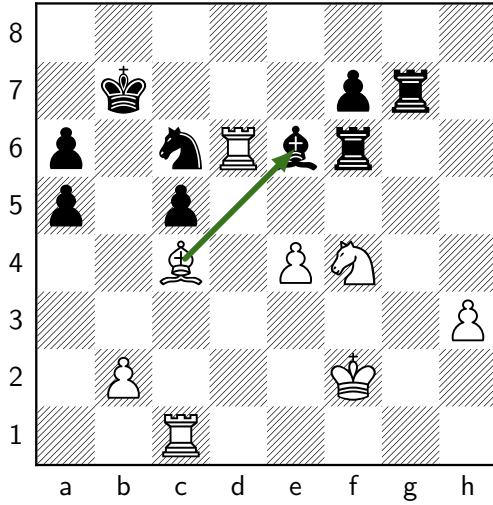
Figure 14: **Move 20, Black: Rb2+.** (a) Our position; Melanie plays Rb2+ (b3b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3160` (ply 49) [structural distance  $d=94$ , QAP obj=44]: the master played Ba4; under  $\mu$  our Rb6 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our bb5 as their Rb8, our Ng8 as their Pf7; transferred move b3b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3268` (ply 43) [structural distance  $d=78$ , QAP obj=51]: the master played Nxd7; under  $\mu$  our Rb6 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our bb5 as their nd7, our Nc6 as their Rb6; transferred move b3b4. (d) Retrieved from `ccr1_top_p0000#2409` (ply 45) [structural distance  $d=75$ , QAP obj=50]: the master played Rxa5; under  $\mu$  our Rb6 plays the role of that game’s Ra3; cross-type roles: our Bg4 as their Pf7, our nf3 as their Pg7; transferred move b3b5. Synthesis: 1 retrieved analog transferred to Rb2+ (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



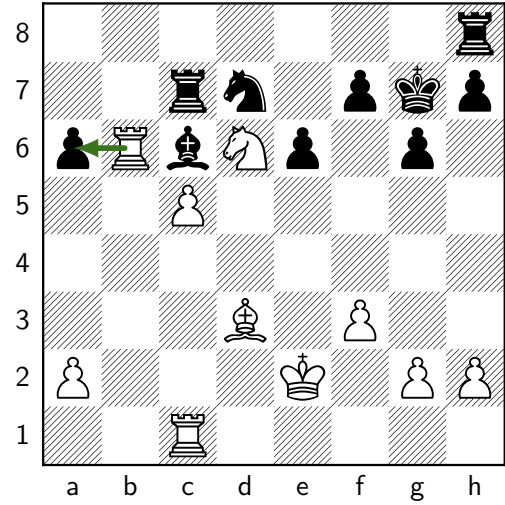
(a)



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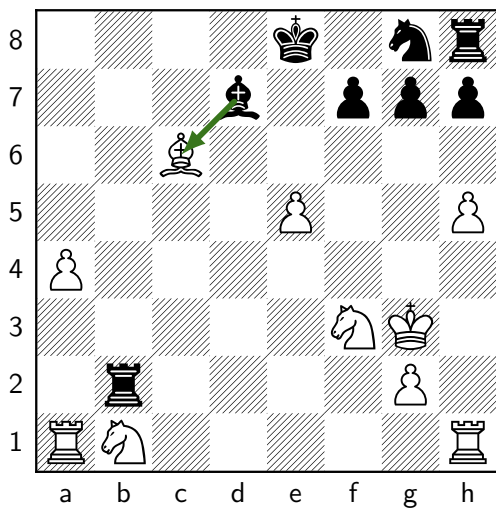


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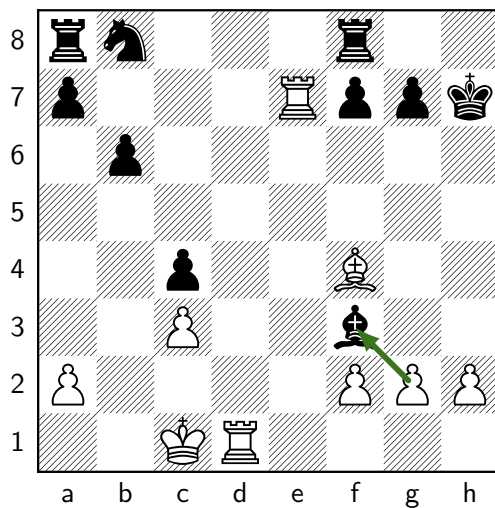


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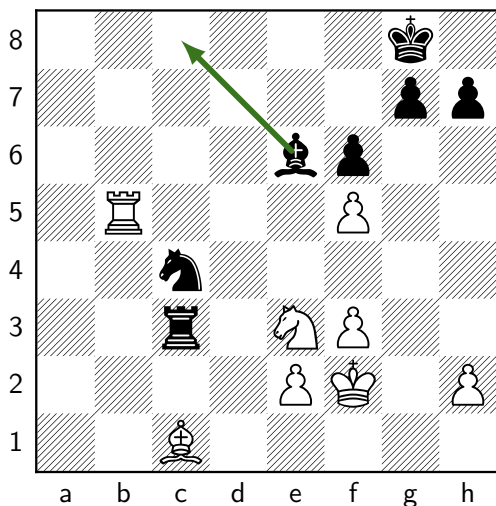
Figure 15: **Move 21, Black: Bd7**. (a) Our position; Melanie plays Bd7 (g4d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1818` (ply 57) [structural distance  $d=90$ , QAP obj=4]: the master played Be5; under  $\mu$  our Bg5 plays the role of that game’s Bf3; cross-type roles: our Nc6 as their Rb8, our Ng8 as their Pa7; transferred move g4d7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2765` (ply 54) [structural distance  $d=78$ , QAP obj=74]: the master played Bxe6; under  $\mu$  our Bg5 plays the role of that game’s Bc4; cross-type roles: our Ng8 as their Pb2, our Rh8 as their Ph3; transferred move g4h5. (d) Retrieved from `ccrl_top_p0001#3787` (ply 54) [structural distance  $d=89$ , QAP obj=57]: the master played Rxa6; under  $\mu$  our Bg5 plays the role of that game’s Rb6 (a **cross-type** correspondence); cross-type roles: our Ng8 as their Pa2, our Nc6 as their Pg2; transferred move g4e6. Synthesis: 1 retrieved analog transferred to Bd7 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



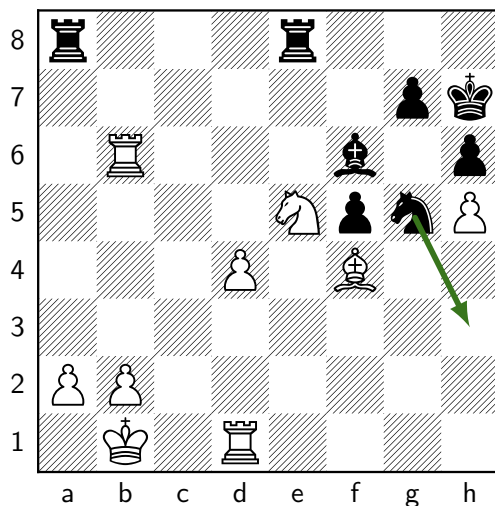
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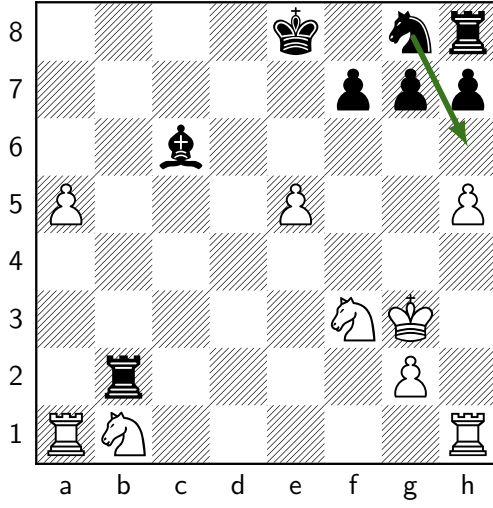


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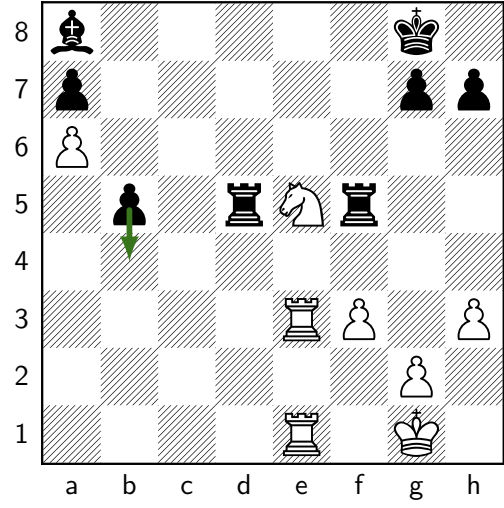


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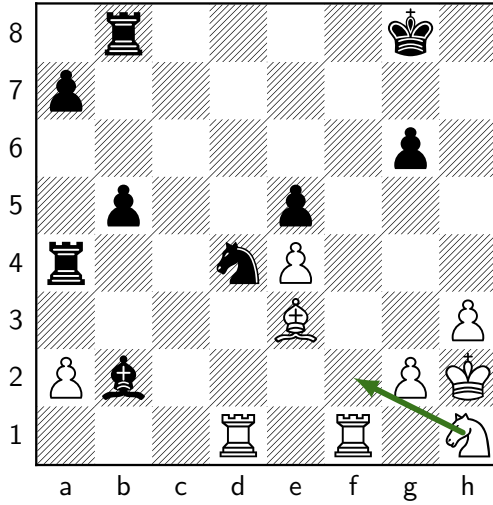
Figure 16: **Move 22, Black: Bxc6**. (a) Our position; Melanie plays Bxc6 (d7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1405` (ply 36) [structural distance  $d=74$ , QAP obj=49]: the master played `gxf3`; under  $\mu$  our *Bd2* plays the role of that game’s *Pg2* (a **cross-type** correspondence); cross-type roles: our *Bd7* as their *Rd1*, our *Ng8* as their *Pa2*; transferred move `d7c6` (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2426` (ply 57) [structural distance  $d=79$ , QAP obj=49]: the master played `Bc8`; under  $\mu$  our *Bd2* plays the role of that game’s *Be3*; cross-type roles: our *nb1* as their *Pg7*, our *Rb2* as their *Pf6*; transferred move `d7c6` (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1699` (ply 57) [structural distance  $d=78$ , QAP obj=42]: the master played `Nh3`; under  $\mu$  our *Bd2* plays the role of that game’s *Ng4* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *Ra8*, our *Ng8* as their *rb6*; transferred move `d7c6` (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Bxc6 (summed QAP objective 460, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



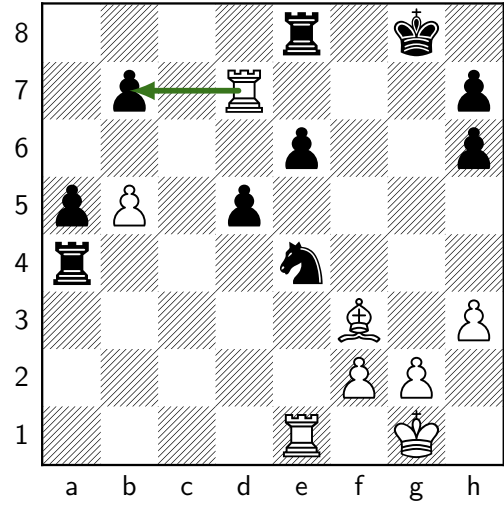
(a)



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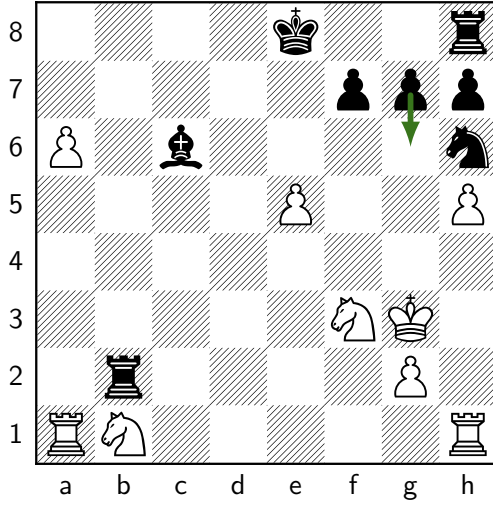


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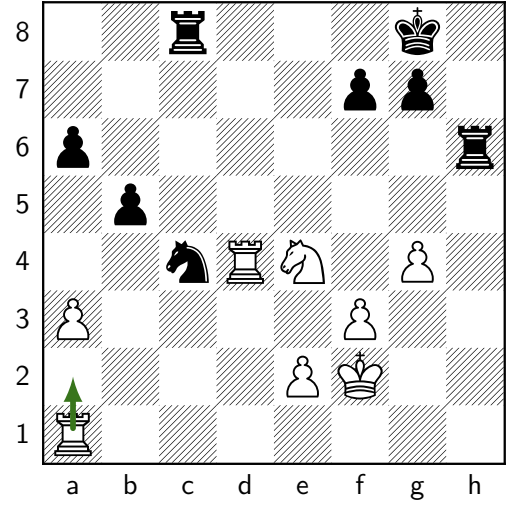


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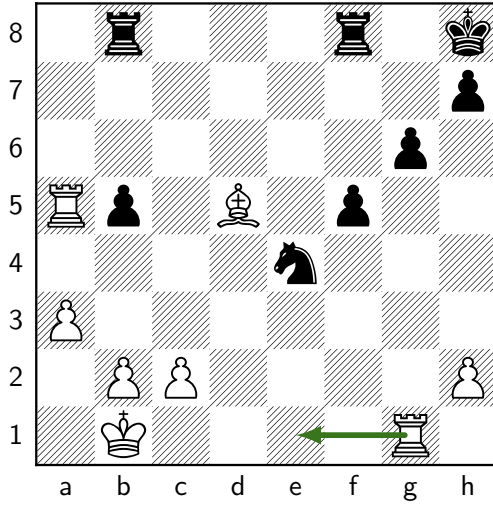
Figure 17: **Move 23, Black: Nh6**. (a) Our position; Melanie plays Nh6 (g8h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1484` (ply 57) [structural distance  $d=55$ , QAP obj=38]: the master played b4; under  $\mu$  our Ng1 plays the role of that game’s Pb4 (a **cross-type** correspondence); cross-type roles: our Pg7 as their Ba8, our nf3 as their pf3; transferred move g8h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1522` (ply 48) [structural distance  $d=58$ , QAP obj=23]: the master played Nf2; under  $\mu$  our Ng1 plays the role of that game’s Nh1; cross-type roles: our pe5 as their Rf1, our Bc6 as their Pa2; transferred move g8h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#610` (ply 60) [structural distance  $d=60$ , QAP obj=22]: the master played Rxb7; under  $\mu$  our Ng1 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our Bc6 as their Pf2, our Rb2 as their Bf3; transferred move g8h6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nh6 (summed QAP objective 83, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



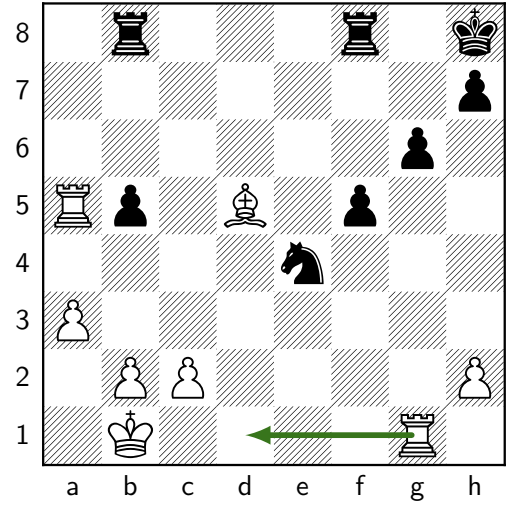
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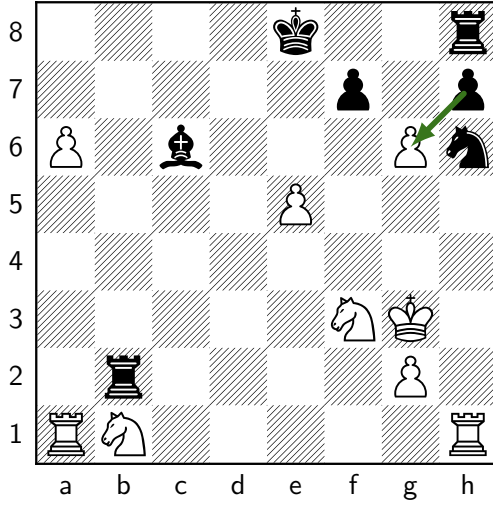


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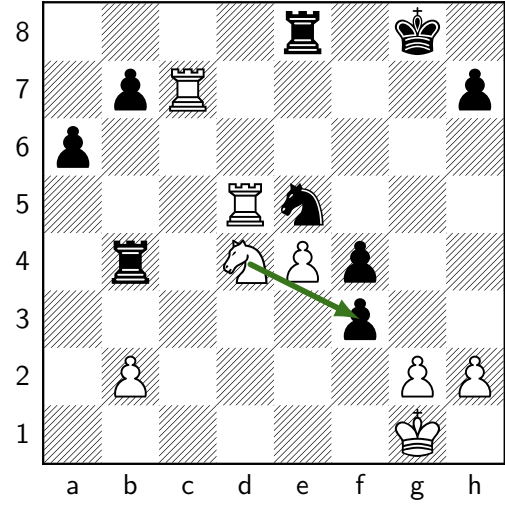


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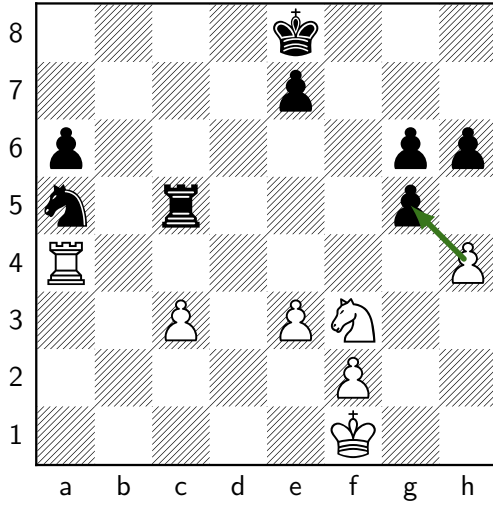
Figure 18: **Move 24, Black: g6** (g7g6). (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2741` (ply 58) [structural distance  $d=59$ , QAP obj=23]: the master played Ra2; under  $\mu$  our Pg2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Bc6 as their Ne4, our nf3 as their pb5; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3893` (ply 58) [structural distance  $d=60$ , QAP obj=28]: the master played Re1; under  $\mu$  our Nh3 plays the role of that game’s Rg1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pc2, our Rh8 as their Ph2; transferred move h6g8. (d) Retrieved from `ccr1_top_p0003#938` (ply 58) [structural distance  $d=60$ , QAP obj=28]: the master played Rd1; under  $\mu$  our Nh3 plays the role of that game’s Rg1 (a **cross-type** correspondence); cross-type roles: our ra1 as their Pc2, our Rh8 as their Ph2; transferred move h6g8. Synthesis: 1 retrieved analog transferred to g6 (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



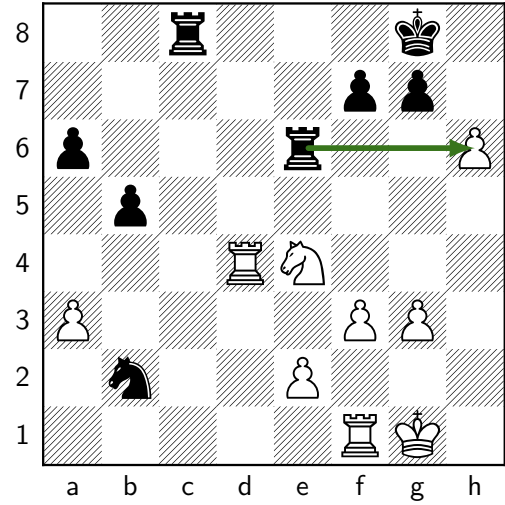
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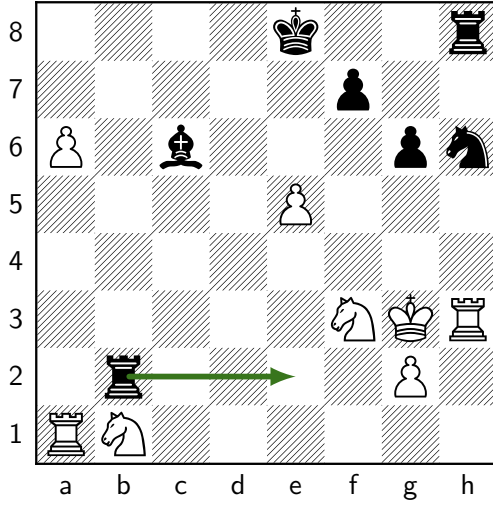


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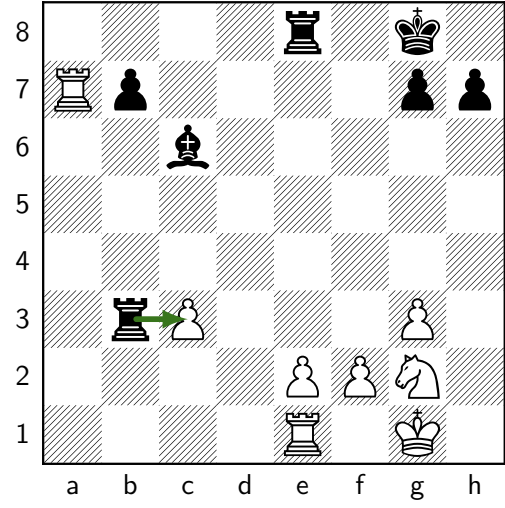


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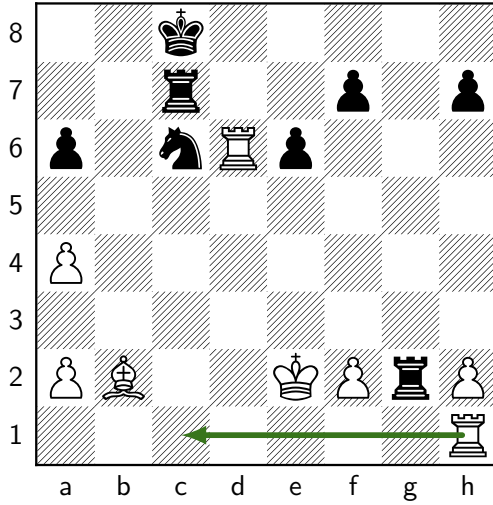
Figure 19: **Move 25, Black: hxg6.** (a) Our position; Melanie plays hxg6 (h7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3172` (ply 62) [structural distance  $d=62$ , QAP obj=46]: the master played Nxf3; under  $\mu$  our Ph2 plays the role of that game’s Nd4 (a **cross-type** correspondence); cross-type roles: our Rh8 as their Pb2, our rh1 as their Ph2; transferred move h7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3048` (ply 62) [structural distance  $d=74$ , QAP obj=41]: the master played hxg5; under  $\mu$  our Ph2 plays the role of that game’s Ph4; cross-type roles: our nf3 as their Pc3, our Rh8 as their Pe3; transferred move h7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2741` (ply 51) [structural distance  $d=68$ , QAP obj=26]: the master played Rxh6; under  $\mu$  our Ph2 plays the role of that game’s Re3 (a **cross-type** correspondence); cross-type roles: our Nh6 as their Pf7, our Bc6 as their Pa6; transferred move h7g6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to hxg6 (summed QAP objective 129, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



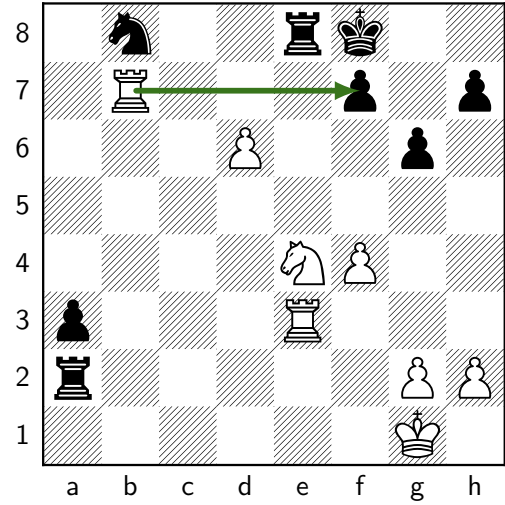
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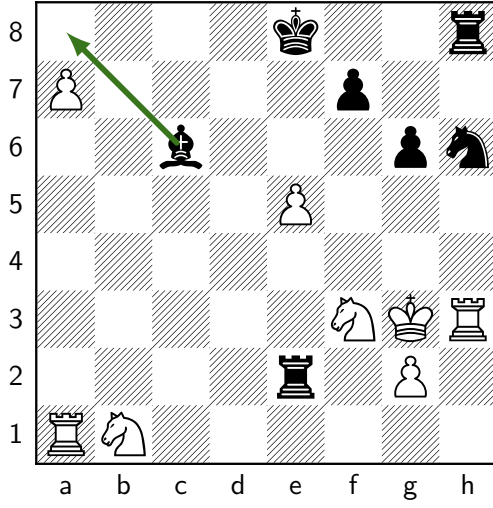


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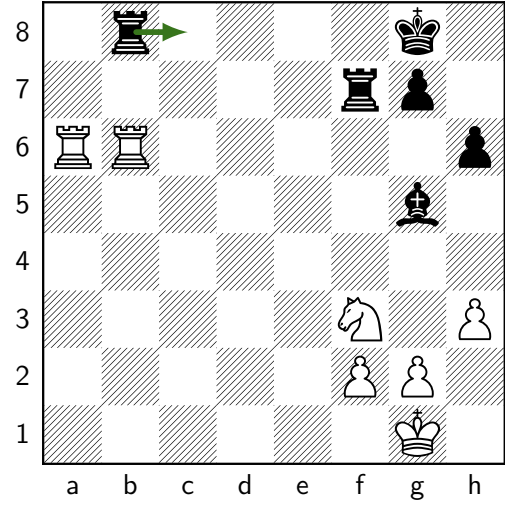


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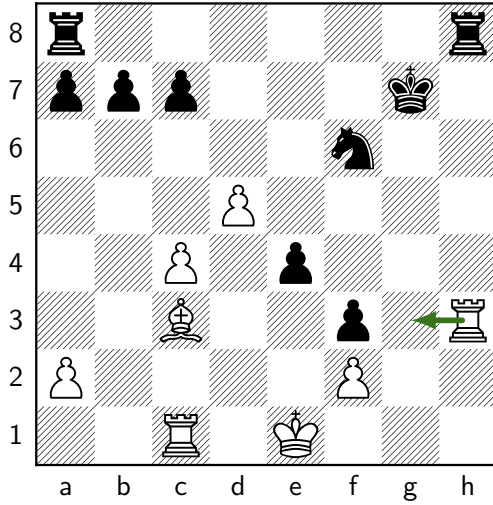
Figure 20: **Move 26, Black: Re2.** (a) Our position; Melanie plays **Re2** (**b2e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#573` (ply 59) [structural distance  $d=47$ , QAP obj=21]: the master played **Rxc3**; under  $\mu$  our **Rb7** plays the role of that game’s **Rb6**; cross-type roles: our **Nh6** as their **Pb7**, our **nb1** as their **pc3**; transferred move **b2e2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3230` (ply 54) [structural distance  $d=57$ , QAP obj=51]: the master played **Rc1**; under  $\mu$  our **Rh1** plays the role of that game’s **Rh1**; cross-type roles: our **Nh6** as their **Pf2**, our **Nh6** as their **Ph2**; transferred move **h8g8**. (d) Retrieved from `ccr1_top_p0002#357` (ply 54) [structural distance  $d=47$ , QAP obj=50]: the master played **Rxf7+**; under  $\mu$  our **Rb7** plays the role of that game’s **Rb7**; cross-type roles: our **nf3** as their **ra2**, our **Bc6** as their **Ph2**; transferred move **b2g2**. Synthesis: 1 retrieved analog transferred to **Re2** (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



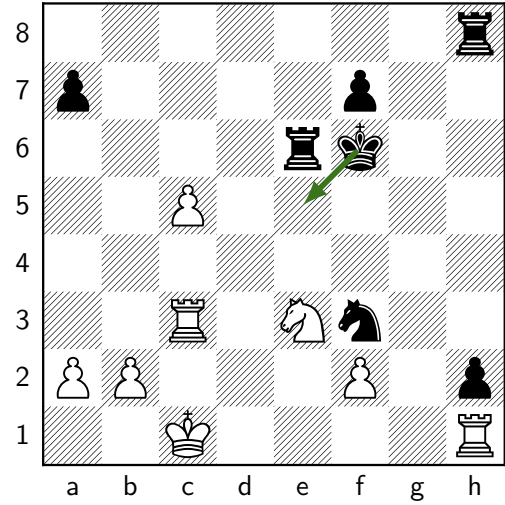
(a)



(b)

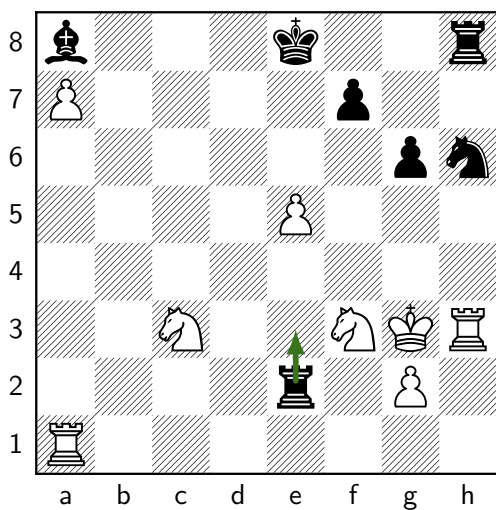


(c)

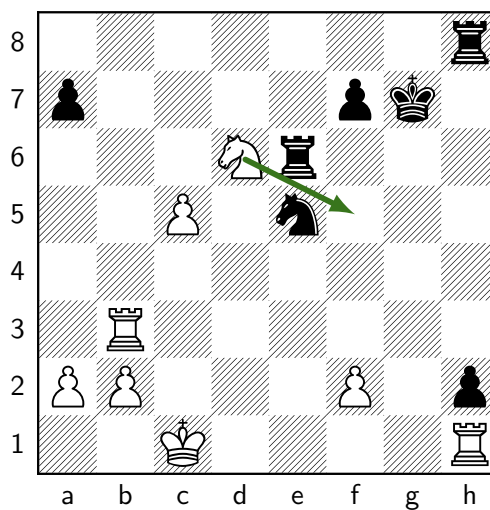


(d)

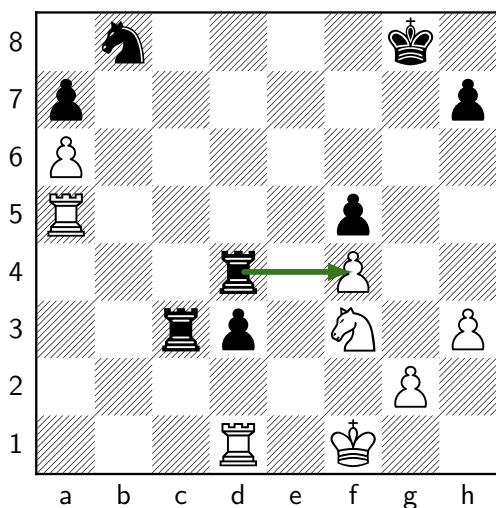
Figure 21: **Move 27, Black: Ba8**. (a) Our position; Melanie plays Ba8 (c6a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1712` (ply 67) [structural distance  $d=51$ , QAP obj=18]: the master played Rc8; under  $\mu$  our Bc3 plays the role of that game’s Rb1 (a **cross-type** correspondence); cross-type roles: our pa7 as their ra6, our nf3 as their rb6; transferred move c6a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2806` (ply 54) [structural distance  $d=56$ , QAP obj=62]: the master played Rg3+; under  $\mu$  our Bc3 plays the role of that game’s Rh3 (a **cross-type** correspondence); cross-type roles: our pg2 as their Rc1, our rh3 as their Bc3; transferred move c6b7. (d) Retrieved from `ccr1_top_p0000#822` (ply 69) [structural distance  $d=56$ , QAP obj=60]: the master played Ke5; under  $\mu$  our Ke1 plays the role of that game’s Kf3; cross-type roles: our ra1 as their Pa7, our Bc6 as their Re6; transferred move e8f8. Synthesis: 1 retrieved analog transferred to Ba8 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



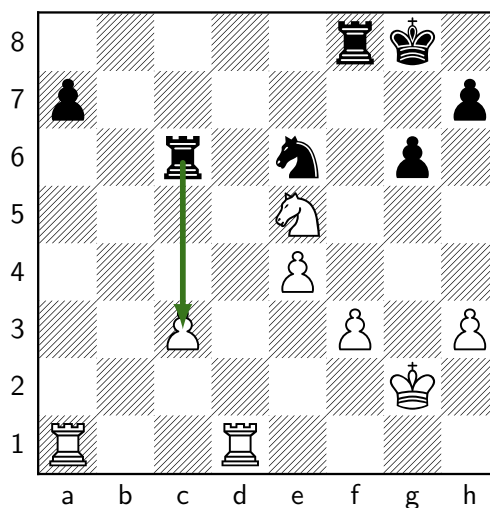
(a)



(b)

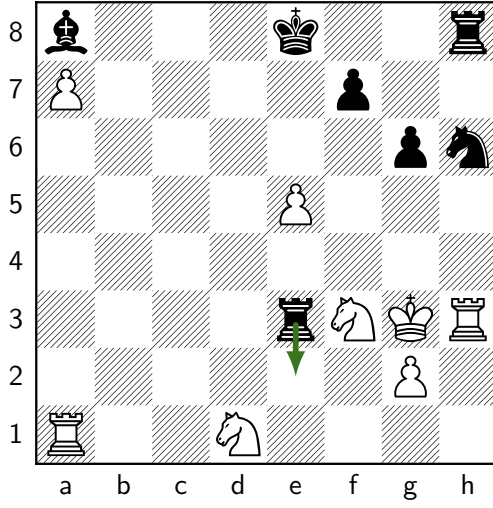


(c)

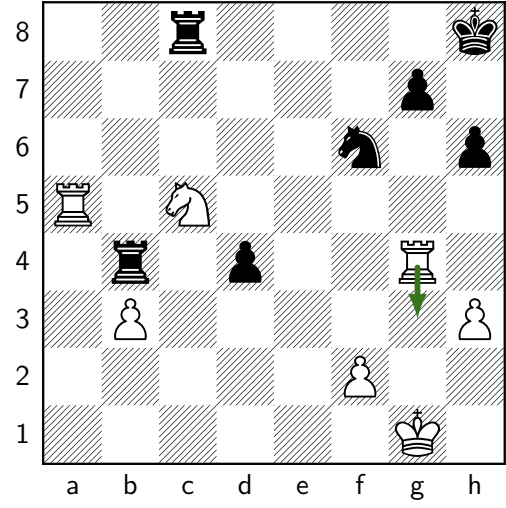


(d)

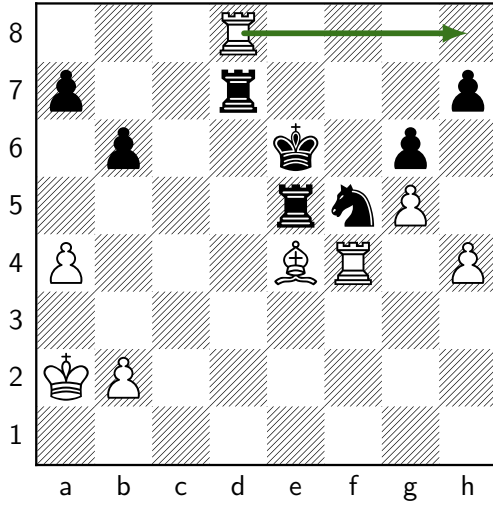
Figure 22: **Move 28, Black: Re3 (e2e3)**. (a) Our position; Melanie plays Re3 (e2e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#822` (ply 64) [structural distance  $d=50$ , QAP obj=53]: the master played Nf5+; under  $\mu$  our Re7 plays the role of that game’s Nd6 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Pa2, our Nh6 as their ph2; transferred move e2e3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3730` (ply 67) [structural distance  $d=54$ , QAP obj=44]: the master played Rxf4; under  $\mu$  our Re7 plays the role of that game’s Rd5; cross-type roles: our Pg6 as their Nb8, our Ba8 as their Pa7; transferred move e2e3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3814` (ply 53) [structural distance  $d=49$ , QAP obj=38]: the master played Rxc3; under  $\mu$  our Re7 plays the role of that game’s Rc3; cross-type roles: our Ba8 as their Pg6, our rh3 as their pe4; transferred move e2e3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Re3 (summed QAP objective 134, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



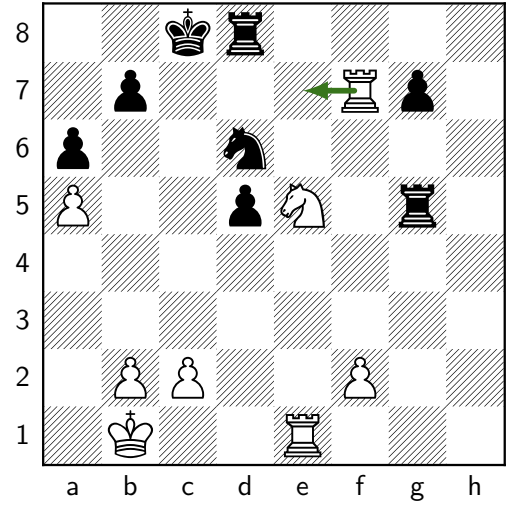
(a)



(b)

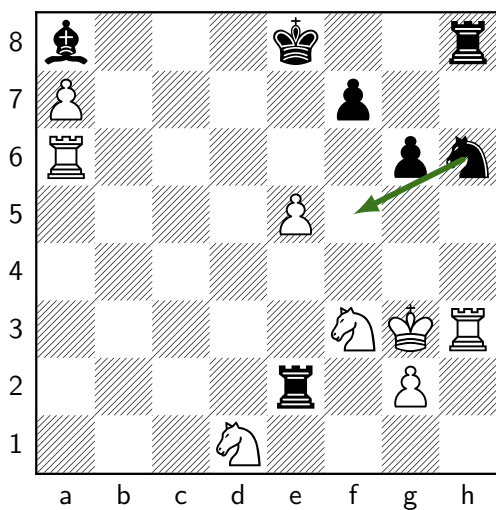


(c)

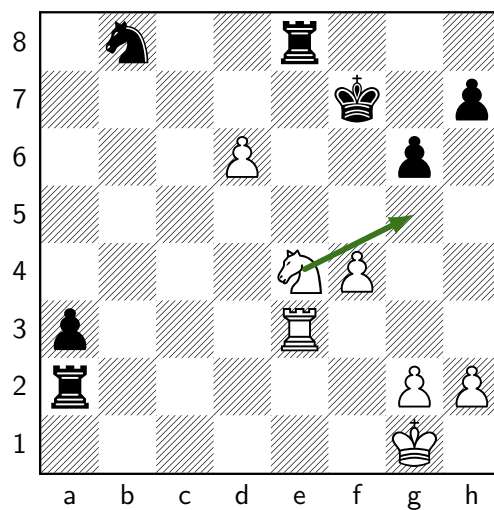


(d)

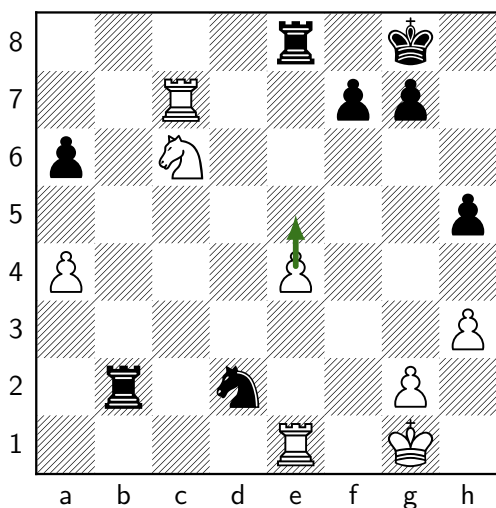
Figure 23: **Move 29, Black: Re2**. (a) Our position; Melanie plays **Re2** (**e3e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2604` (ply 72) [structural distance  $d=52$ , QAP obj=41]: the master played **Rg3**; under  $\mu$  our *Re6* plays the role of that game’s *Rg4*; cross-type roles: our *Ba8* as their *Ph3*, our *Pf7* as their *rb4*; transferred move **e3e2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#637` (ply 72) [structural distance  $d=54$ , QAP obj=38]: the master played **Rh8**; under  $\mu$  our *Re6* plays the role of that game’s *Rd8*; cross-type roles: our *Ba8* as their *Pb2*, our *rh3* as their *Pa4*; transferred move **e3e2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#17` (ply 62) [structural distance  $d=50$ , QAP obj=37]: the master played **Re7**; under  $\mu$  our *Re6* plays the role of that game’s *Rf7*; cross-type roles: our *Ba8* as their *Pb2*, our *ra1* as their *pd5*; transferred move **e3e2** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Re2** (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



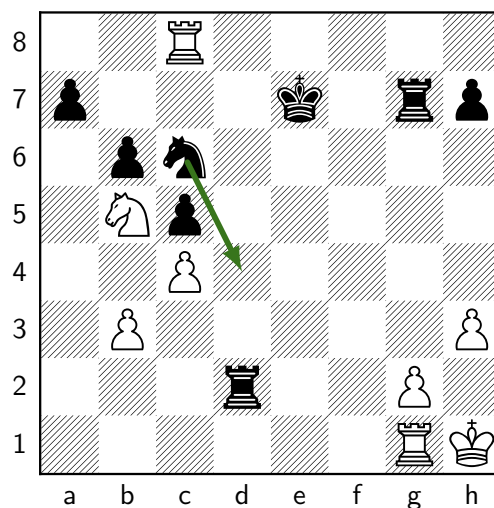
(a)



(b)



(c)



(d)

Figure 24: **Move 30, Black: Nf5+.** (a) Our position; Melanie plays Nf5+ (h6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#357` (ply 56) [structural distance  $d=50$ , QAP obj=40]: the master played Ng5+; under  $\mu$  our Nh3 plays the role of that game’s Ne4; cross-type roles: our nf3 as their ra2, our Ba8 as their Pg2; transferred move h6f5 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#1188` (ply 54) [structural distance  $d=51$ , QAP obj=27]: the master played e5; under  $\mu$  our Nh3 plays the role of that game’s Pe4 (a **cross-type** correspondence); cross-type roles: our pe5 as their rb2, our rh3 as their nd2; transferred move h6f5 (**endorses this move**). (d) Retrieved from `ccl1_top_p0002#1022` (ply 67) [structural distance  $d=54$ , QAP obj=20]: the master played Nd4; under  $\mu$  our Nh3 plays the role of that game’s Nc3; cross-type roles: our nf3 as their Pa7, our Ba8 as their nb5; transferred move h6f5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nf5+ (summed QAP objective 116, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

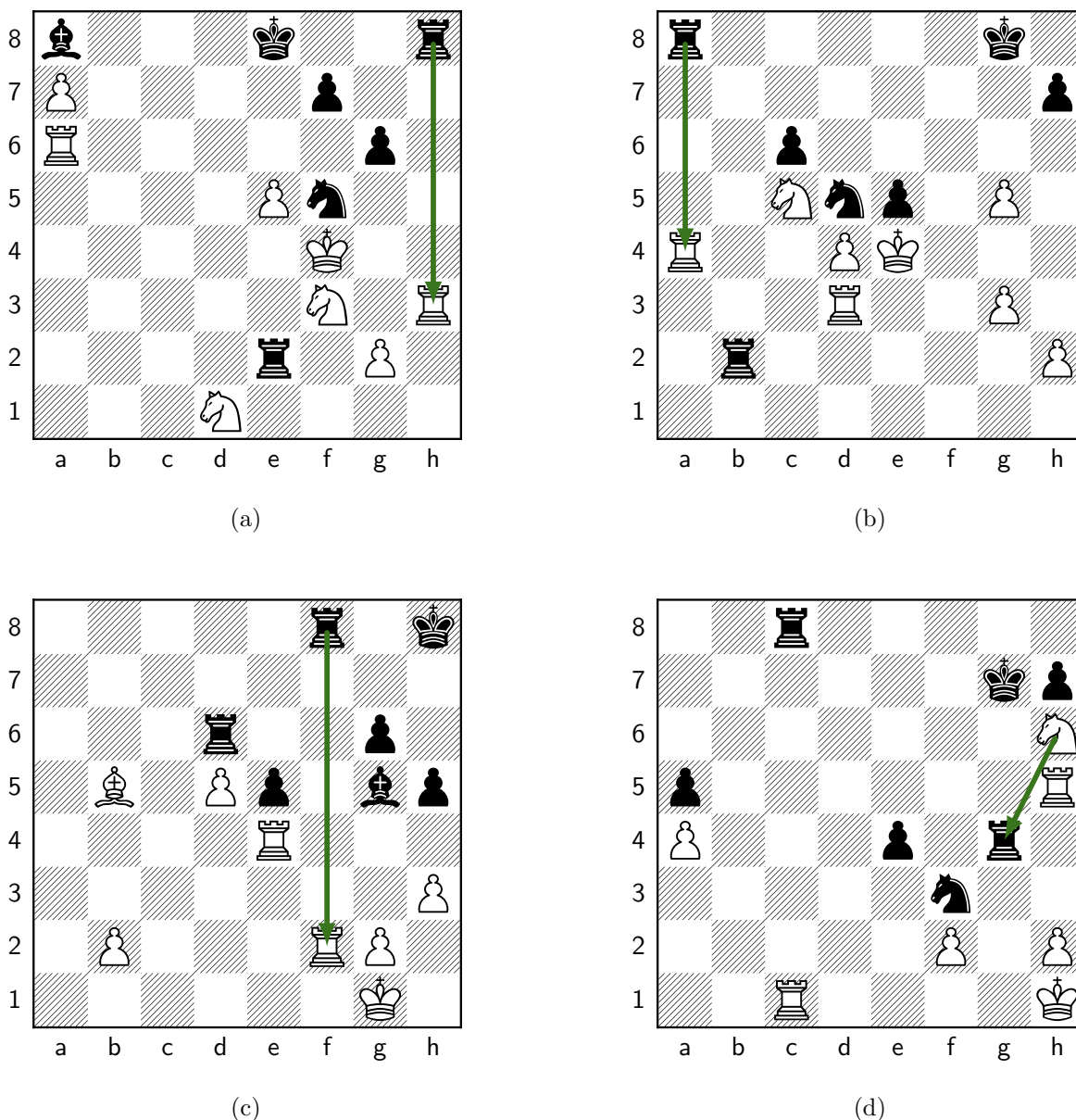
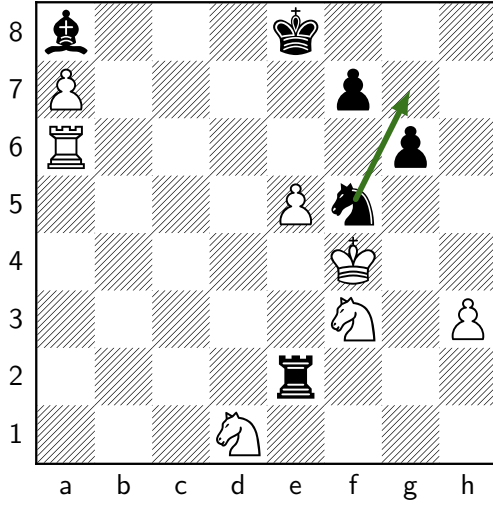
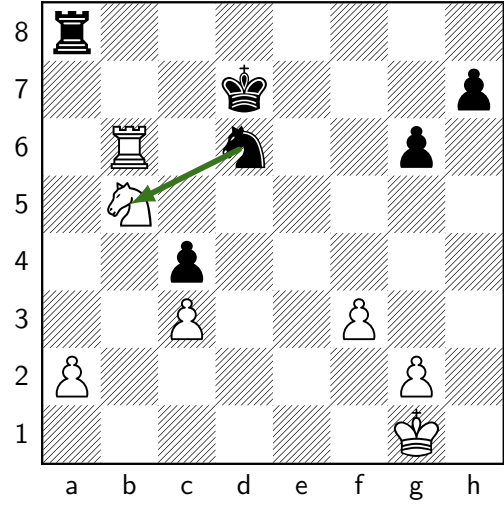


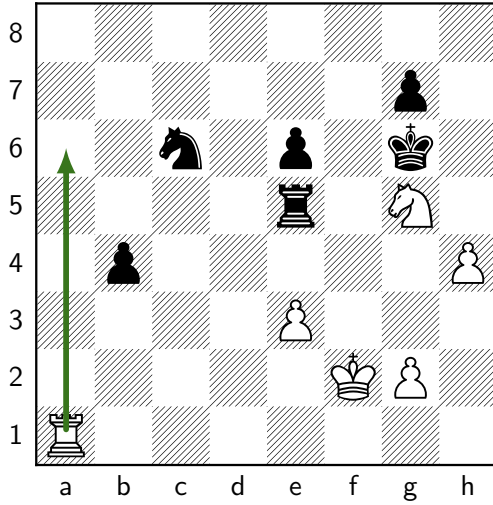
Figure 25: **Move 31, Black: Rxf3**. (a) Our position; Melanie plays **Rxf3** (**h8h3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3174` (ply 65) [structural distance  $d=65$ , QAP obj=41]: the master played **Rxa4**; under  $\mu$  our *Rh1* plays the role of that game's *Ra1*; cross-type roles: our *Re2* as their *Ph7*, our *pg2* as their *nc5*; transferred move **h8h3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1022` (ply 71) [structural distance  $d=66$ , QAP obj=19]: the master played **Rxf2**; under  $\mu$  our *Rh1* plays the role of that game's *Rf1*; cross-type roles: our *ra6* as their *bb5*, our *nd1* as their *pd5*; transferred move **h8h3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1285` (ply 76) [structural distance  $d=52$ , QAP obj=32]: the master played **Nxg4**; under  $\mu$  our *Nf4* plays the role of that game's *Nh6*; cross-type roles: our *Re2* as their *Pf2*, our *Pf7* as their *nf3*; transferred move **f5h6**. Synthesis: 2 retrieved analogs transferred to **Rxf3** (summed QAP objective 59, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



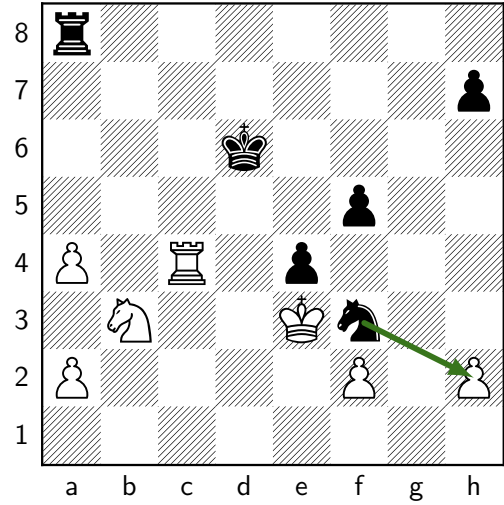
(a)



(b)

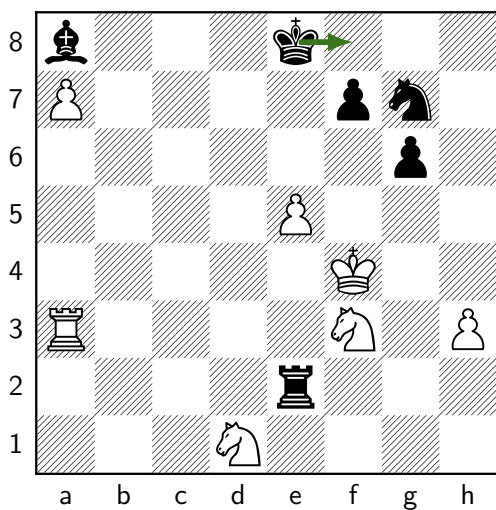


(c)

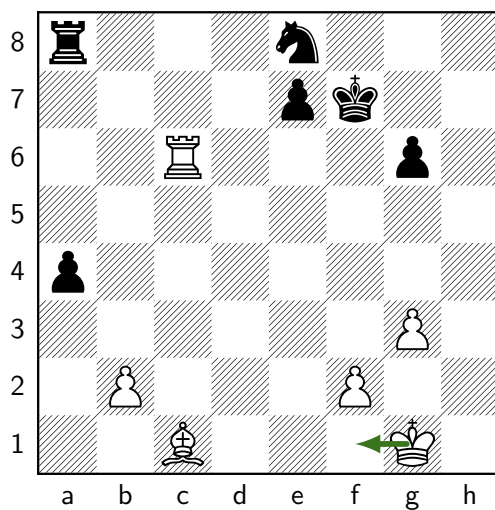


(d)

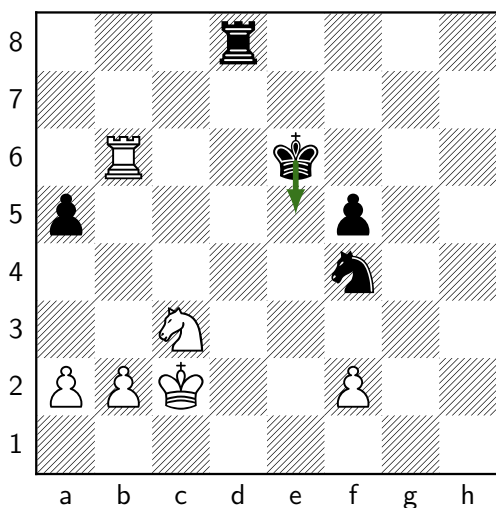
Figure 26: **Move 32, Black: Ng7 (f5g7).** (a) Our position; Melanie plays Ng7 (f5g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#441` (ply 79) [structural distance  $d=32$ , QAP obj=11]: the master played Nxb5; under  $\mu$  our Nf4 plays the role of that game’s Nd3; cross-type roles: our Ba8 as their Ra8, our kf4 as their rb6; transferred move f5g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1881` (ply 66) [structural distance  $d=38$ , QAP obj=27]: the master played Ra6; under  $\mu$  our Ba1 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Re2 as their Pe3, our kf4 as their re5; transferred move a8b7. (d) Retrieved from `ccr1_top_p0002#132` (ply 77) [structural distance  $d=35$ , QAP obj=23]: the master played Nxh2; under  $\mu$  our Nf4 plays the role of that game’s Nf6; cross-type roles: our nf3 as their rc4, our Nf5 as their Pe4; transferred move f5h6. Synthesis: 1 retrieved analog transferred to Ng7 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



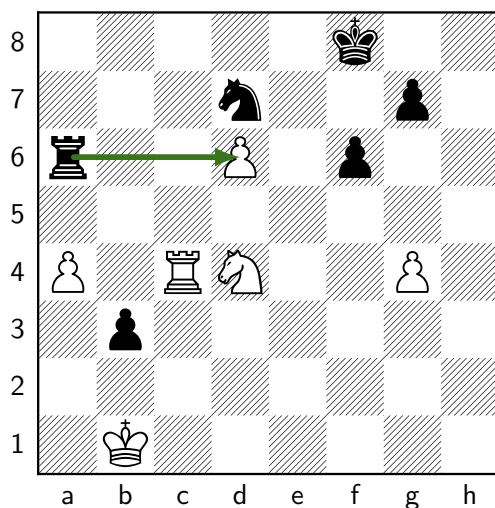
(a)



(b)



(c)



(d)

Figure 27: **Move 33, Black: Kf8**. (a) Our position; Melanie plays Kf8 (e8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1603` (ply 62) [structural distance  $d=32$ , QAP obj=11]: the master played Kf1; under  $\mu$  our Ke1 plays the role of that game’s Kg1; cross-type roles: our Ng7 as their Bc1, our Ba8 as their Pb2; transferred move e8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2508` (ply 81) [structural distance  $d=31$ , QAP obj=4]: the master played Ke5; under  $\mu$  our Ke1 plays the role of that game’s Ke3; cross-type roles: our Ba8 as their rb6, our ra3 as their pf2; transferred move e8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2448` (ply 81) [structural distance  $d=33$ , QAP obj=33]: the master played Rxd6; under  $\mu$  our Ba1 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our nf3 as their pd6, our nf3 as their pa4; transferred move a8c6. Synthesis: 2 retrieved analogs transferred to Kf8 (summed QAP objective 15, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

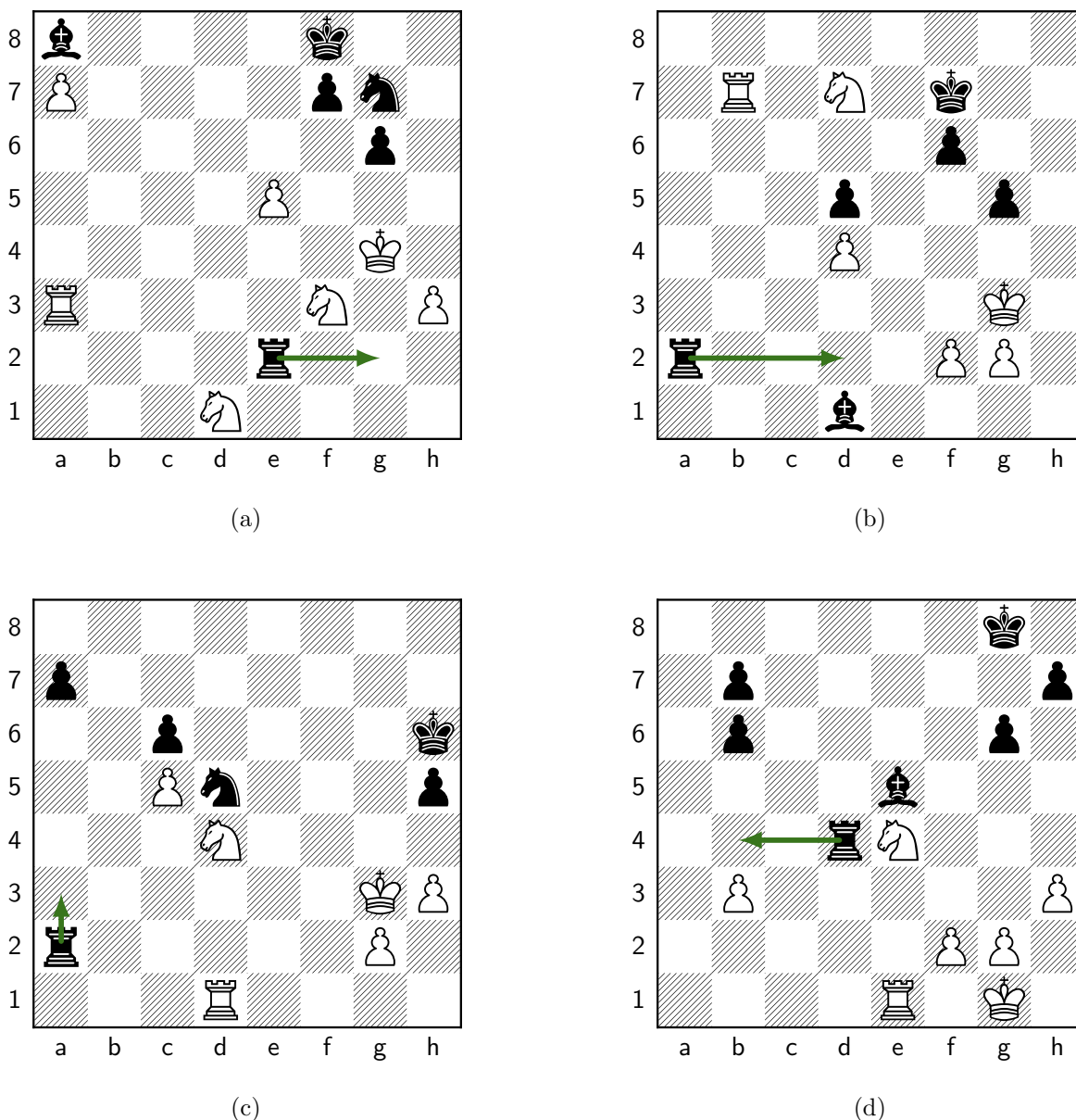


Figure 28: **Move 34, Black: Rg2+.** (a) Our position; Melanie plays Rg2+ (e2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1699` (ply 77) [structural distance  $d=32$ , QAP obj=22]: the master played Rd2; under  $\mu$  our Re7 plays the role of that game’s Ra7; cross-type roles: our pa7 as their rb7, our Ba8 as their nd7; transferred move e2g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3632` (ply 79) [structural distance  $d=32$ , QAP obj=21]: the master played Ra3+; under  $\mu$  our Re7 plays the role of that game’s Ra7; cross-type roles: our ra3 as their Pa7, our nf3 as their Pc6; transferred move e2g2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2437` (ply 77) [structural distance  $d=31$ , QAP obj=1]: the master played Rb4; under  $\mu$  our Re7 plays the role of that game’s Rd5; cross-type roles: our Ng7 as their Pb6, our ra3 as their Be5; transferred move e2g2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rg2+ (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

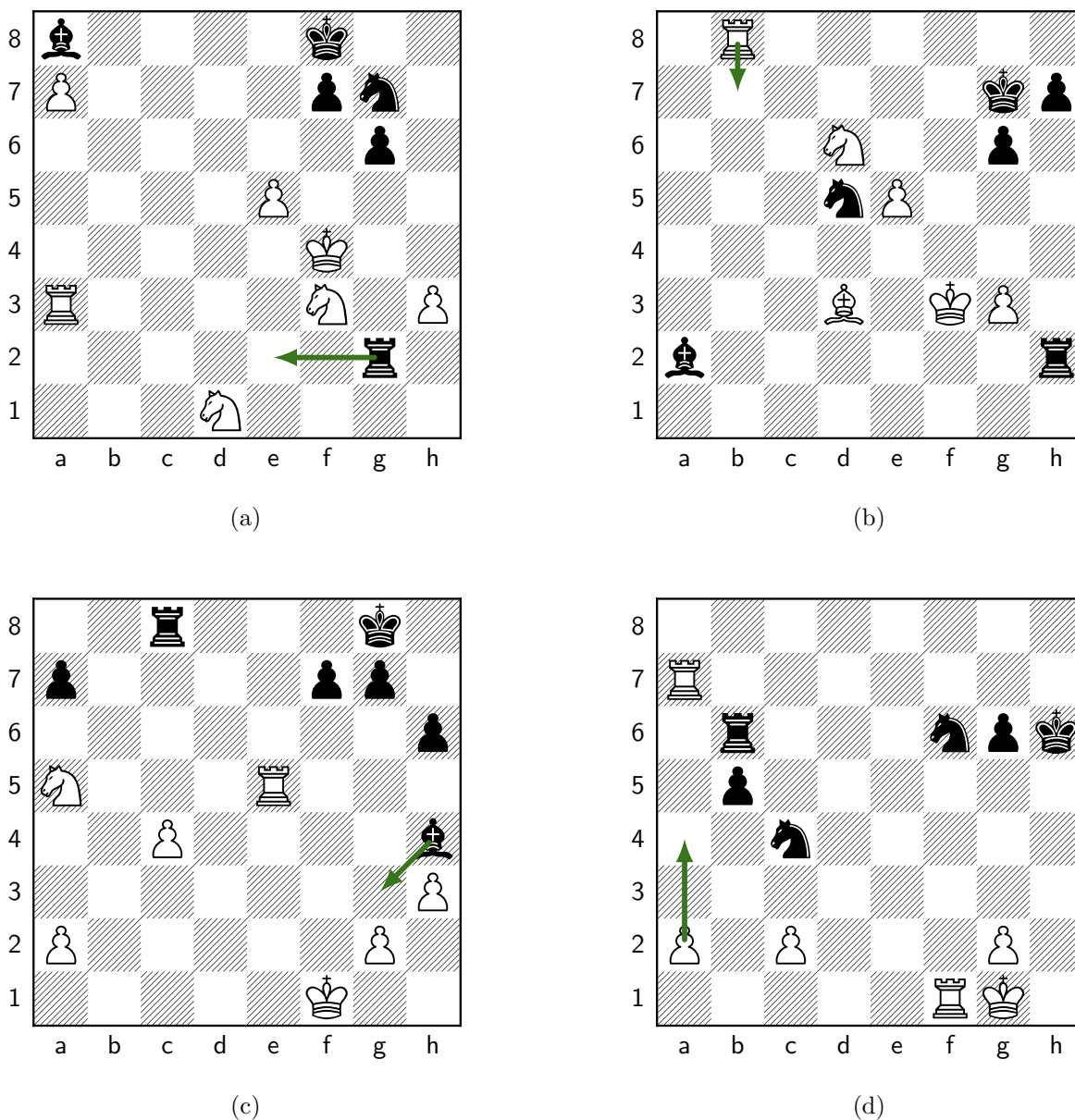
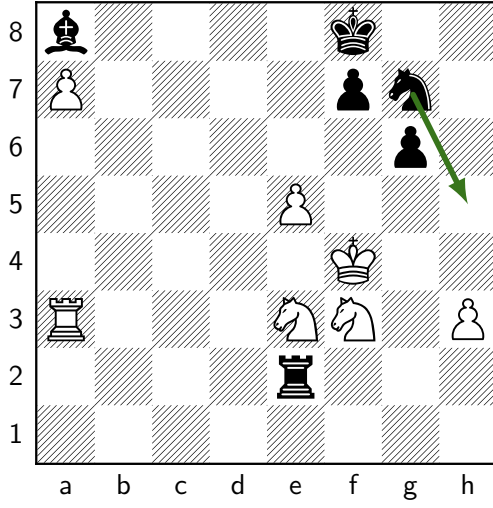
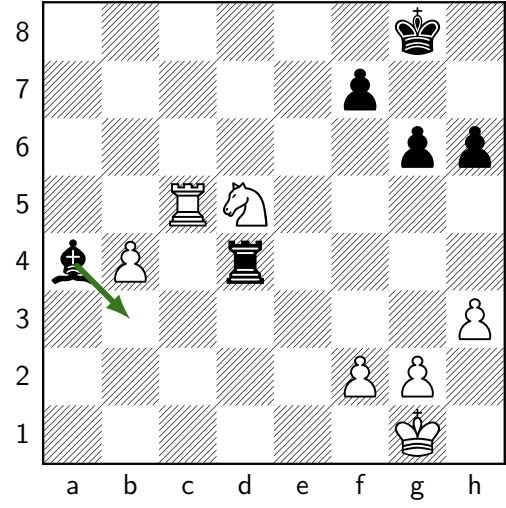


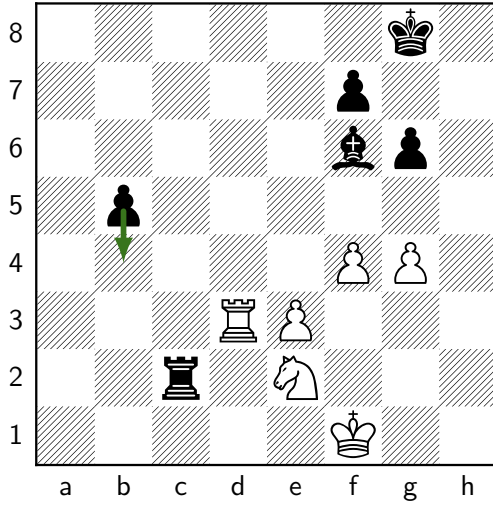
Figure 29: **Move 35, Black: Re2**. (a) Our position; Melanie plays **Re2** (**g2e2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1815` (ply 82) [structural distance  $d=31$ , QAP obj=14]: the master played **Rb7+**; under  $\mu$  our *Rg7* plays the role of that game’s *Rb8*; cross-type roles: our *nd1* as their *ba2*, our *pe5* as their *nd5*; transferred move **g2e2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3465` (ply 63) [structural distance  $d=30$ , QAP obj=9]: the master played **Bg3**; under  $\mu$  our *Rg7* plays the role of that game’s *Bh5* (a **cross-type** correspondence); cross-type roles: our *Ba8* as their *Rc8*, our *ra3* as their *na5*; transferred move **g2e2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2684` (ply 64) [structural distance  $d=30$ , QAP obj=21]: the master played **a4**; under  $\mu$  our *Pf2* plays the role of that game’s *Pa2*; cross-type roles: our *Ba8* as their *Rf1*, our *Ng7* as their *Pc2*; transferred move **f7f6**. Synthesis: 2 retrieved analogs transferred to **Re2** (summed QAP objective 23, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



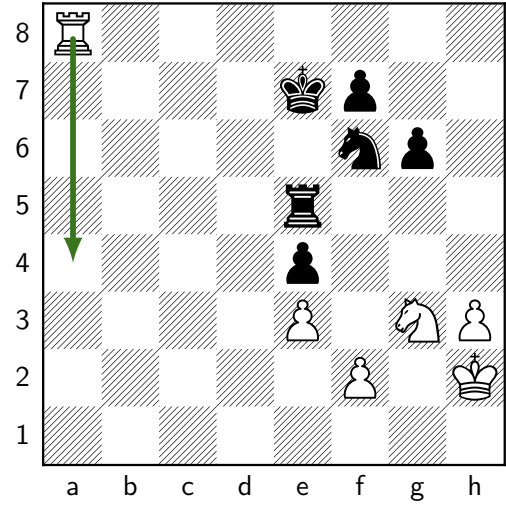
(a)



(b)

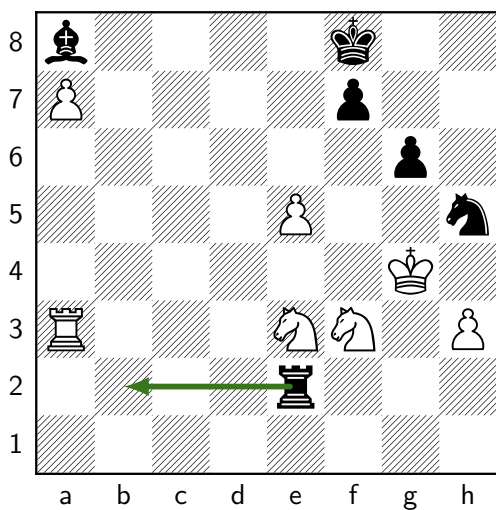


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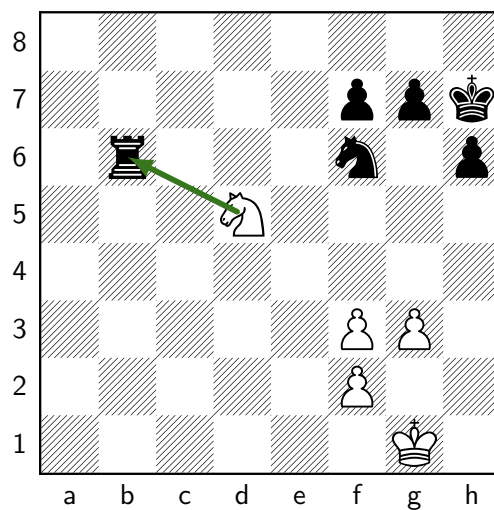


(d)

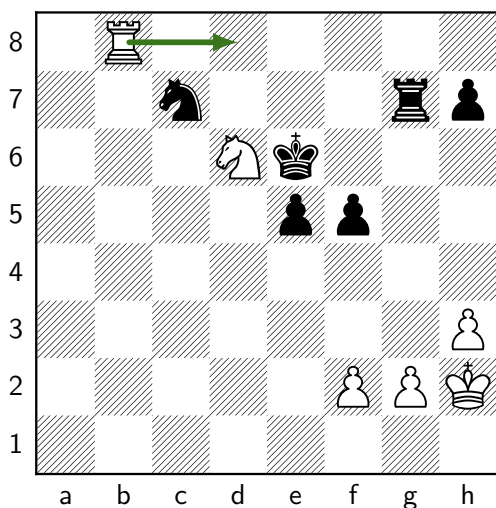
Figure 30: **Move 36, Black: Nh5+.** (a) Our position; Melanie plays Nh5+ (g7h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1049` (ply 75) [structural distance  $d=36$ , QAP obj=20]: the master played Bb3; under  $\mu$  our Ng2 plays the role of that game’s Ba5 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Ph6, our nf3 as their ph3; transferred move g7h5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1449` (ply 85) [structural distance  $d=38$ , QAP obj=14]: the master played b4; under  $\mu$  our Ng2 plays the role of that game’s Pb4 (a **cross-type** correspondence); cross-type roles: our nf3 as their pg4; transferred move g7h5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1683` (ply 80) [structural distance  $d=35$ , QAP obj=3]: the master played Ra4; under  $\mu$  our Ng2 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Pe3, our Re2 as their Ng3; transferred move g7h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nh5+ (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



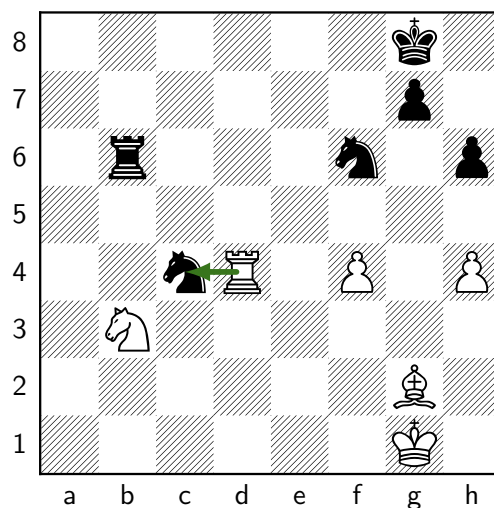
(a)



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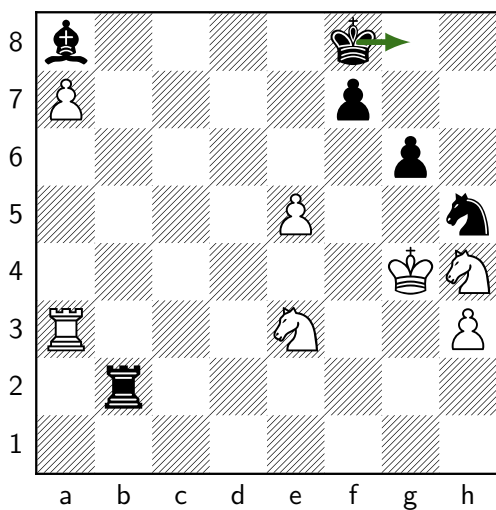


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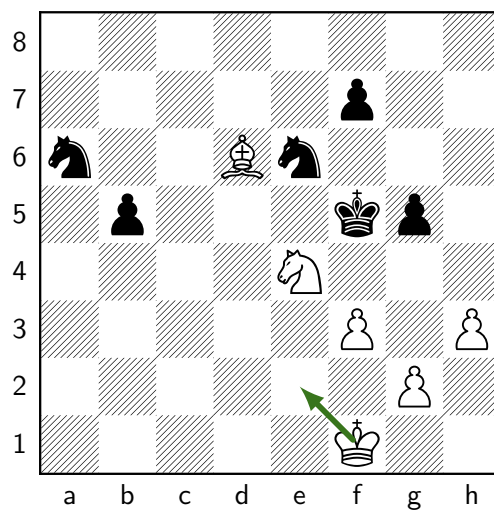


(d)

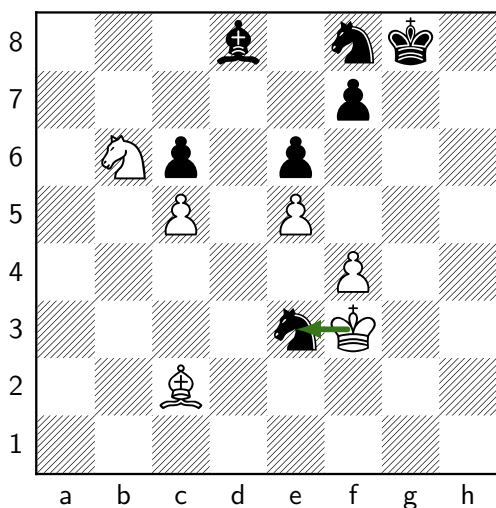
Figure 31: **Move 37, Black: Rb2**. (a) Our position; Melanie plays Rb2 (e2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2047` (ply 86) [structural distance  $d=41$ , QAP obj=23]: the master played **Nxb6**; under  $\mu$  our *Re7* plays the role of that game’s *Nd5* (a **cross-type** correspondence); cross-type roles: our *ne3* as their *rb6*, our *Ba8* as their *ph6*; transferred move **e2b2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3419` (ply 82) [structural distance  $d=35$ , QAP obj=22]: the master played **Rd8**; under  $\mu$  our *Re7* plays the role of that game’s *Rb8*; cross-type roles: our *Ba8* as their *Pf2*, our *nf3* as their *pf5*; transferred move **e2b2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1877` (ply 74) [structural distance  $d=41$ , QAP obj=22]: the master played **Rxc4**; under  $\mu$  our *Re7* plays the role of that game’s *Rd4*; cross-type roles: our *kg4* as their *rb6*, our *ph3* as their *nf6*; transferred move **e2b2** (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Rb2 (summed QAP objective 166, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



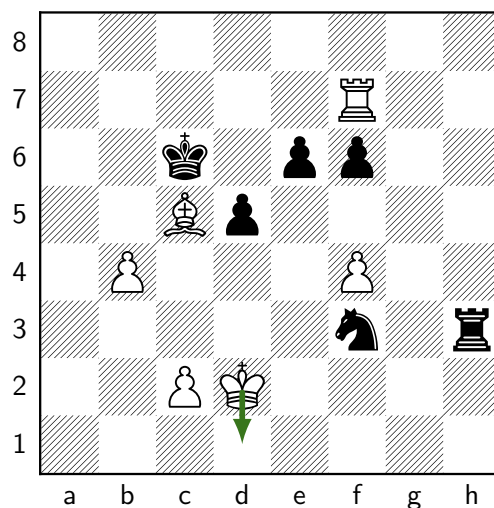
(a)



(b)

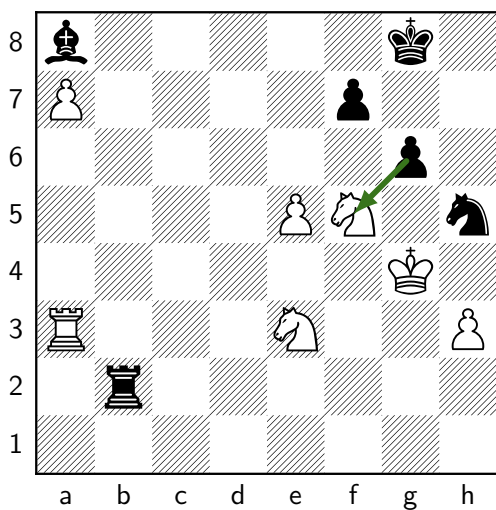


(c)

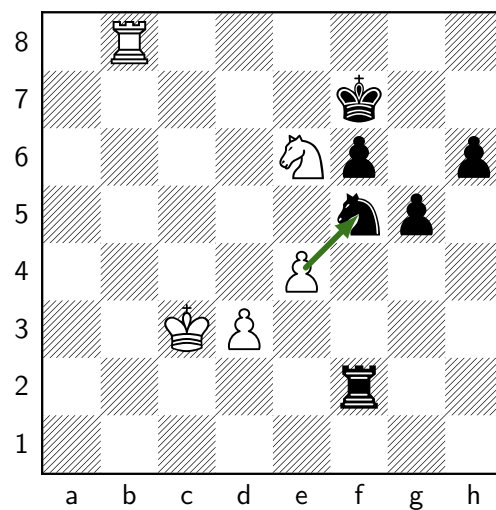


(d)

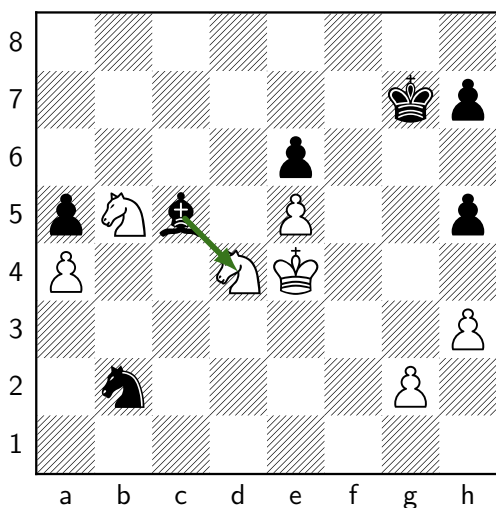
Figure 32: **Move 38, Black: Kg8. (a)** Our position; Melanie plays Kg8 (f8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3078` (ply 90) [structural distance  $d=37$ , QAP obj=24]: the master played K<sub>e2</sub>; under  $\mu$  our K<sub>f1</sub> plays the role of that game's K<sub>f1</sub>; cross-type roles: our B<sub>a8</sub> as their p<sub>b5</sub>, our n<sub>h4</sub> as their p<sub>g5</sub>; transferred move f8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2726` (ply 84) [structural distance  $d=43$ , QAP obj=7]: the master played K<sub>x</sub>e3; under  $\mu$  our K<sub>f1</sub> plays the role of that game's K<sub>f3</sub>; cross-type roles: our R<sub>b2</sub> as their P<sub>e5</sub>, our n<sub>e3</sub> as their p<sub>f7</sub>; transferred move f8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2655` (ply 80) [structural distance  $d=45$ , QAP obj=3]: the master played K<sub>d</sub>1; under  $\mu$  our K<sub>f1</sub> plays the role of that game's K<sub>d2</sub>; cross-type roles: our p<sub>e5</sub> as their n<sub>f3</sub>, our B<sub>a8</sub> as their r<sub>h3</sub>; transferred move f8g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



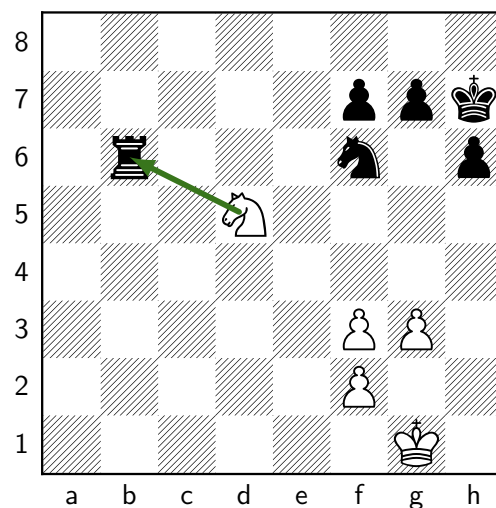
(a)



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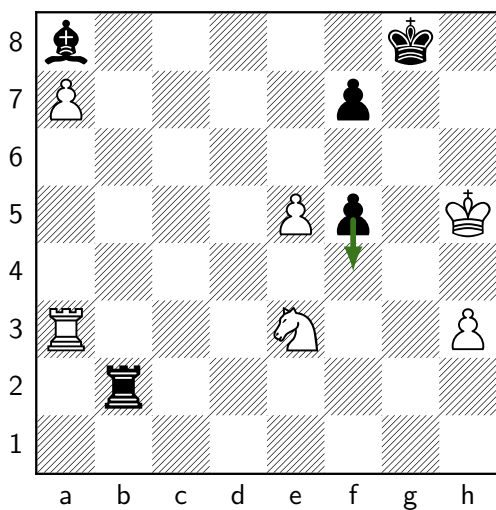


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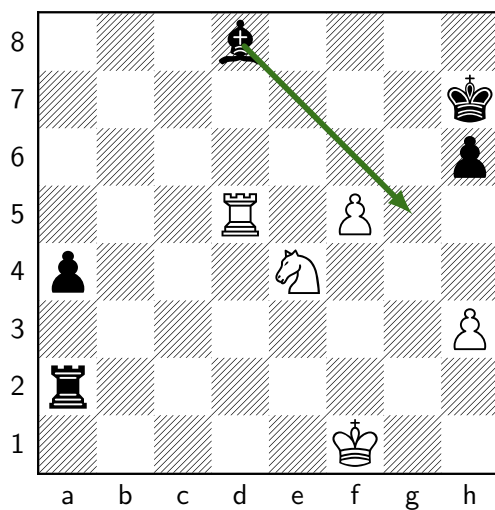


(d)

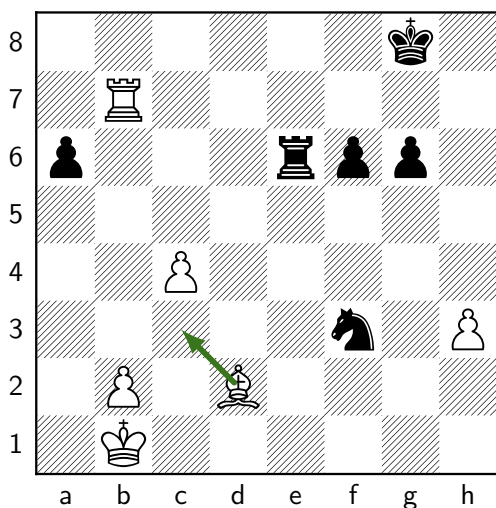
Figure 33: **Move 39, Black: gxf5+.** (a) Our position; Melanie plays gxf5+ (g6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#1089 (ply 92) [structural distance  $d=52$ , QAP obj=39]: the master played exf5; under  $\mu$  our Pg3 plays the role of that game’s Pe4; cross-type roles: our Ba8 as their rf2, our ne3 as their ph6; transferred move g6f5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#2150 (ply 83) [structural distance  $d=50$ , QAP obj=21]: the master played Bxd4; under  $\mu$  our Pg3 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Ph7, our Nh5 as their Pe6; transferred move g6f5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#2047 (ply 86) [structural distance  $d=51$ , QAP obj=21]: the master played Nxb6; under  $\mu$  our Pg3 plays the role of that game’s Nd5 (a **cross-type** correspondence); cross-type roles: our Nh5 as their Pf3, our Ba8 as their Pg3; transferred move g6f5 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to gxf5+ (summed QAP objective 160, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



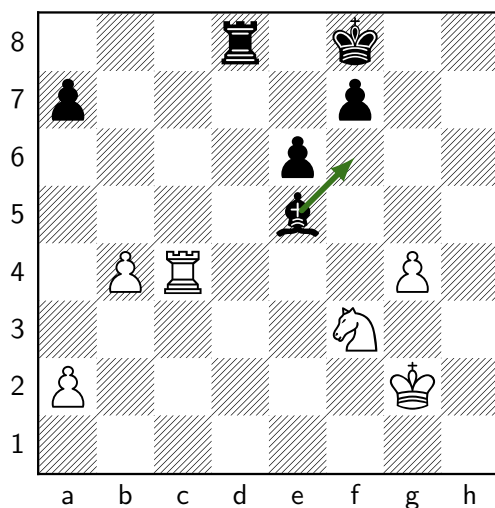
(a)



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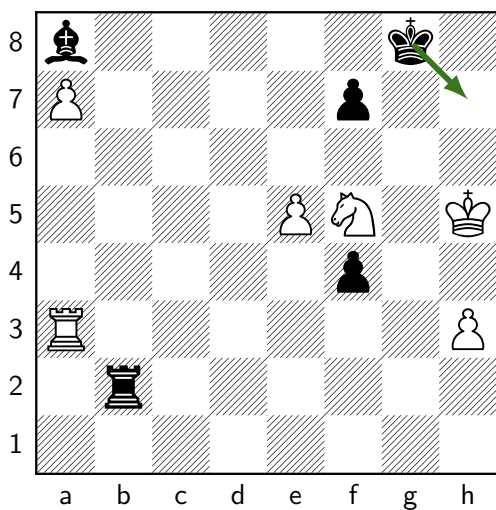


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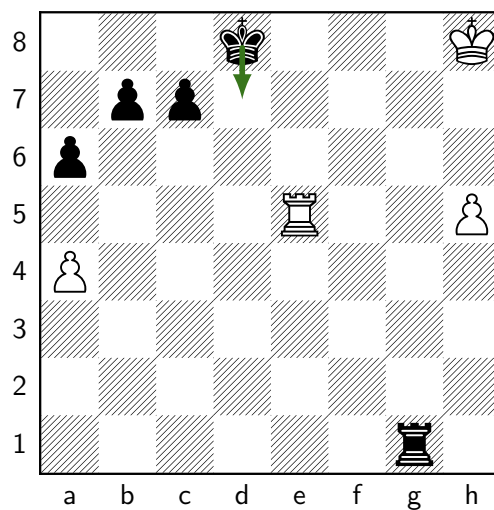


(d)

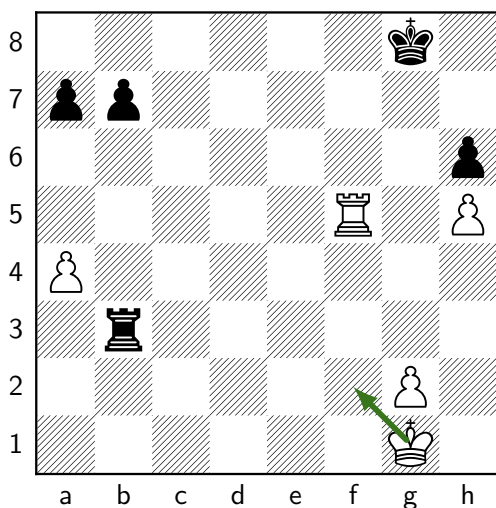
Figure 34: **Move 40, Black: f4 (f5f4)**. (a) Our position; Melanie plays f4 (f5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1840` (ply 95) [structural distance  $d=18$ , QAP obj=12]: the master played Bg5; under  $\mu$  our Pf4 plays the role of that game’s Bd1 (a **cross-type** correspondence); cross-type roles: our ne3 as their rd5, our Ba8 as their pf5; transferred move f5f4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2262` (ply 82) [structural distance  $d=15$ , QAP obj=10]: the master played Bc3; under  $\mu$  our Pf4 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Ph3, our Rb2 as their Pc4; transferred move f5f4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#290` (ply 73) [structural distance  $d=20$ , QAP obj=9]: the master played Bf6; under  $\mu$  our Pf4 plays the role of that game’s Be4 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Rd8, our ra3 as their pb4; transferred move f5f4 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to f4 (summed QAP objective 47, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



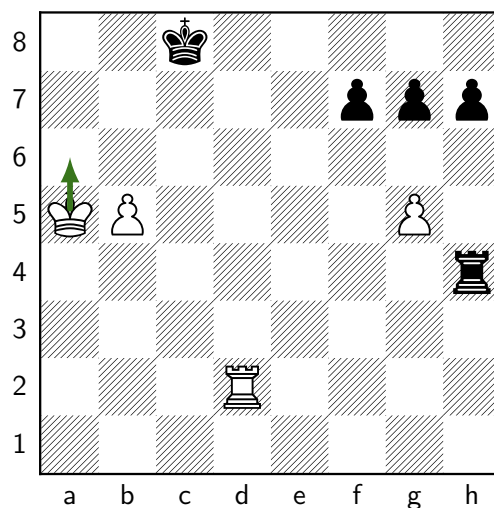
(a)



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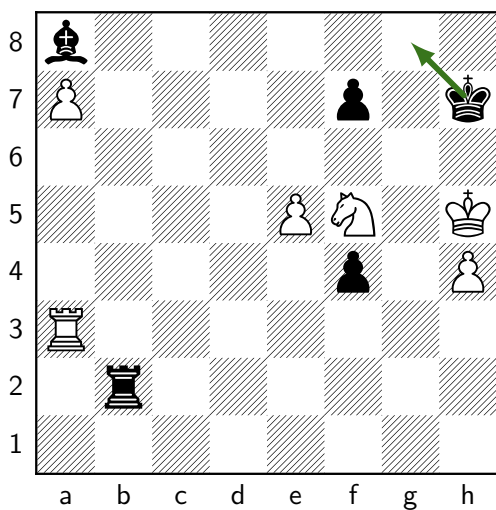


(c)

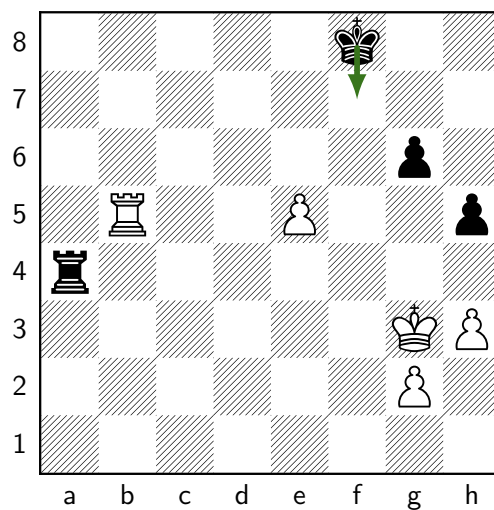


(d)

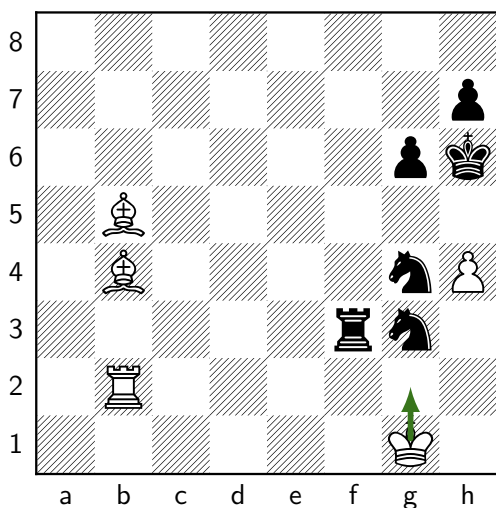
Figure 35: **Move 41, Black: Kh7**. (a) Our position; Melanie plays Kh7 (g8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3795` (ply 91) [structural distance  $d=17$ , QAP obj=3]: the master played Kd7; under  $\mu$  our *Kg1* plays the role of that game’s *Kd1*; cross-type roles: our *Ba8* as their *Pb7*, our *nf5* as their *pa4*; transferred move g8h7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2688` (ply 64) [structural distance  $d=15$ , QAP obj=3]: the master played Kf2; under  $\mu$  our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *pe5* as their *rb3*, our *nf5* as their *Pa4*; transferred move g8h7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3526` (ply 90) [structural distance  $d=13$ , QAP obj=3]: the master played Ka6; under  $\mu$  our *Kg1* plays the role of that game’s *Ka5*; cross-type roles: our *Ba8* as their *Rd2*, our *nf5* as their *pg7*; transferred move g8h7 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kh7 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



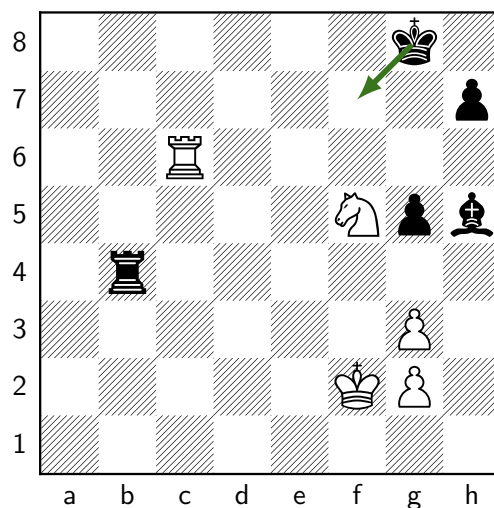
(a)



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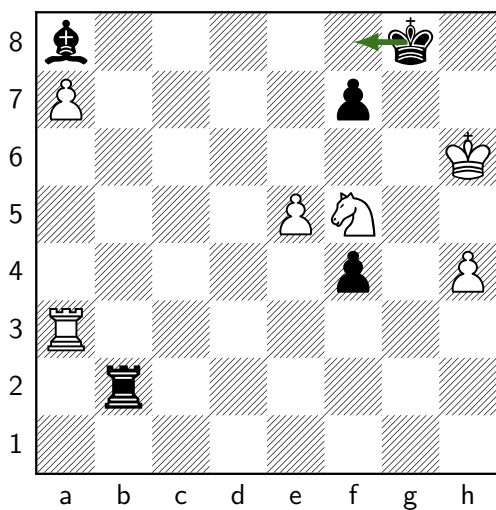


(c)

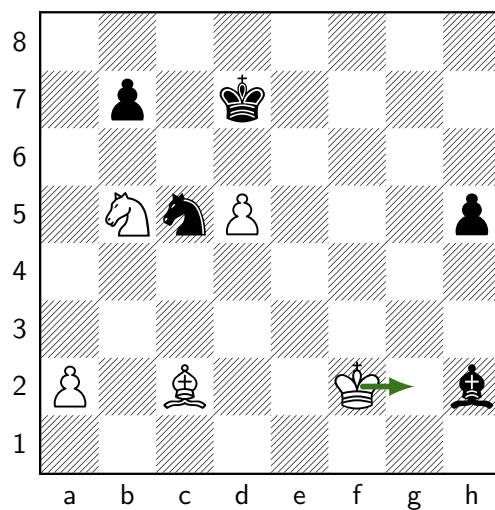


(d)

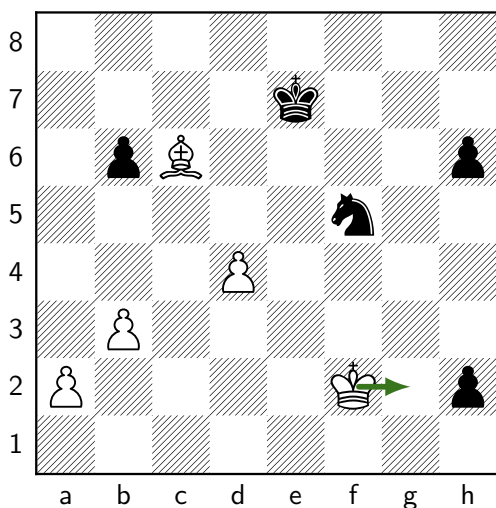
Figure 36: **Move 42, Black: Kg8 (h7g8)**. (a) Our position; Melanie plays Kg8 (h7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#716` (ply 81) [structural distance  $d=20$ , QAP obj=3]: the master played Kf7; under  $\mu$  our Kh2 plays the role of that game’s Kf1; cross-type roles: our Ba8 as their rb5, our nf5 as their Ph5; transferred move h7g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2482` (ply 84) [structural distance  $d=20$ , QAP obj=2]: the master played Kg2; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our Ba8 as their Rb2, our pe5 as their rf3; transferred move h7g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#2945` (ply 83) [structural distance  $d=18$ , QAP obj=2]: the master played Kf7; under  $\mu$  our Kh2 plays the role of that game’s Kg1; cross-type roles: our Ba8 as their nf5, our nf5 as their Pg5; transferred move h7g8 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kg8 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



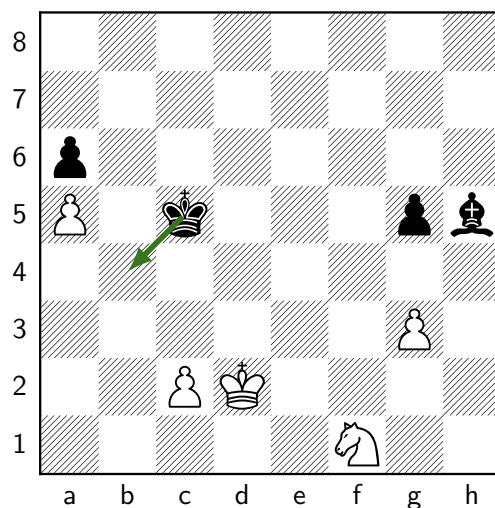
(a)



(b)

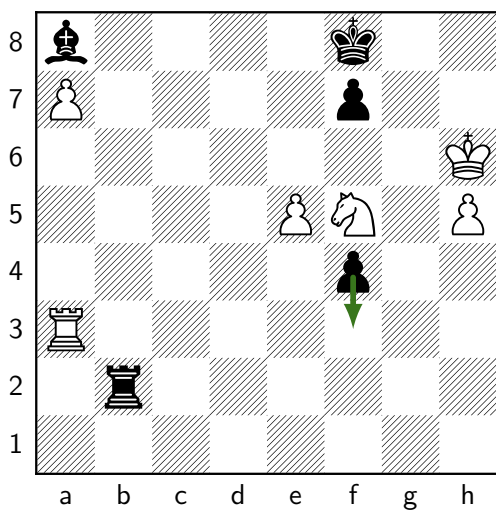


(c)

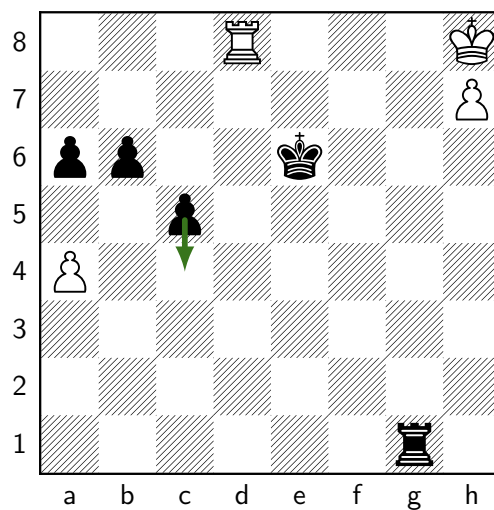


(d)

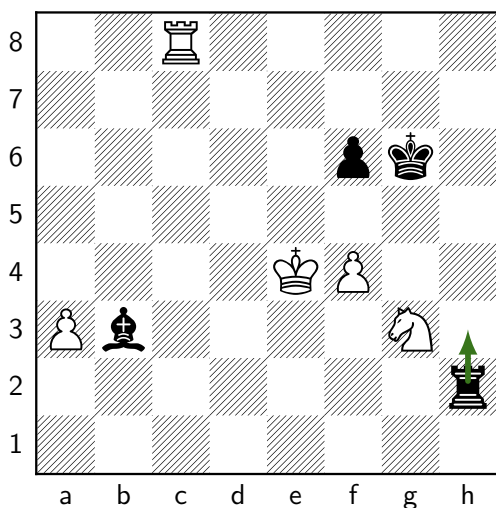
Figure 37: **Move 43, Black: Kf8.** (a) Our position; Melanie plays Kf8 (g8f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1021` (ply 84) [structural distance  $d=21$ , QAP obj=3]: the master played Kg2; under  $\mu$  our Kg1 plays the role of that game’s Kf2; cross-type roles: our ph4 as their Bc2, our Rb2 as their bh2; transferred move g8f8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#389` (ply 86) [structural distance  $d=21$ , QAP obj=3]: the master played Kg2; under  $\mu$  our Kg1 plays the role of that game’s Kf2; cross-type roles: our ra3 as their Pb3; transferred move g8f8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1287` (ply 95) [structural distance  $d=21$ , QAP obj=3]: the master played Kb4; under  $\mu$  our Kg1 plays the role of that game’s Kc4; cross-type roles: our Rb2 as their pa5, our ra3 as their Pg5; transferred move g8f8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf8 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



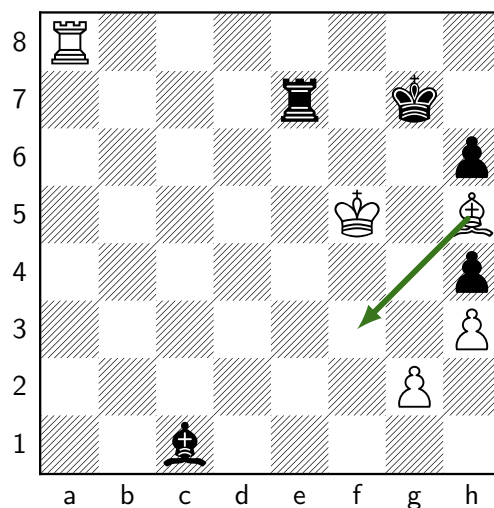
(a)



(b)



(c)



(d)

Figure 38: **Move 44, Black: f3**. (a) Our position; Melanie plays f3 (f4f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3795` (ply 99) [structural distance  $d=21$ , QAP obj=3]: the master played c4; under  $\mu$  our *Pf5* plays the role of that game’s *Pc4*; cross-type roles: our *nf5* as their *rd8*, our *Rb2* as their *Pa6*; transferred move f4f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1750` (ply 97) [structural distance  $d=15$ , QAP obj=3]: the master played Rh3; under  $\mu$  our *Pf5* plays the role of that game’s *Rh7* (a **cross-type** correspondence); cross-type roles: our *Ba8* as their *rc8*, our *Rb2* as their *Bb3*; transferred move f4f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#402` (ply 92) [structural distance  $d=20$ , QAP obj=3]: the master played Bf3; under  $\mu$  our *Pf5* plays the role of that game’s *Bh5* (a **cross-type** correspondence); cross-type roles: our *nf5* as their *re7*; transferred move f4f3 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to f3 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

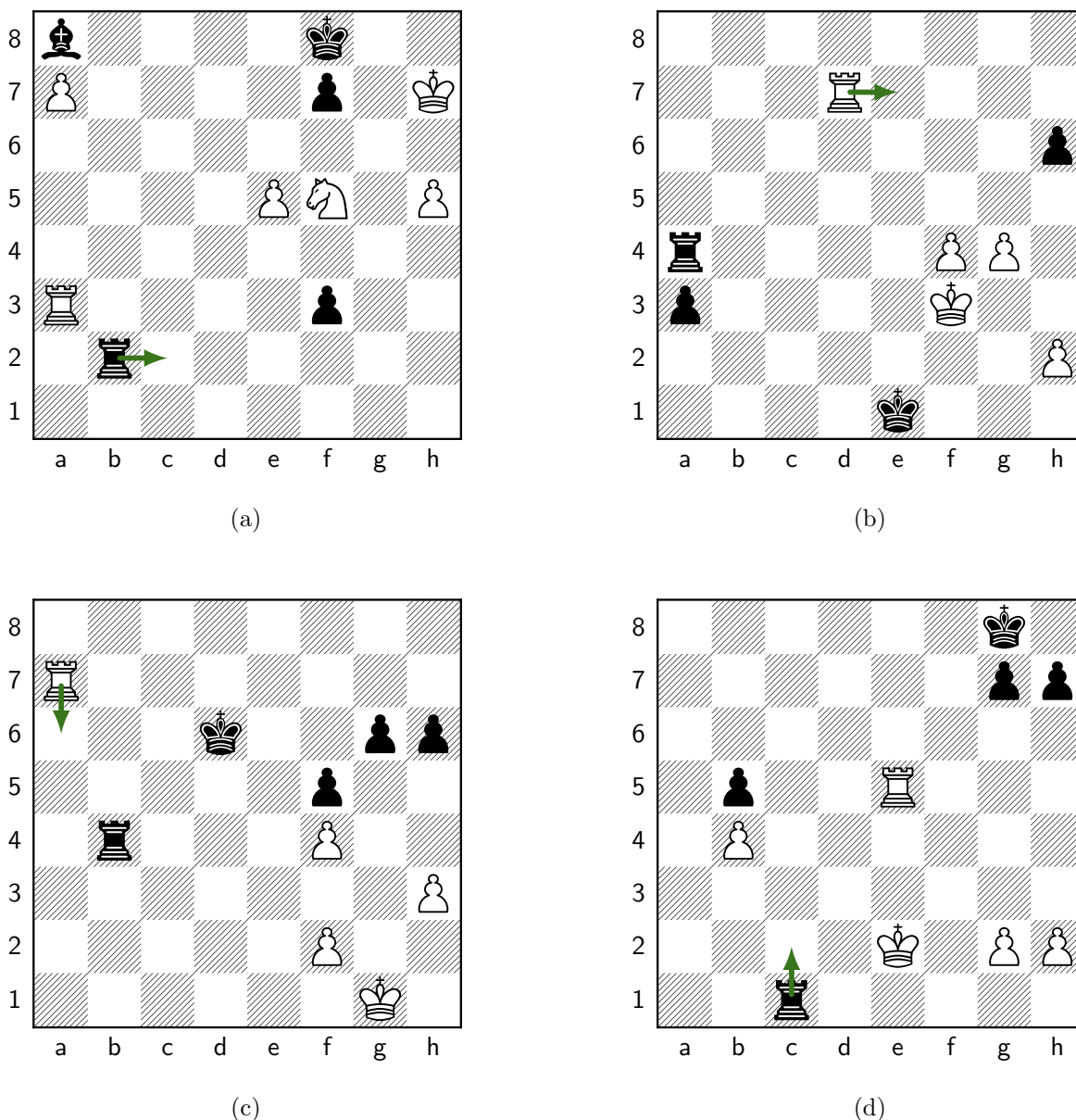
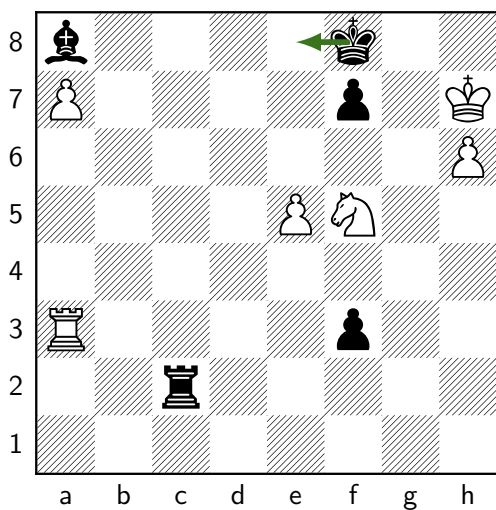
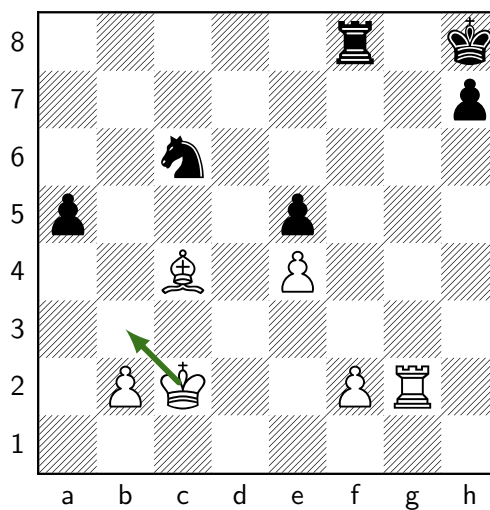


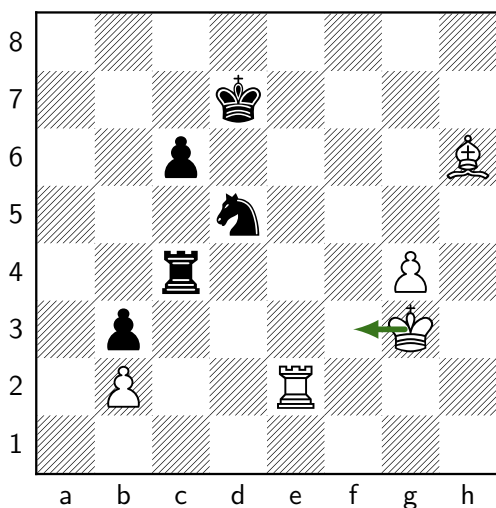
Figure 39: **Move 45, Black: Rc2**. (a) Our position; Melanie plays **Rc2** (b2c2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#603` (ply 96) [structural distance  $d=14$ , QAP obj=24]: the master played **Re7+**; under  $\mu$  our *Rb7* plays the role of that game’s *Rd7*; cross-type roles: our *Ba8* as their *Ph2*; transferred move **b2c2** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1281` (ply 82) [structural distance  $d=16$ , QAP obj=18]: the master played **Ra6+**; under  $\mu$  our *Rb7* plays the role of that game’s *Ra7*; cross-type roles: our *Ba8* as their *pf5*, our *nf5* as their *ph6*; transferred move **b2c2** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2468` (ply 93) [structural distance  $d=16$ , QAP obj=12]: the master played **Rc2+**; under  $\mu$  our *Rb7* plays the role of that game’s *Rc8*; cross-type roles: our *nf5* as their *Ph7*, our *Ba8* as their *pg2*; transferred move **b2c2** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **Rc2** (summed QAP objective 194, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



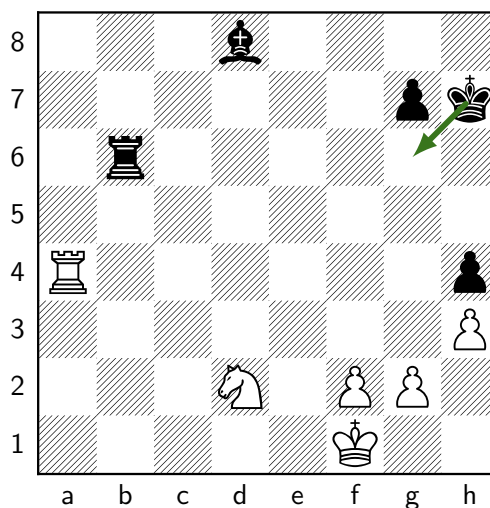
(a)



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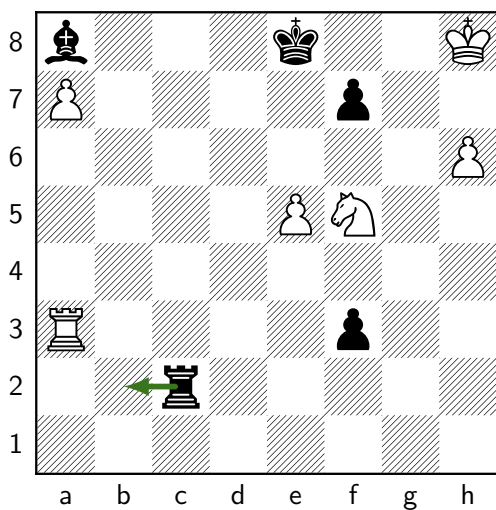


(c)

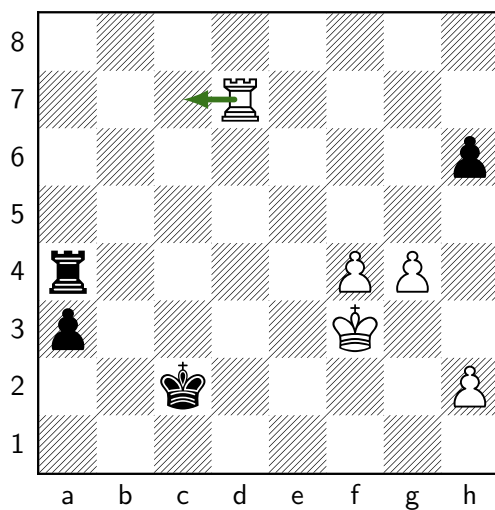


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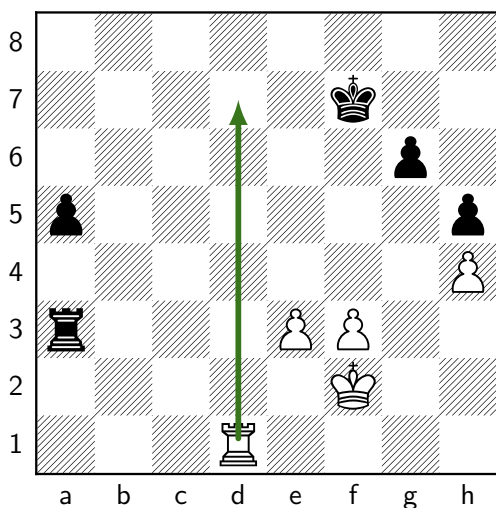
Figure 40: **Move 46, Black: Ke8 (f8e8).** (a) Our position; Melanie plays Ke8 (f8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1485` (ply 94) [structural distance  $d=17$ , QAP obj=17]: the master played Kb3; under  $\mu$  our *Kf1* plays the role of that game’s *Kc2*; cross-type roles: our *Ba8* as their *Rg2*, our *Rc2* as their *Bc4*; transferred move f8e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#490` (ply 88) [structural distance  $d=22$ , QAP obj=17]: the master played Kf3; under  $\mu$  our *Kf1* plays the role of that game’s *Kg3*; cross-type roles: our *ph6* as their *Re2*, our *Ba8* as their *nd5*; transferred move f8e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1712` (ply 89) [structural distance  $d=19$ , QAP obj=17]: the master played Kg6; under  $\mu$  our *Kf1* plays the role of that game’s *Kh2*; cross-type roles: our *ph6* as their *nd2*, our *nf5* as their *pg2*; transferred move f8e8 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Ke8 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



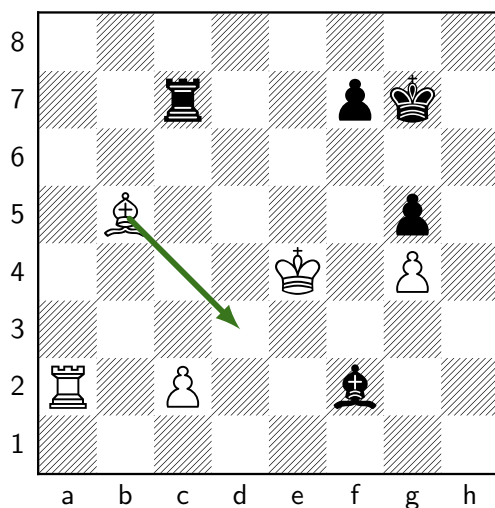
(a)



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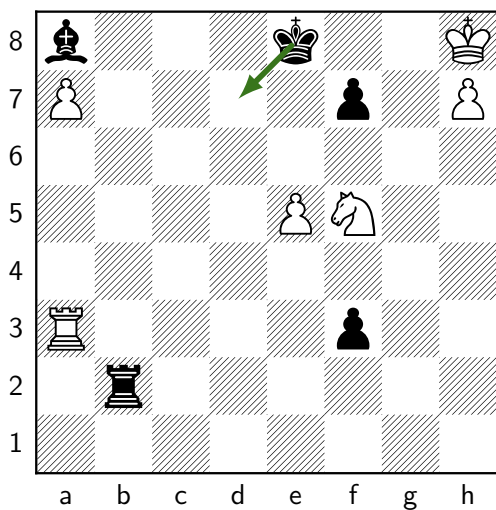


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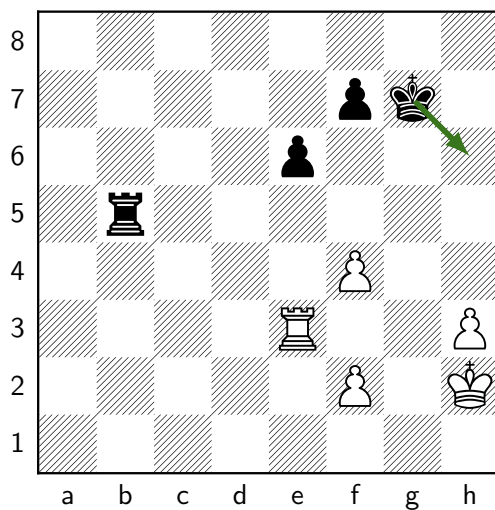


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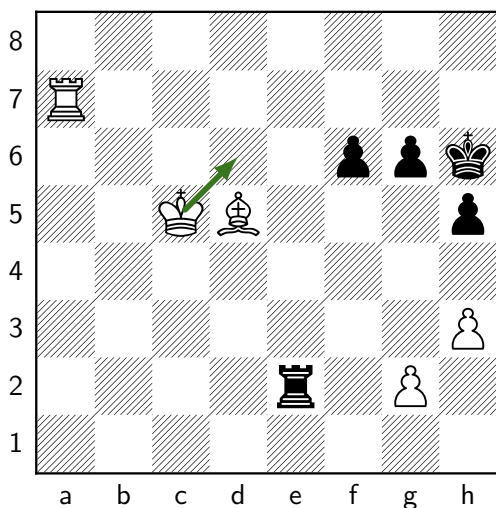
Figure 41: **Move 47, Black: Rb2**. (a) Our position; Melanie plays Rb2 (c2b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#603` (ply 104) [structural distance  $d=18$ , QAP obj=23]: the master played Rc7+; under  $\mu$  our Rc7 plays the role of that game’s Rd7; cross-type roles: our Ba8 as their Ph2; transferred move c2b2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2355` (ply 82) [structural distance  $d=19$ , QAP obj=19]: the master played Rd7+; under  $\mu$  our Rc7 plays the role of that game’s Rd1; cross-type roles: our nf5 as their Ph4; transferred move c2b2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1699` (ply 108) [structural distance  $d=17$ , QAP obj=23]: the master played Bd3; under  $\mu$  our Rc7 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Ra2, our nf5 as their bf2; transferred move c2c6. Synthesis: 9 retrieved analogs transferred to Rb2 (summed QAP objective 203, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



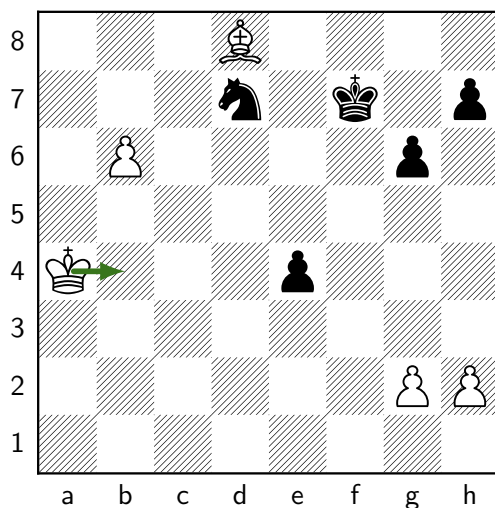
(a)



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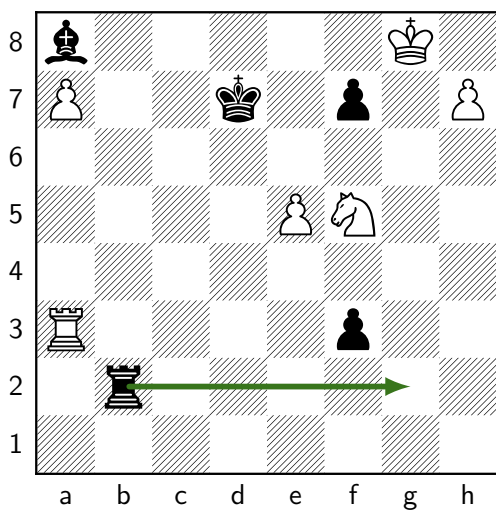


(c)

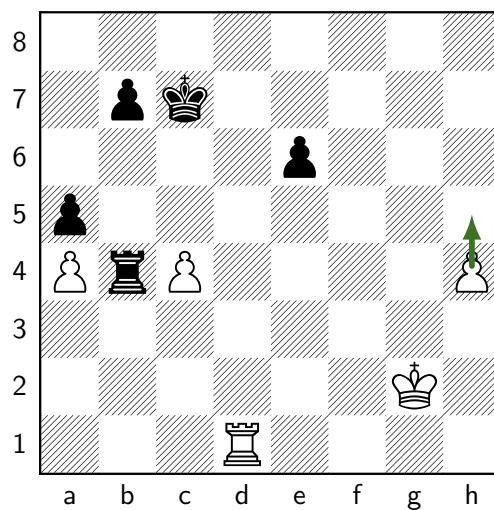


(d)

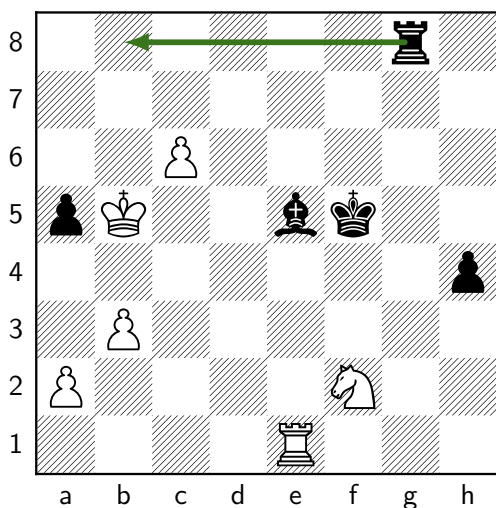
Figure 42: **Move 48, Black: Kd7.** (a) Our position; Melanie plays Kd7 (e8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `crrl_top_p0001#110` (ply 91) [structural distance  $d=17$ , QAP obj=13]: the master played Kh6; under  $\mu$  our *Ke1* plays the role of that game’s *Kg2*; cross-type roles: our *Ba8* as their *Pf7*; transferred move e8d7 (**endorses this move**). (c) Retrieved from `crrl_top_p0003#1808` (ply 102) [structural distance  $d=16$ , QAP obj=10]: the master played Kd6; under  $\mu$  our *Ke1* plays the role of that game’s *Kc5*; transferred move e8d7 (**endorses this move**). (d) Retrieved from `crrl_top_p0002#969` (ply 110) [structural distance  $d=18$ , QAP obj=9]: the master played Kb4; under  $\mu$  our *Ke1* plays the role of that game’s *Ka4*; cross-type roles: our *ra3* as their *nd7*, our *nf5* as their *ph7*; transferred move e8d7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kd7 (summed QAP objective 49, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



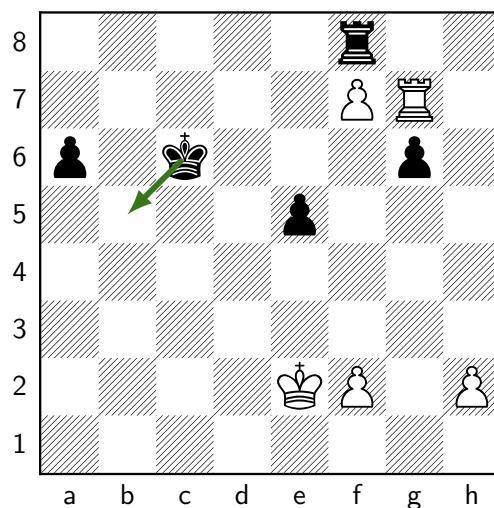
(a)



(b)



(c)



(d)

Figure 43: **Move 49, Black: Rg2+.** (a) Our position; Melanie plays Rg2+ (b2g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1781` (ply 88) [structural distance  $d=18$ , QAP obj=31]: the master played h5; under  $\mu$  our *Ba1* plays the role of that game’s *Ph4* (a **cross-type** correspondence); cross-type roles: our *kg8* as their *rb4*; transferred move a8e4. (c) Retrieved from `ccrl_top_p0000#2372` (ply 103) [structural distance  $d=18$ , QAP obj=25]: the master played Rb8+; under  $\mu$  our *Ba1* plays the role of that game’s *Rg1* (a **cross-type** correspondence); cross-type roles: our *Pf7* as their *Be5*, our *nf5* as their *pa2*; transferred move a8b7. (d) Retrieved from `ccrl_top_p0002#941` (ply 85) [structural distance  $d=19$ , QAP obj=23]: the master played Kb5; under  $\mu$  our *Kd2* plays the role of that game’s *Kc3*; cross-type roles: our *kg8* as their *Rf8*, our *kg8* as their *rg7*; transferred move d7c7. Synthesis: no retrieved analog transferred to Rg2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

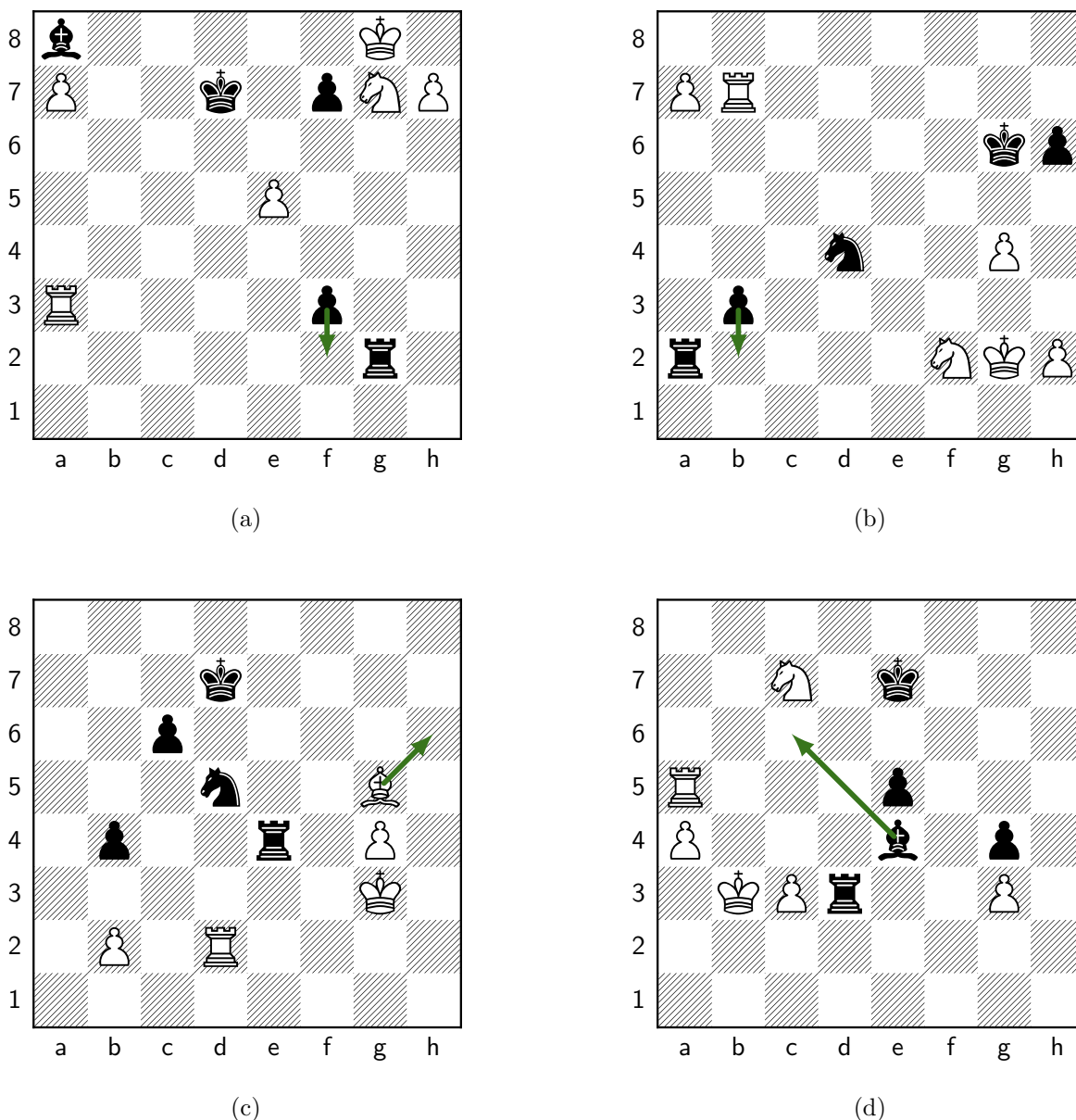
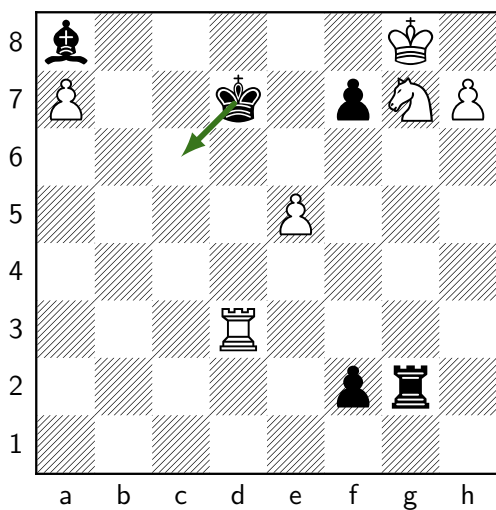
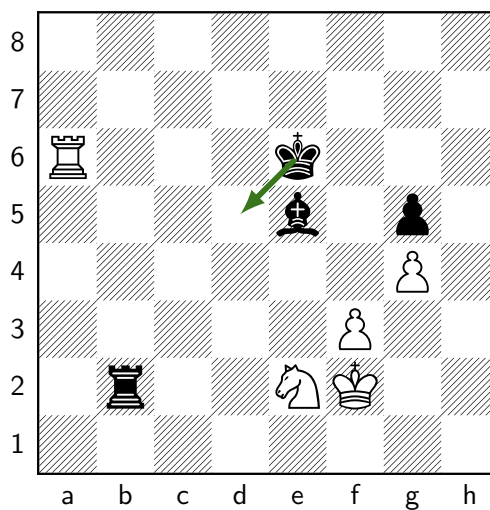


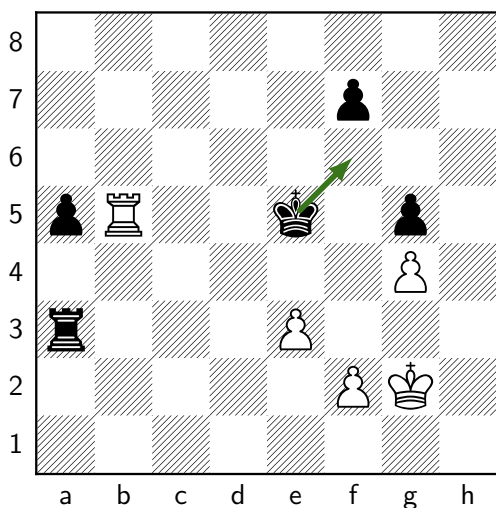
Figure 44: **Move 50, Black: f2**. (a) Our position; Melanie plays f2 (f3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#527` (ply 93) [structural distance  $d=23$ , QAP obj=39]: the master played b2; under  $\mu$  our *Pf6* plays the role of that game’s *Pb6*; cross-type roles: our *Ba8* as their *Nd4*; transferred move f3f2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#490` (ply 84) [structural distance  $d=22$ , QAP obj=31]: the master played Bh6; under  $\mu$  our *Pf6* plays the role of that game’s *Bg5* (a **cross-type** correspondence); cross-type roles: our *Ba8* as their *Pb2*, our *kg8* as their *re4*; transferred move f3f2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3373` (ply 95) [structural distance  $d=24$ , QAP obj=29]: the master played Bc6; under  $\mu$  our *Pf6* plays the role of that game’s *Be5* (a **cross-type** correspondence); cross-type roles: our *pe5* as their *nc7*, our *kg8* as their *ra5*; transferred move f3f2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to f2 (summed QAP objective 167, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



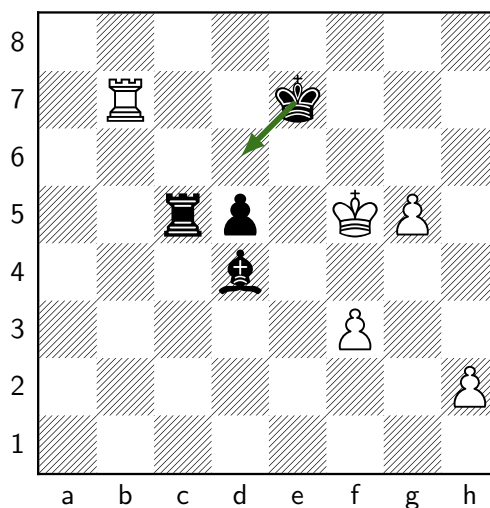
(a)



(b)

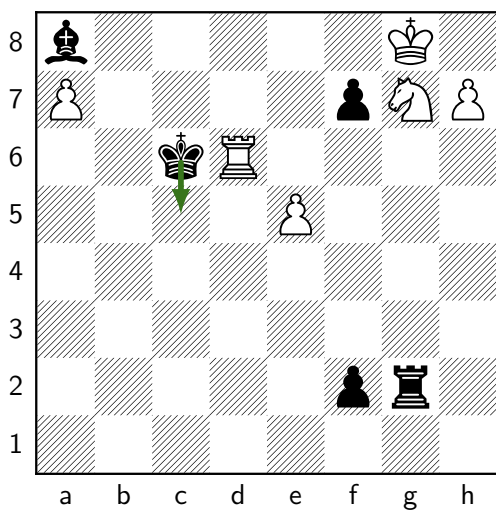


(c)

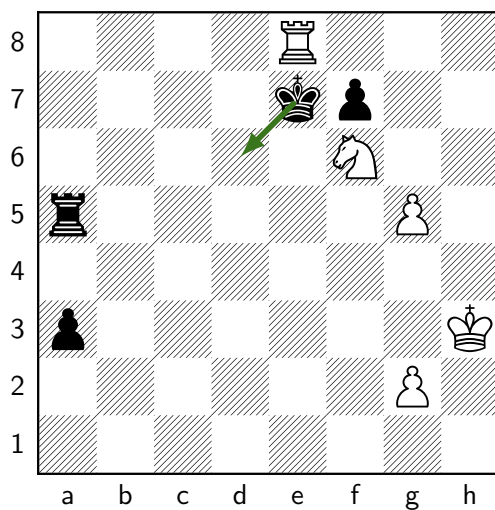


(d)

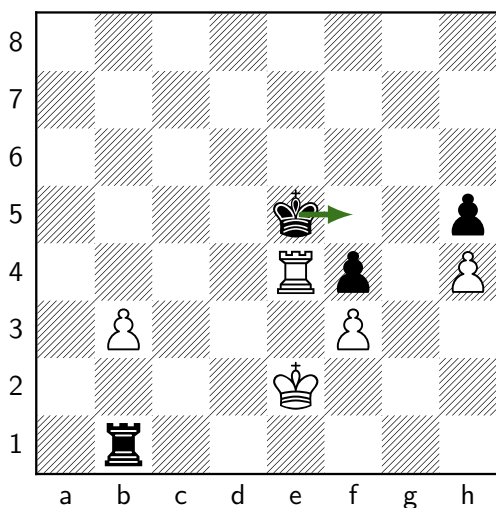
Figure 45: **Move 51, Black: Kc6.** (a) Our position; Melanie plays Kc6 (d7c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1160` (ply 97) [structural distance  $d=21$ , QAP obj=84]: the master played Kd5; under  $\mu$  our Kd2 plays the role of that game’s Ke3; transferred move d7c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1146` (ply 99) [structural distance  $d=22$ , QAP obj=54]: the master played Kf6; under  $\mu$  our Kd2 plays the role of that game’s Ke4; cross-type roles: our Ba8 as their Pf7, our ng7 as their pe3; transferred move d7c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3211` (ply 87) [structural distance  $d=25$ , QAP obj=51]: the master played Kd6; under  $\mu$  our Kd2 plays the role of that game’s Ke2; cross-type roles: our Ba8 as their Rc5, our ng7 as their pg5; transferred move d7c6 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kc6 (summed QAP objective 432, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



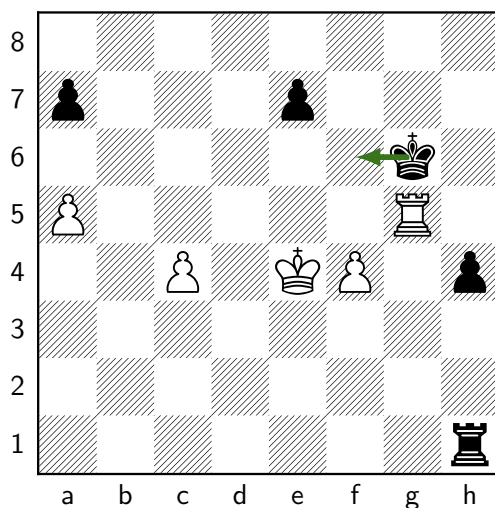
(a)



(b)

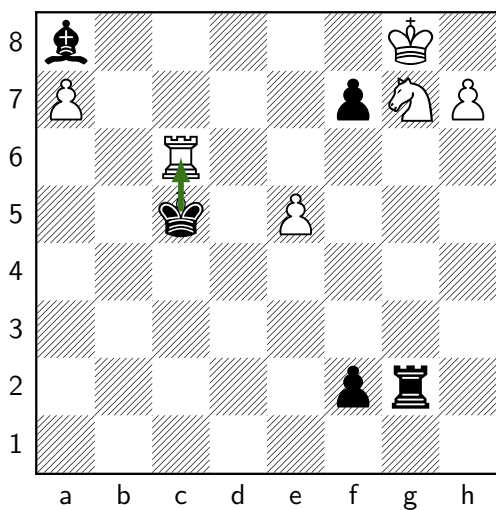


(c)

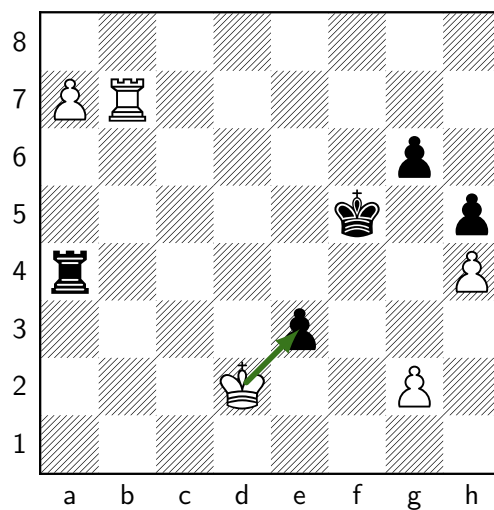


(d)

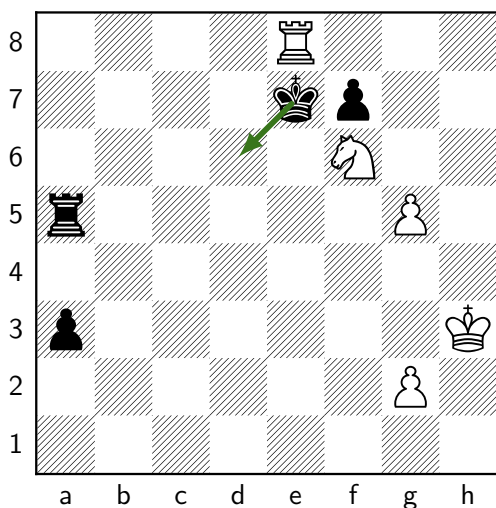
Figure 46: **Move 52, Black: Kc5.** (a) Our position; Melanie plays Kc5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2931` (ply 113) [structural distance  $d=30$ , QAP obj=68]: the master played Kd6; under  $\mu$  our Kc3 plays the role of that game’s Ke2; cross-type roles: our Ba8 as their nf6, our pa7 as their Ra5; transferred move c6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#471` (ply 119) [structural distance  $d=30$ , QAP obj=68]: the master played Kf5; under  $\mu$  our Kc3 plays the role of that game’s Ke4; cross-type roles: our Ba8 as their Ph5, our ng7 as their pb3; transferred move c6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2350` (ply 113) [structural distance  $d=29$ , QAP obj=67]: the master played Kf6; under  $\mu$  our Kc3 plays the role of that game’s Kg3; cross-type roles: our Ba8 as their Pa7, our ng7 as their pc4; transferred move c6c5 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Kc5 (summed QAP objective 451, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



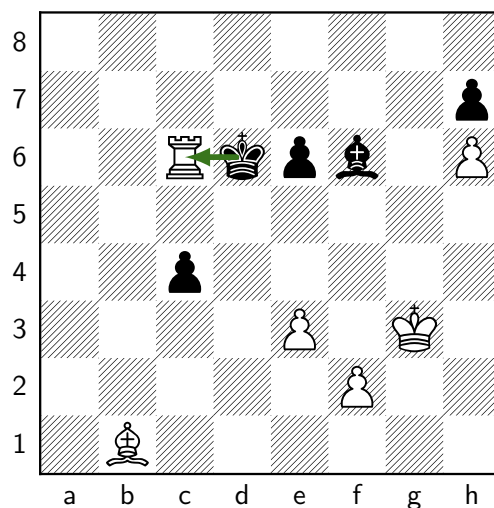
(a)



(b)

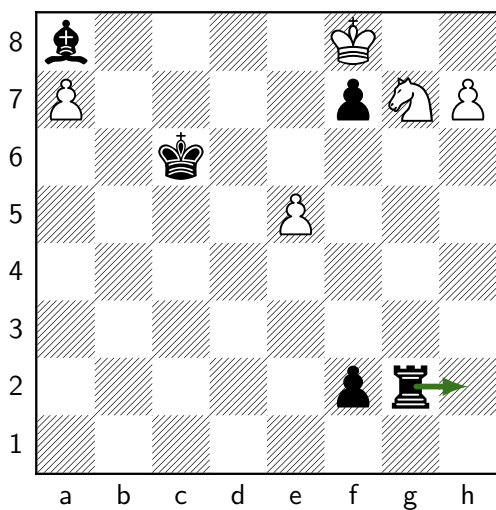


(c)

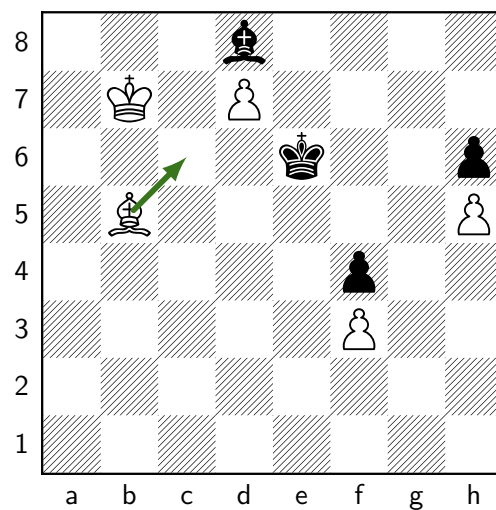


(d)

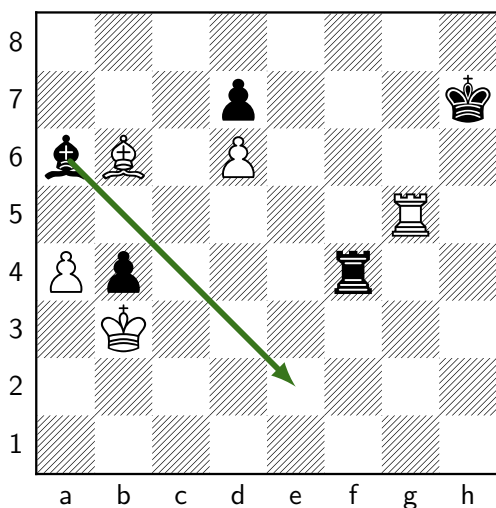
Figure 47: **Move 53, Black: Kxc6**. (a) Our position; Melanie plays Kxc6 (c5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1738` (ply 116) [structural distance  $d=28$ , QAP obj=69]: the master played Kxe3; under  $\mu$  our Kc4 plays the role of that game’s Kd2; cross-type roles: our Ba8 as their Pg2, our rc6 as their pe3; transferred move c5c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2931` (ply 113) [structural distance  $d=29$ , QAP obj=68]: the master played Kd6; under  $\mu$  our Kc4 plays the role of that game’s Ke2; cross-type roles: our pa7 as their Ra5, our Ba8 as their Pa3; transferred move c5c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2362` (ply 119) [structural distance  $d=28$ , QAP obj=68]: the master played Kxc6; under  $\mu$  our Kc4 plays the role of that game’s Kd3; cross-type roles: our ng7 as their Pe6, our Rg2 as their pe3; transferred move c5c6 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kxc6 (summed QAP objective 272, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



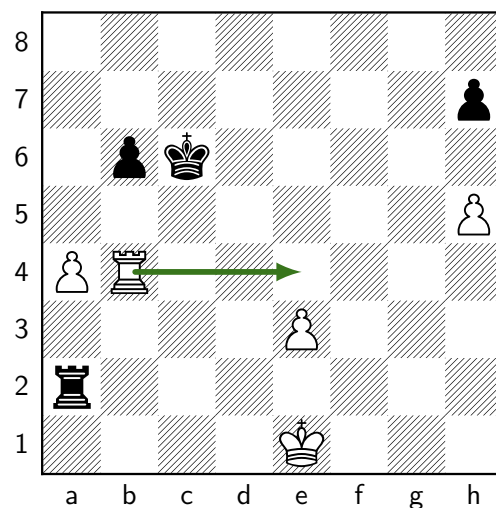
(a)



(b)

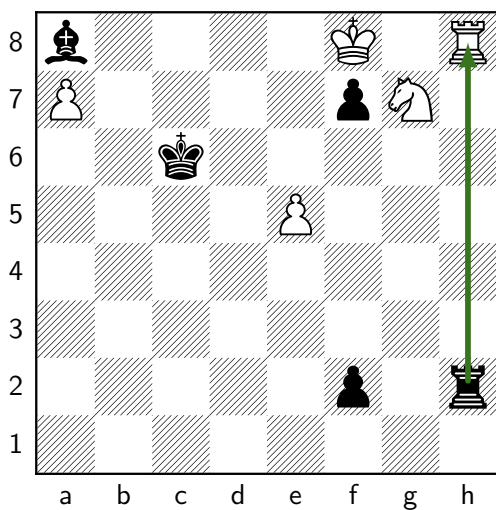


(c)

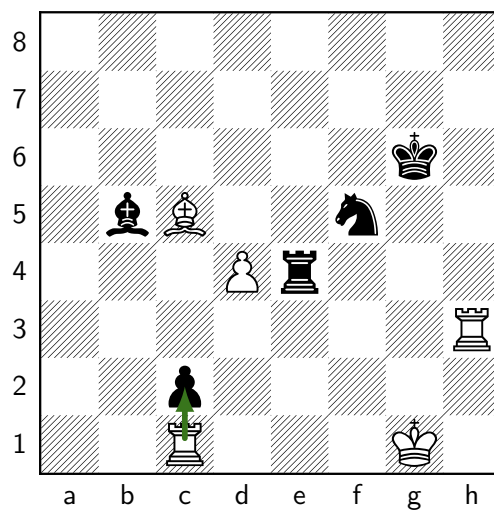


(d)

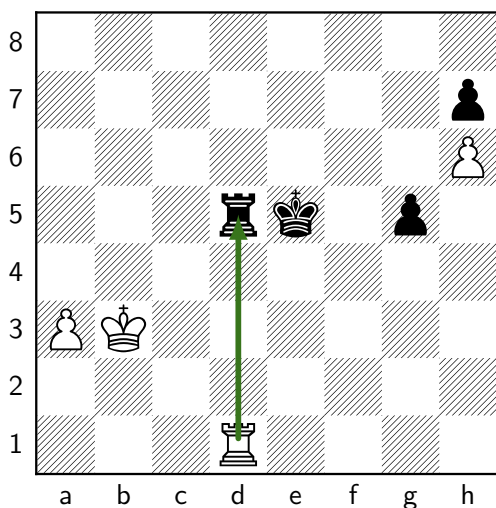
Figure 48: **Move 54, Black: Rh2.** (a) Our position; Melanie plays Rh2 (**g2h2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2795` (ply 92) [structural distance  $d=17$ , QAP obj=19]: the master played Bc6; under  $\mu$  our Rg7 plays the role of that game’s Bb5 (a **cross-type** correspondence); cross-type roles: our ng7 as their Pf3, our Ba8 as their Ph5; transferred move g2h2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3257` (ply 111) [structural distance  $d=19$ , QAP obj=19]: the master played Be2; under  $\mu$  our Rg7 plays the role of that game’s Ba3 (a **cross-type** correspondence); cross-type roles: our Ba8 as their Pd7, our ph7 as their bb6; transferred move g2h2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#127` (ply 108) [structural distance  $d=19$ , QAP obj=15]: the master played Re4; under  $\mu$  our Rg7 plays the role of that game’s Rb4; cross-type roles: our kf8 as their ra2, our ng7 as their pb6; transferred move g2h2 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rh2 (summed QAP objective 96, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



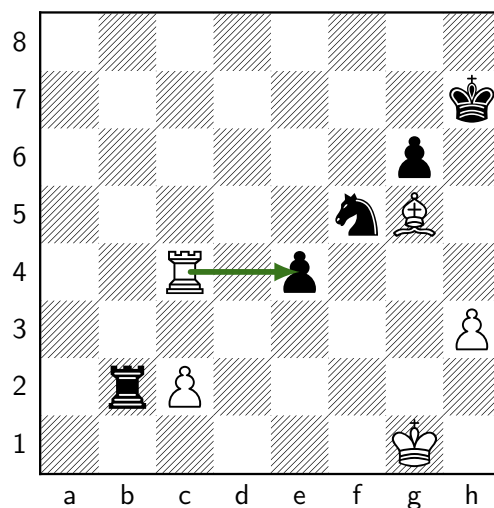
(a)



(b)

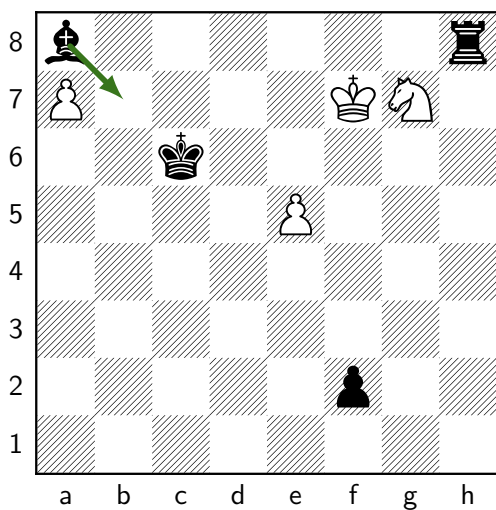


(c)

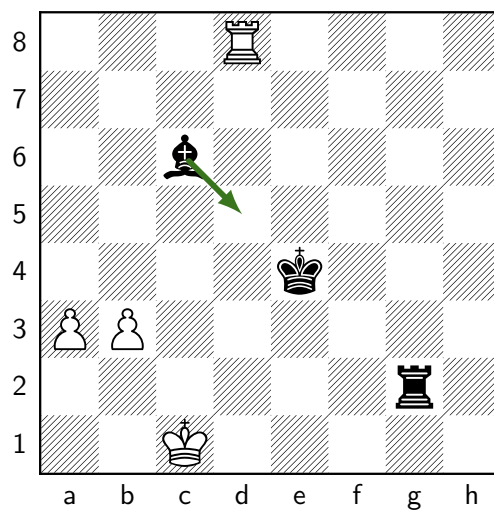


(d)

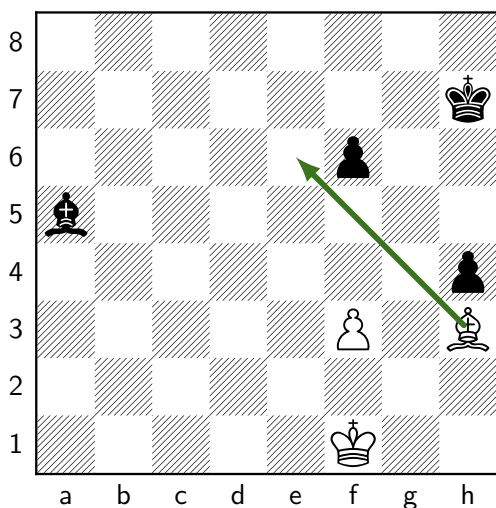
Figure 49: **Move 55, Black: Rxxh8+.** (a) Our position; Melanie plays Rxxh8+ (h2h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2867` (ply 108) [structural distance  $d=21$ , QAP obj=33]: the master played Rxc2; under  $\mu$  our Rh7 plays the role of that game’s Rc1; cross-type roles: our rh8 as their pc2, our Ba8 as their Rh3; transferred move h2h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#244` (ply 98) [structural distance  $d=23$ , QAP obj=28]: the master played Rxd5+; under  $\mu$  our Rh7 plays the role of that game’s Rd1; cross-type roles: our Ba8 as their pg5, our ng7 as their Ph6; transferred move h2h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2261` (ply 102) [structural distance  $d=21$ , QAP obj=23]: the master played Rxe4; under  $\mu$  our Rh7 plays the role of that game’s Rc4; cross-type roles: our kf8 as their rb2, our rh8 as their pe4; transferred move h2h8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rxxh8+ (summed QAP objective 100, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



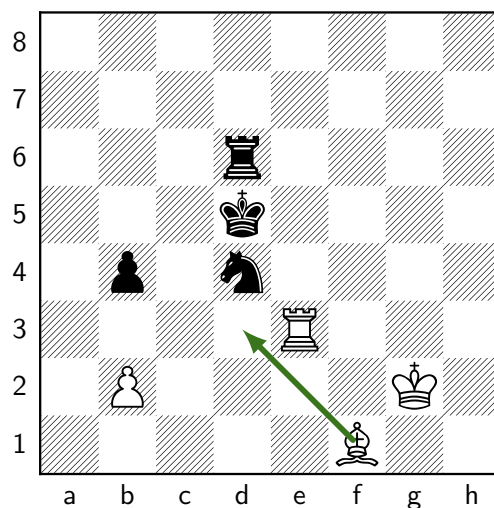
(a)



(b)

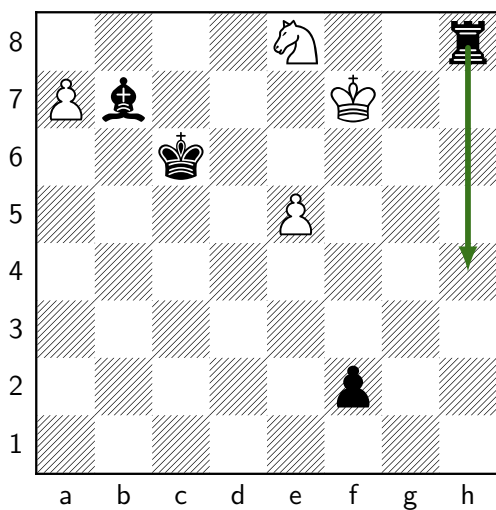


(c)

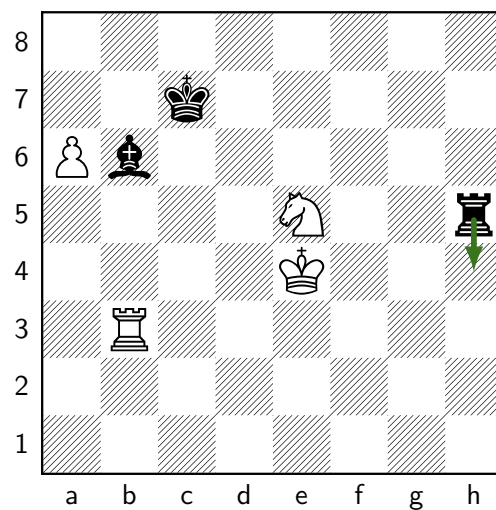


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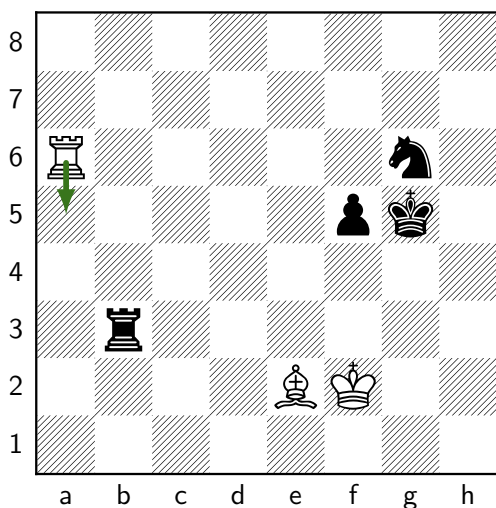
Figure 50: **Move 56, Black: Bb7 (a8b7)**. (a) Our position; Melanie plays Bb7 (a8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1625` (ply 121) [structural distance  $d=16$ , QAP obj=2]: the master played Bd5; under  $\mu$  our *Ba1* plays the role of that game’s *Bc3*; cross-type roles: our *pe5* as their *rd8*, our *ng7* as their *pb3*; transferred move a8b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2086` (ply 110) [structural distance  $d=18$ , QAP obj=2]: the master played Be6; under  $\mu$  our *Ba1* plays the role of that game’s *Bh3*; cross-type roles: our *Rh8* as their *Pf3*, our *ng7* as their *ba5*; transferred move a8b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3242` (ply 122) [structural distance  $d=18$ , QAP obj=2]: the master played Bd3; under  $\mu$  our *Ba1* plays the role of that game’s *Bf1*; cross-type roles: our *Pf2* as their *rd6*; transferred move a8b7 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Bb7 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



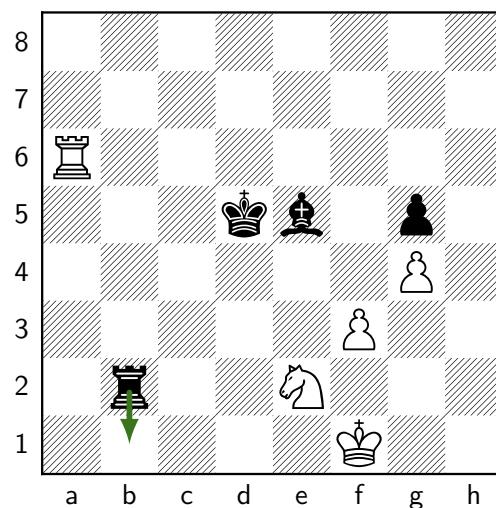
(a)



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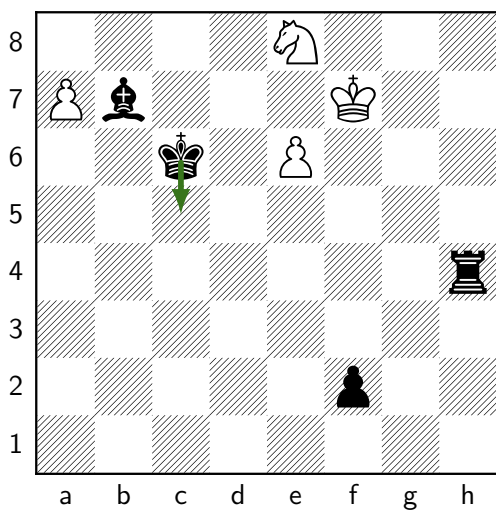


(c)

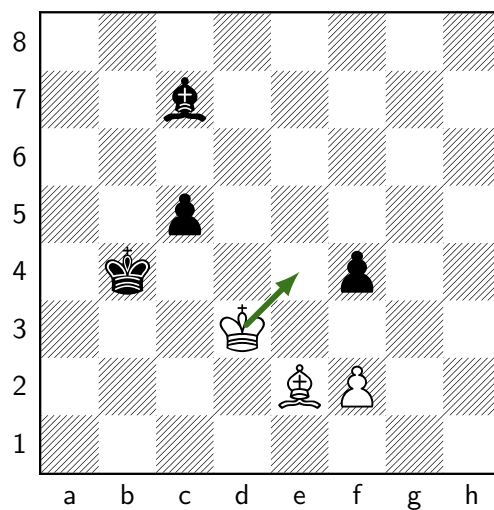


(d)

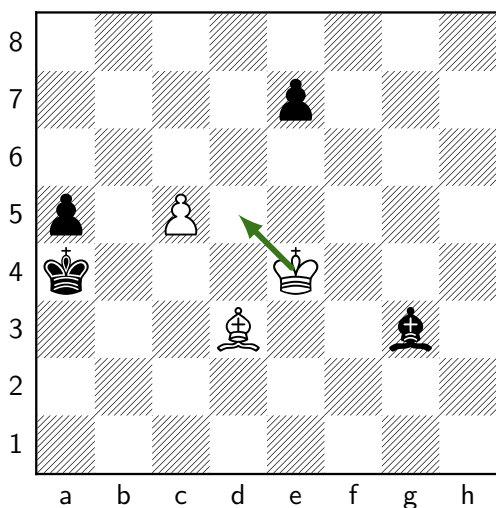
Figure 51: **Move 57, Black: Rh4**. (a) Our position; Melanie plays Rh4 (h8h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3744` (ply 125) [structural distance  $d=10$ , QAP obj=31]: the master played Rh4+; under  $\mu$  our Rh1 plays the role of that game’s Rh4; cross-type roles: our ne8 as their Bb6; transferred move h8h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2132` (ply 112) [structural distance  $d=15$ , QAP obj=11]: the master played Ra5; under  $\mu$  our Rh1 plays the role of that game’s Ra6; cross-type roles: our pe5 as their rb3; transferred move h8h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1160` (ply 103) [structural distance  $d=13$ , QAP obj=21]: the master played Rb1+; under  $\mu$  our Rh1 plays the role of that game’s Rb7; cross-type roles: our pe5 as their ra6, our Pf2 as their Be5; transferred move h8h1. Synthesis: 2 retrieved analogs transferred to Rh4 (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



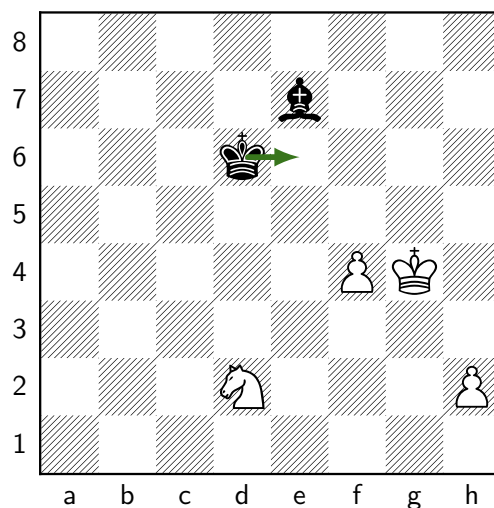
(a)



(b)

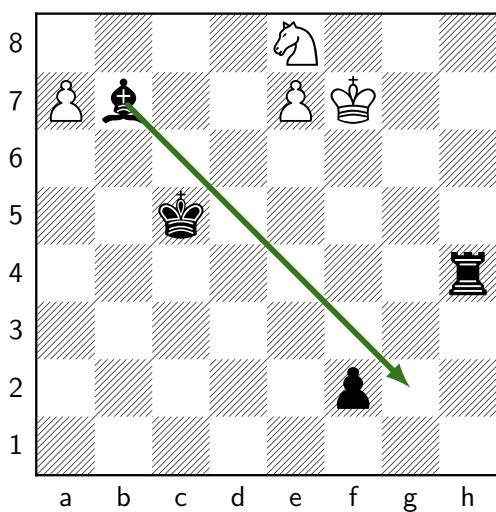


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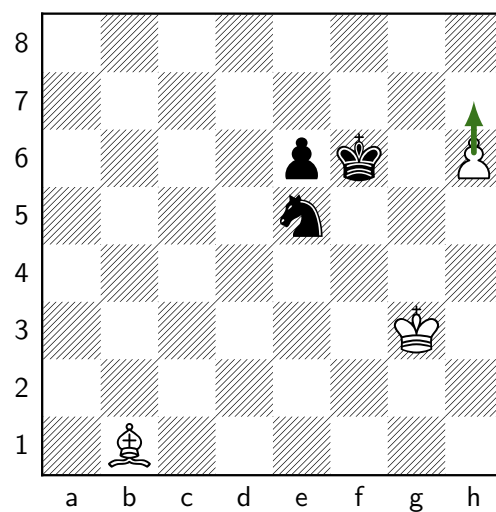


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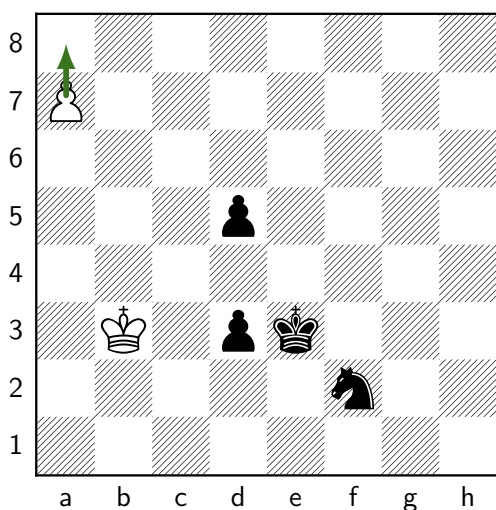
Figure 52: **Move 58, Black: Kc5.** (a) Our position; Melanie plays Kc5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1866` (ply 112) [structural distance  $d=16$ , QAP obj=3]: the master played Ke4; under  $\mu$  our Kc3 plays the role of that game’s Kd3; cross-type roles: our ne8 as their Pf2, our Rh4 as their bc7; transferred move c6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2104` (ply 104) [structural distance  $d=14$ , QAP obj=3]: the master played Kd5; under  $\mu$  our Kc3 plays the role of that game’s Ke4; cross-type roles: our ne8 as their bg3, our Rh4 as their pe7; transferred move c6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1672` (ply 129) [structural distance  $d=13$ , QAP obj=3]: the master played Ke6; under  $\mu$  our Kc3 plays the role of that game’s Kd3; cross-type roles: our ne8 as their pf4, our pa7 as their nd2; transferred move c6c5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kc5 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



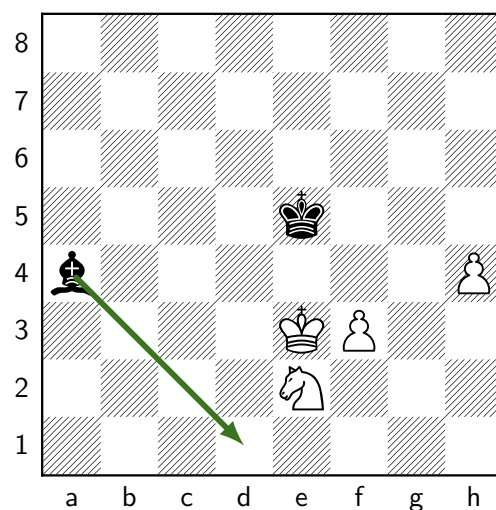
(a)



(b)

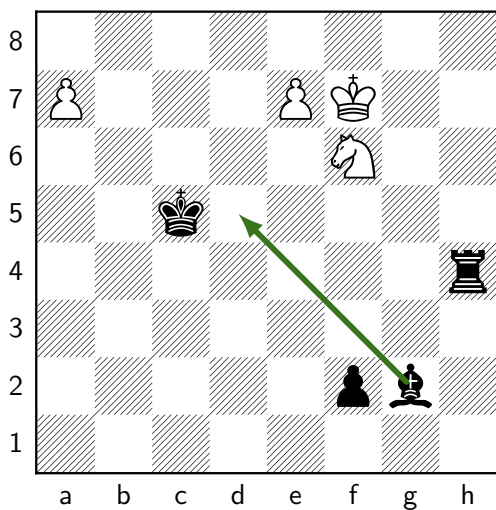


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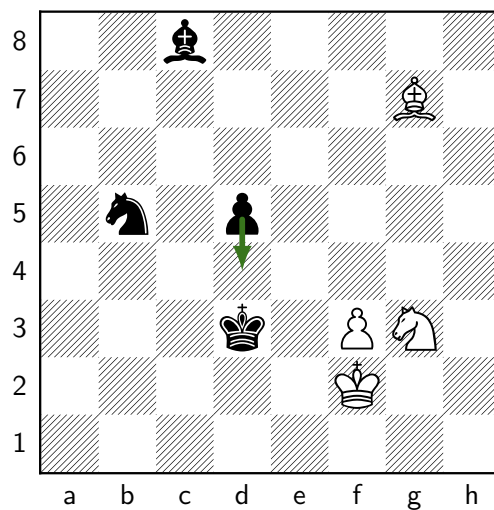


(d)

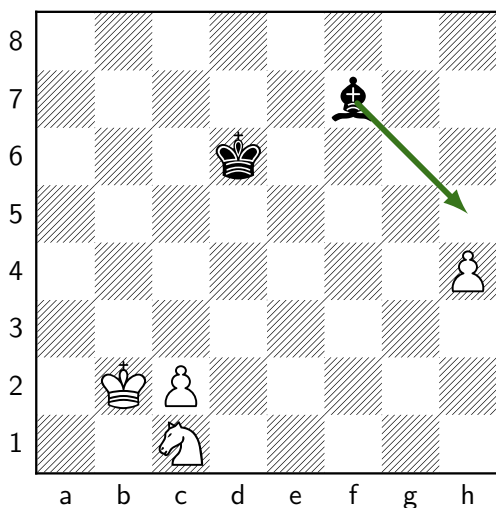
Figure 53: **Move 59, Black: Bg2.** (a) Our position; Melanie plays Bg2 (b7g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3310` (ply 106) [structural distance  $d=8$ , QAP obj=5]: the master played h7; under  $\mu$  our *Bb2* plays the role of that game’s *Ph6* (a **cross-type** correspondence); cross-type roles: our *pa7* as their *Bb1*; transferred move b7g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3825` (ply 120) [structural distance  $d=13$ , QAP obj=3]: the master played a8=Q; under  $\mu$  our *Bb2* plays the role of that game’s *Pa7* (a **cross-type** correspondence); transferred move b7g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3784` (ply 115) [structural distance  $d=10$ , QAP obj=3]: the master played Bd1; under  $\mu$  our *Bb2* plays the role of that game’s *Ba5*; transferred move b7g2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bg2 (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



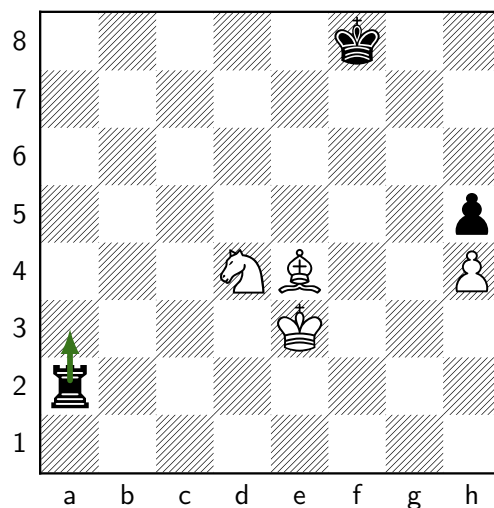
(a)



(b)



(c)



(d)

Figure 54: **Move 60, Black: Bd5+.** (a) Our position; Melanie plays Bd5+ (g2d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#198` (ply 109) [structural distance  $d=9$ , QAP obj=4]: the master played d4; under  $\mu$  our Bg7 plays the role of that game’s Pd4 (a **cross-type** correspondence); cross-type roles: our Rh4 as their Bc8, our pa7 as their bg7; transferred move g2d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2546` (ply 127) [structural distance  $d=8$ , QAP obj=4]: the master played Bh5; under  $\mu$  our Bg7 plays the role of that game’s Bf2; transferred move g2b7. (d) Retrieved from `ccr1_top_p0000#2943` (ply 127) [structural distance  $d=10$ , QAP obj=4]: the master played Ra3+; under  $\mu$  our Rh5 plays the role of that game’s Ra7; cross-type roles: our pe7 as their nd4, our nf6 as their be4; transferred move h4a4. Synthesis: 1 retrieved analog transferred to Bd5+ (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

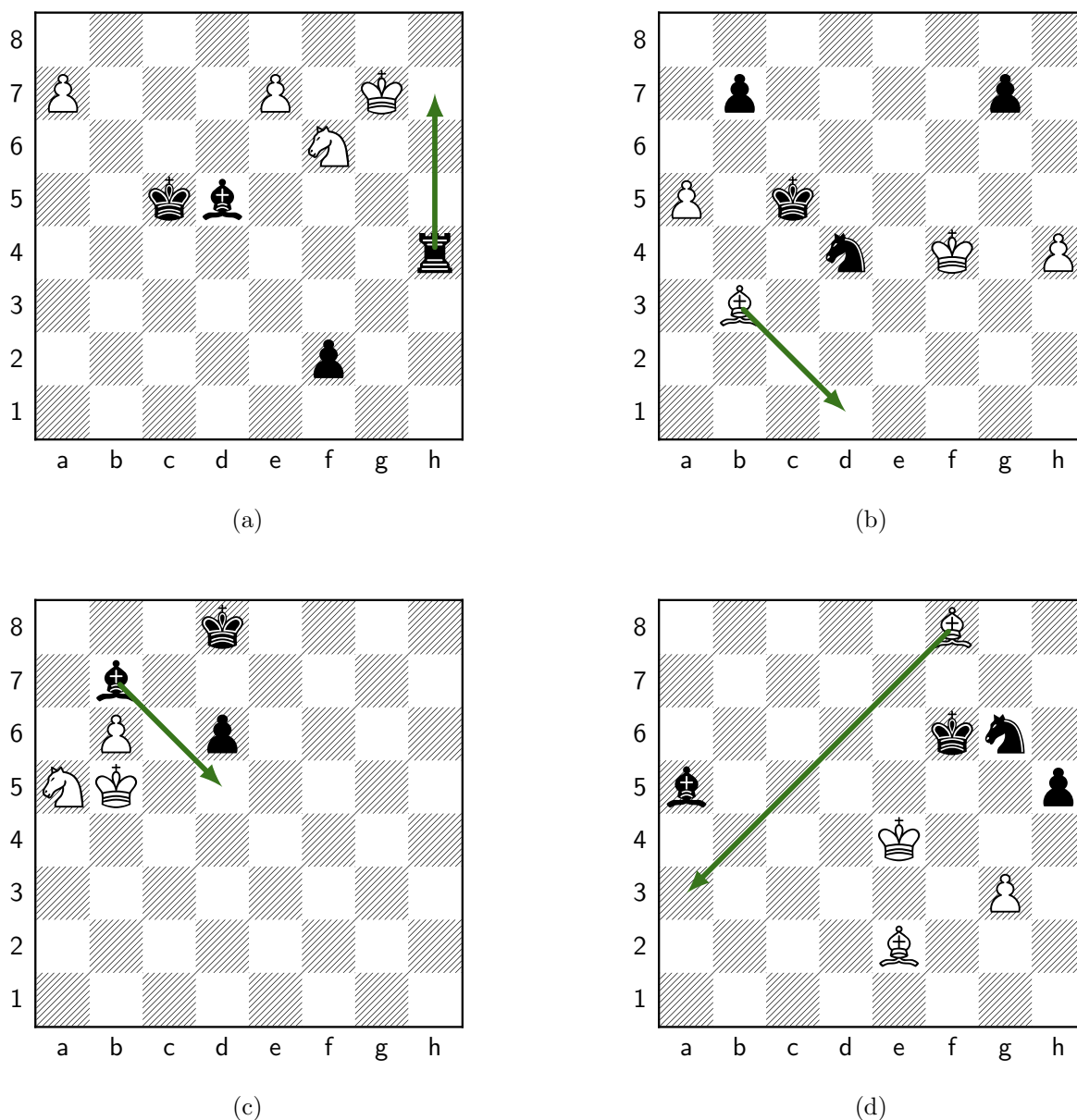
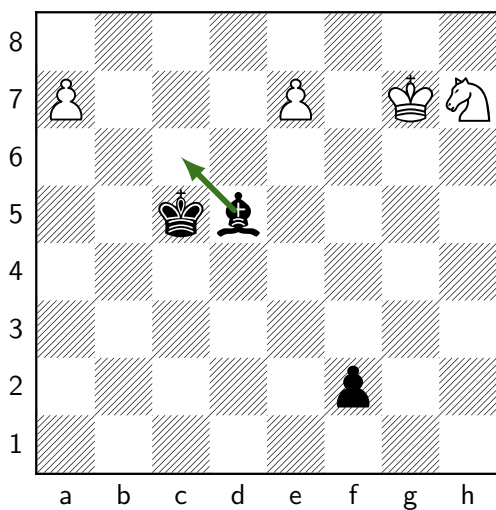
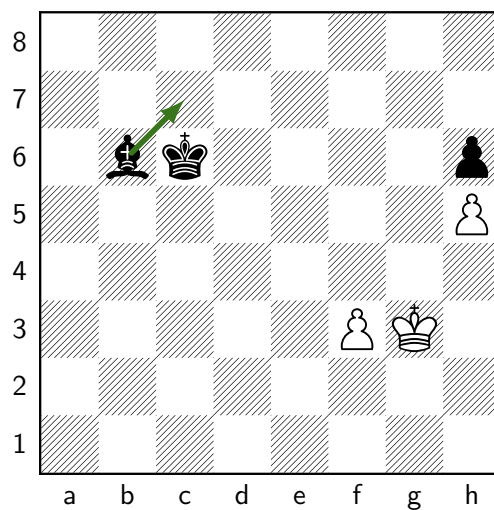


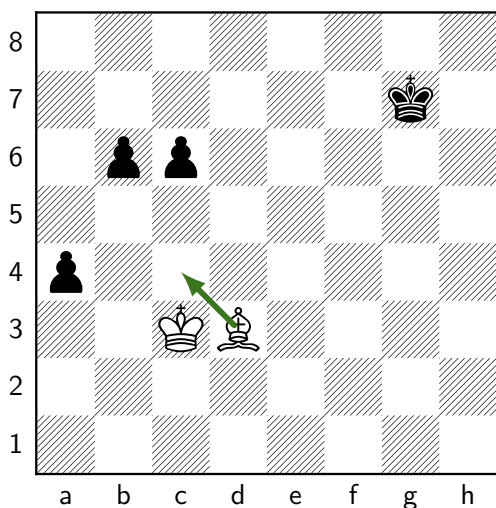
Figure 55: **Move 61, Black: Rh7+.** (a) Our position; Melanie plays Rh7+ (h4h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top.p0000#1625 (ply 122) [structural distance  $d=7$ , QAP obj=23]: the master played Bd1; under  $\mu$  our Bd4 plays the role of that game's Bb3; cross-type roles: our Rh4 as their Pa5; transferred move d5a8. (c) Retrieved from ccr1\_top.p0002#1991 (ply 105) [structural distance  $d=14$ , QAP obj=23]: the master played Bd5; under  $\mu$  our Bd4 plays the role of that game's Bb2; cross-type roles: our Rh4 as their Pd6; transferred move d5e6. (d) Retrieved from ccr1\_top.p0002#210 (ply 118) [structural distance  $d=13$ , QAP obj=23]: the master played Ba3; under  $\mu$  our Bd4 plays the role of that game's Bf8; cross-type roles: our Pf2 as their Be2, our Rh4 as their Pg3; transferred move d5b7. Synthesis: no retrieved analog transferred to Rh7+ – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



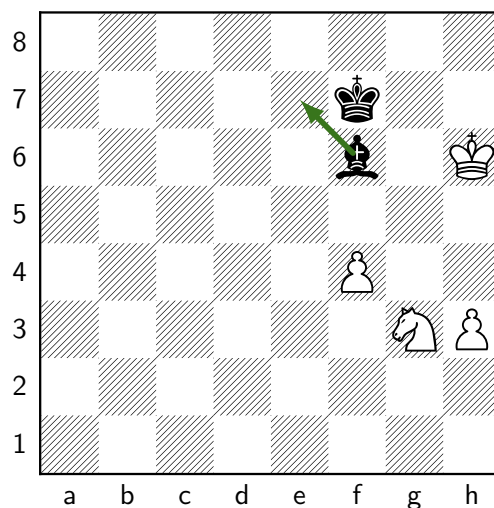
(a)



(b)

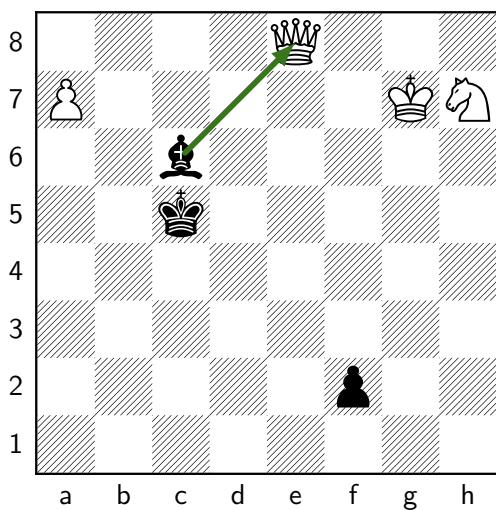


(c)

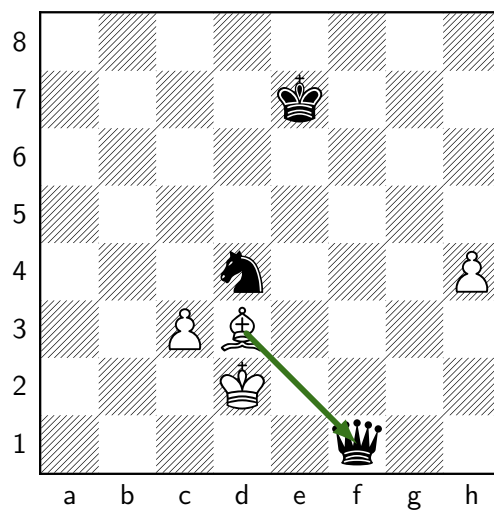


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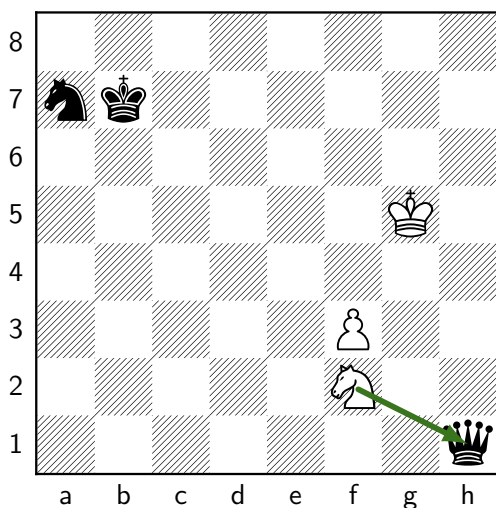
Figure 56: **Move 62, Black: Bc6**. (a) Our position; Melanie plays Bc6 (d5c6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1059` (ply 137) [structural distance  $d=7$ , QAP obj=3]: the master played Bc7+; under  $\mu$  our Bd4 plays the role of that game’s Bb3; cross-type roles: our nh7 as their pf3; transferred move d5c6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2036` (ply 116) [structural distance  $d=9$ , QAP obj=3]: the master played Bc4; under  $\mu$  our Bd4 plays the role of that game’s Bd3; cross-type roles: our nh7 as their pb6; transferred move d5c6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1672` (ply 139) [structural distance  $d=8$ , QAP obj=3]: the master played Be7; under  $\mu$  our Bd4 plays the role of that game’s Bf3; cross-type roles: our pe7 as their ng3, our nh7 as their ph3; transferred move d5c6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc6 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



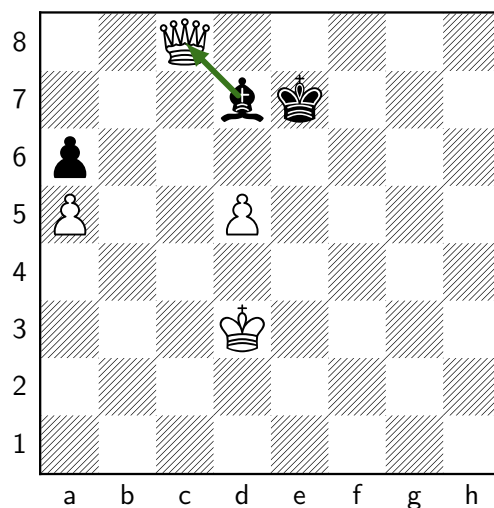
(a)



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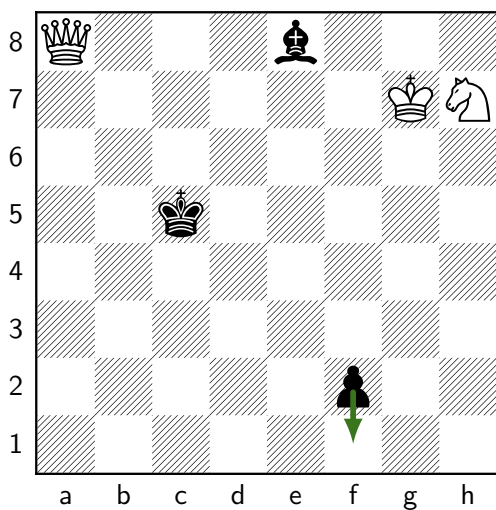


(c)

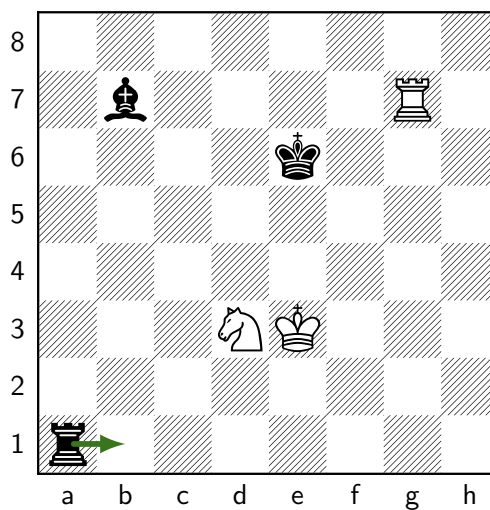


(d)

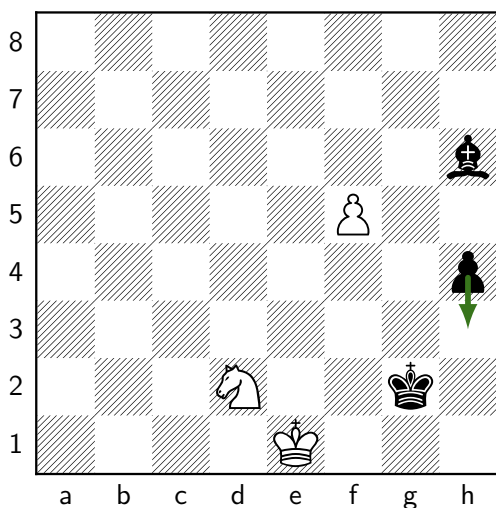
Figure 57: **Move 63, Black: Bxe8.** (a) Our position; Melanie plays Bxe8 (c6e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3208` (ply 114) [structural distance  $d=16$ , QAP obj=31]: the master played Bxf1; under  $\mu$  our Bc3 plays the role of that game’s Bd3; cross-type roles: our Bc6 as their Pc3, our qe8 as their nd4; transferred move c6e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3653` (ply 130) [structural distance  $d=12$ , QAP obj=21]: the master played Nxh1; under  $\mu$  our Bc3 plays the role of that game’s Nf2 (a **cross-type** correspondence); transferred move c6e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3803` (ply 129) [structural distance  $d=13$ , QAP obj=12]: the master played Bxc8; under  $\mu$  our Bc3 plays the role of that game’s Bd2; cross-type roles: our nh7 as their Pa6; transferred move c6e8 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bxe8 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



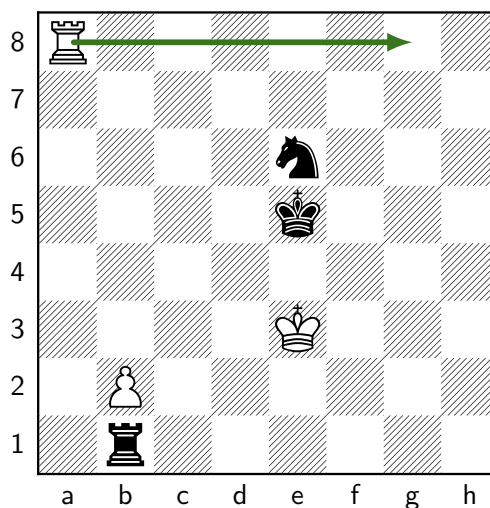
(a)



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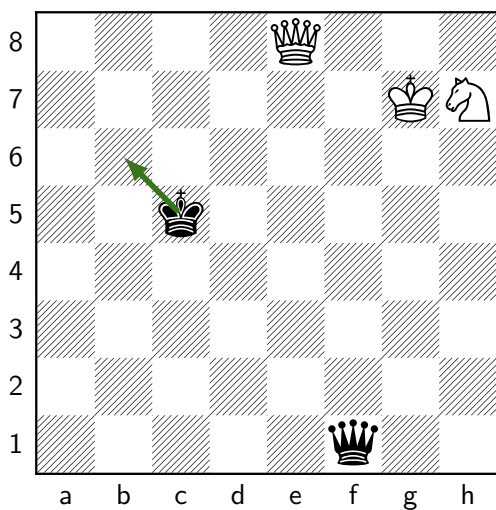


(c)

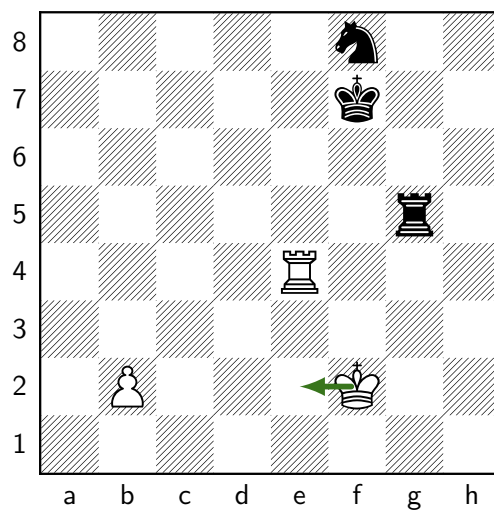


(d)

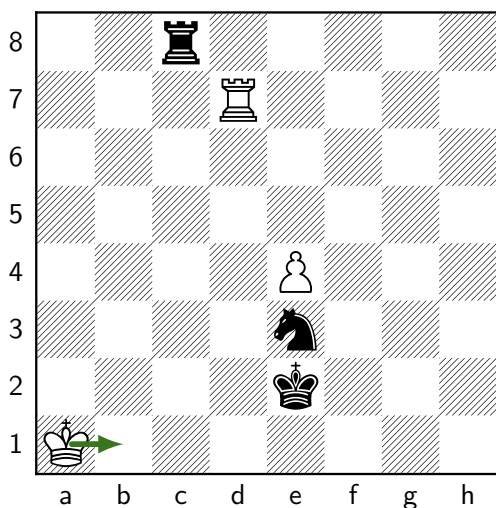
Figure 58: **Move 64, Black: f1=Q**. (a) Our position; Melanie plays f1=Q (f2f1q). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2150` (ply 131) [structural distance  $d=8$ , QAP obj=19]: the master played Rb1; under  $\mu$  our Pf7 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our qa8 as their rg7; transferred move f2f1r. (c) Retrieved from `ccr1_top_p0000#1510` (ply 127) [structural distance  $d=9$ , QAP obj=18]: the master played h3; under  $\mu$  our Pf7 plays the role of that game’s Ph5; cross-type roles: our qa8 as their Bh6, our nh7 as their pf5; transferred move f2f1n. (d) Retrieved from `ccr1_top_p0000#3379` (ply 134) [structural distance  $d=10$ , QAP obj=15]: the master played Rg8; under  $\mu$  our Pf7 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our qa8 as their rb1, our Be8 as their Pb2; transferred move f2f1r. Synthesis: no retrieved analog transferred to f1=Q – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



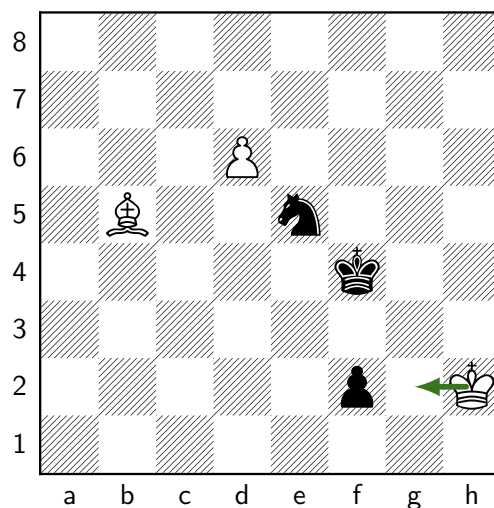
(a)



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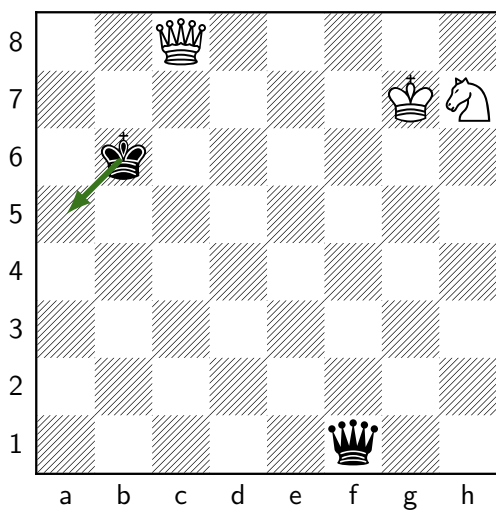


(c)

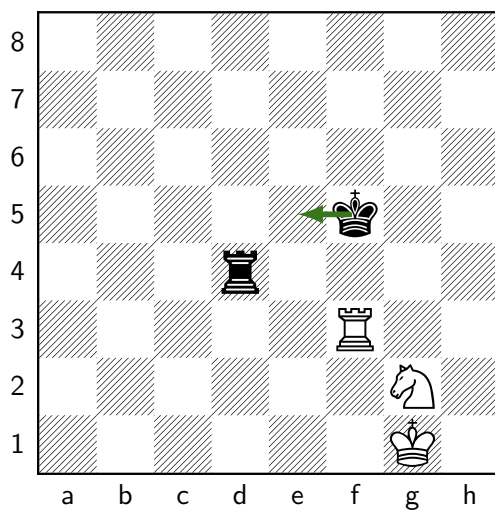


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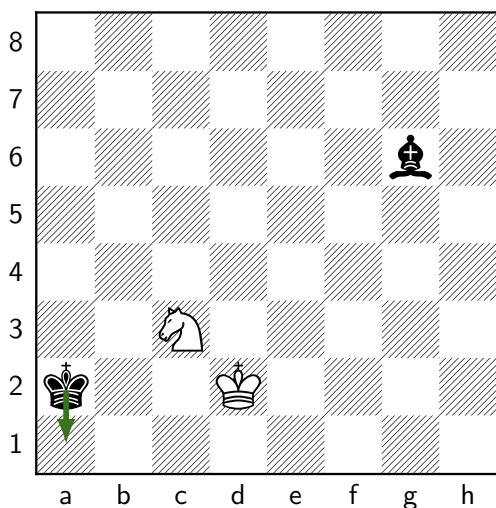
Figure 59: **Move 65, Black: Kb6.** (a) Our position; Melanie plays Kb6 (c5b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3379` (ply 120) [structural distance  $d=10$ , QAP obj=3]: the master played Ke2; under  $\mu$  our Kc4 plays the role of that game’s Kf2; cross-type roles: our Qf1 as their Re4, our qe8 as their rg5; transferred move c5b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1724` (ply 128) [structural distance  $d=10$ , QAP obj=3]: the master played Kb1; under  $\mu$  our Kc4 plays the role of that game’s Ka1; cross-type roles: our Qf1 as their Rd7, our qe8 as their rc8; transferred move c5b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2048` (ply 134) [structural distance  $d=10$ , QAP obj=3]: the master played Kg2; under  $\mu$  our Kc4 plays the role of that game’s Kh2; cross-type roles: our qe8 as their pf2, our Qf1 as their Bb5; transferred move c5b6 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kb6 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



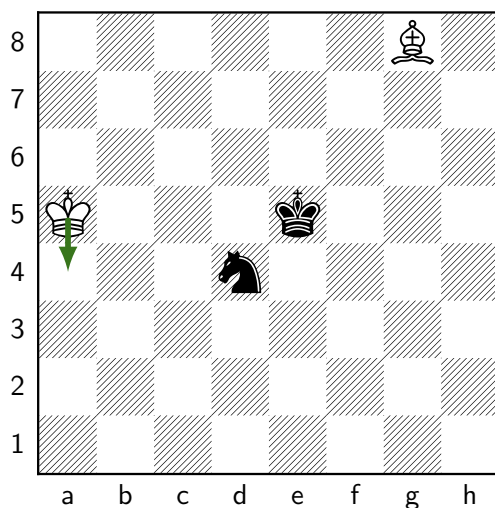
(a)



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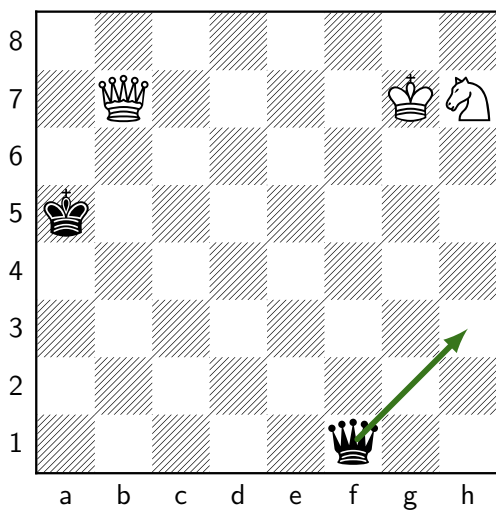


(c)

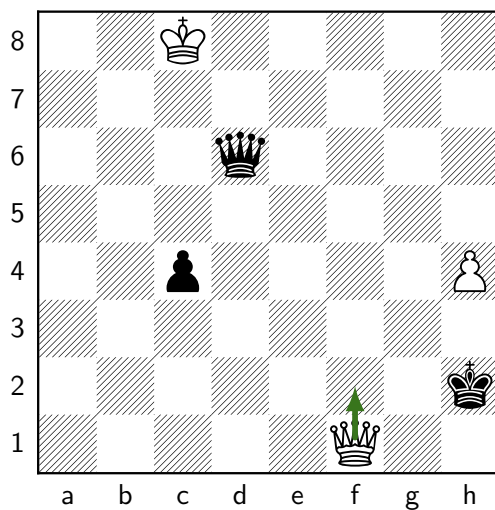


(d)

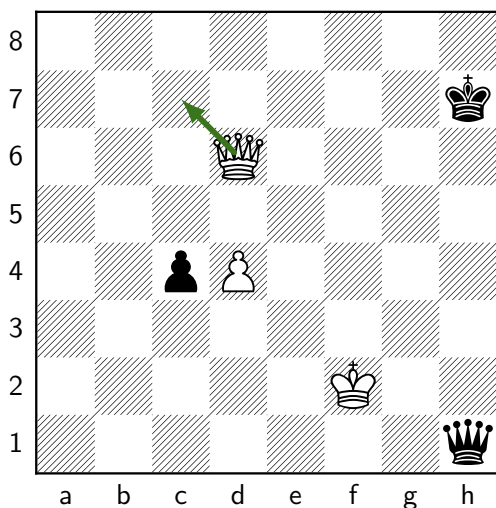
Figure 60: **Move 66, Black: Ka5.** (a) Our position; Melanie plays Ka5 (b6a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#931` (ply 143) [structural distance  $d=11$ , QAP obj=3]: the master played Ke5; under  $\mu$  our *Kb3* plays the role of that game’s *Kf4*; cross-type roles: our *Qf1* as their *Rd4*, our *qc8* as their *rf3*; transferred move b6a5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#140` (ply 135) [structural distance  $d=11$ , QAP obj=2]: the master played Ka1; under  $\mu$  our *Kb3* plays the role of that game’s *Ka7*; cross-type roles: our *Qf1* as their *Bg6*; transferred move b6a5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#1618` (ply 138) [structural distance  $d=8$ , QAP obj=2]: the master played Ka4; under  $\mu$  our *Kb3* plays the role of that game’s *Ka5*; cross-type roles: our *Qf1* as their *Bg8*; transferred move b6a5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Ka5 (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



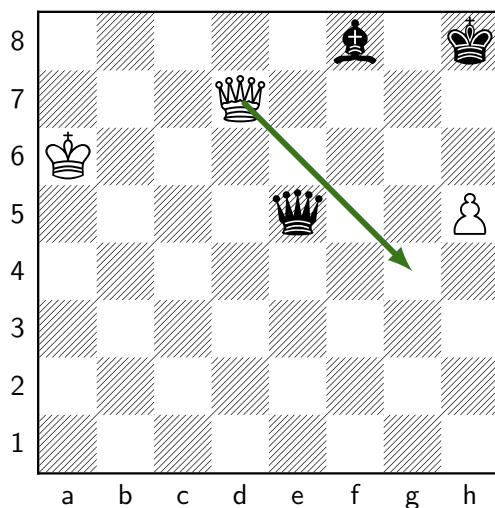
(a)



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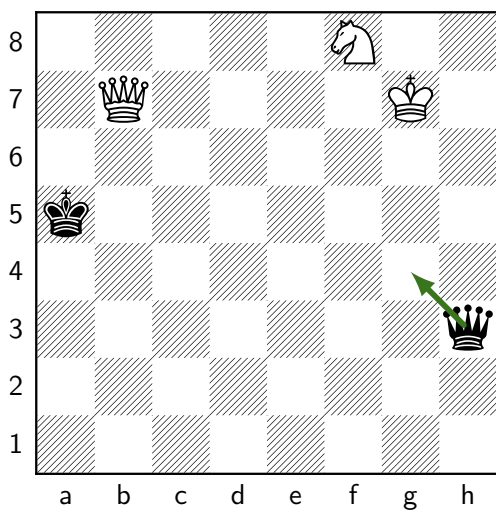


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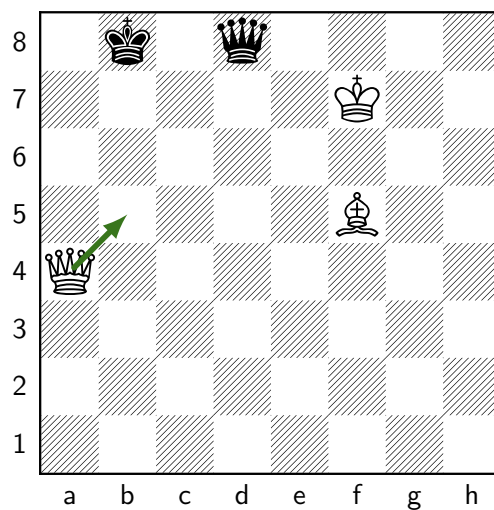


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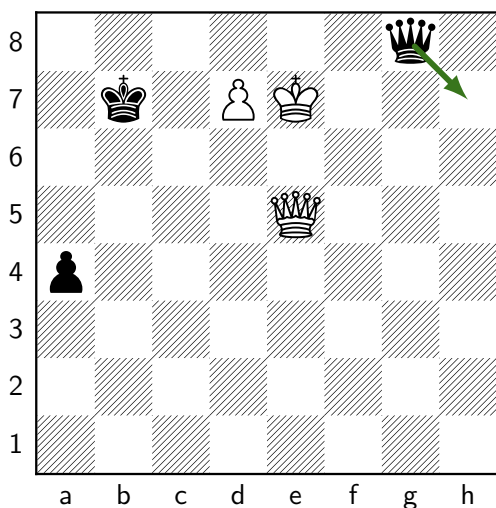
Figure 61: **Move 67, Black: Qh3**. (a) Our position; Melanie plays Qh3 (f1h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#845` (ply 144) [structural distance  $d=12$ , QAP obj=3]: the master played Qf2+; under  $\mu$  our Qf8 plays the role of that game’s Qf1; cross-type roles: our nh7 as their pc4; transferred move f1h3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1662` (ply 138) [structural distance  $d=13$ , QAP obj=3]: the master played Qc7+; under  $\mu$  our Qf8 plays the role of that game’s Qd6; cross-type roles: our nh7 as their Pd4; transferred move f1h3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#705` (ply 140) [structural distance  $d=13$ , QAP obj=2]: the master played Qg4; under  $\mu$  our Qf8 plays the role of that game’s Qd7; cross-type roles: our nh7 as their bf8; transferred move f1h3 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Qh3 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



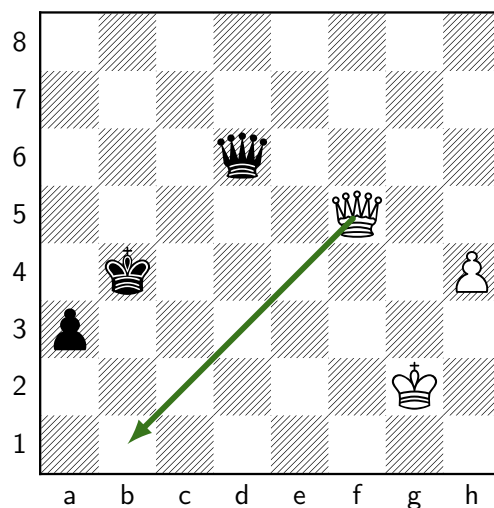
(a)



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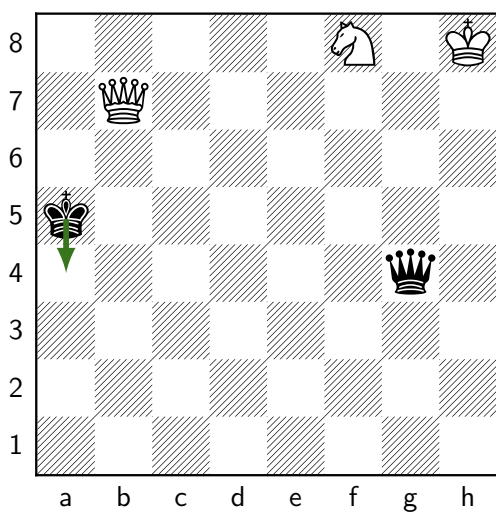


(c)

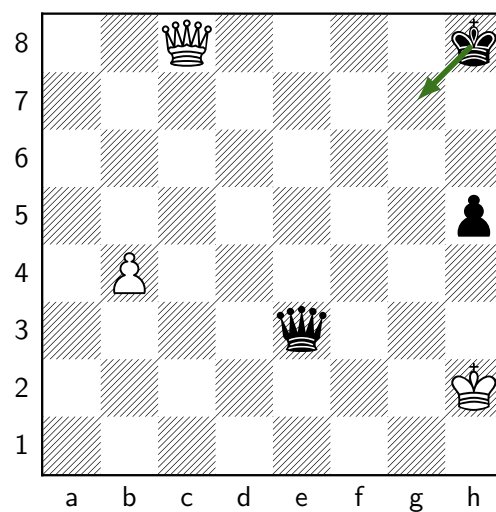


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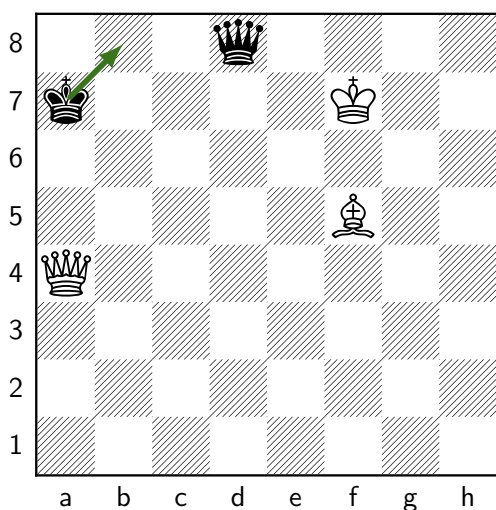
Figure 62: **Move 68, Black: Qg4+.** (a) Our position; Melanie plays Qg4+ (h3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3607` (ply 134) [structural distance  $d=8$ , QAP obj=2]: the master played Qb5+; under  $\mu$  our Qh6 plays the role of that game’s Qa4; cross-type roles: our nf8 as their Bf5; transferred move h3g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3920` (ply 137) [structural distance  $d=11$ , QAP obj=2]: the master played Qh7+; under  $\mu$  our Qh6 plays the role of that game’s Qg1; cross-type roles: our nf8 as their pd7; transferred move h3g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3376` (ply 134) [structural distance  $d=10$ , QAP obj=2]: the master played Qb1+; under  $\mu$  our Qh6 plays the role of that game’s Qf5; cross-type roles: our nf8 as their pa3; transferred move h3g4 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Qg4+ (summed QAP objective 18, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



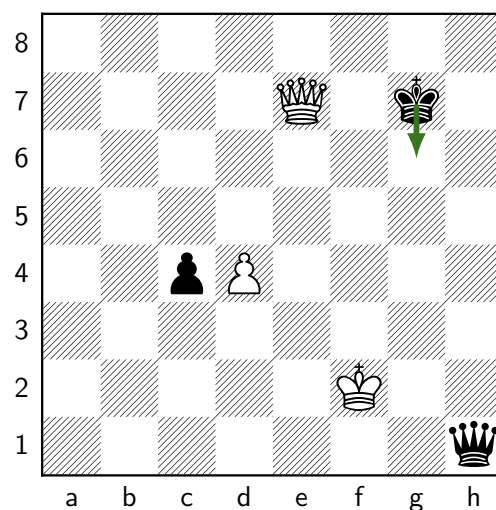
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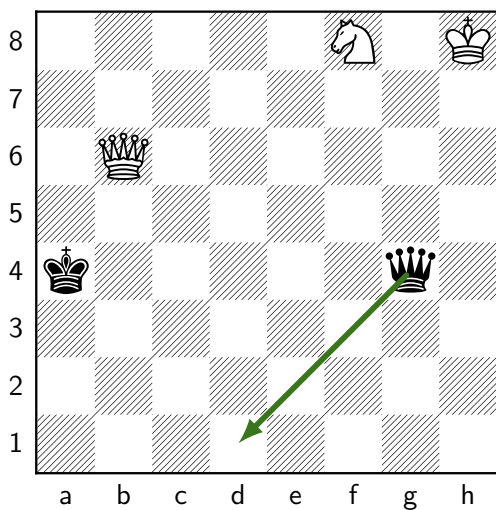


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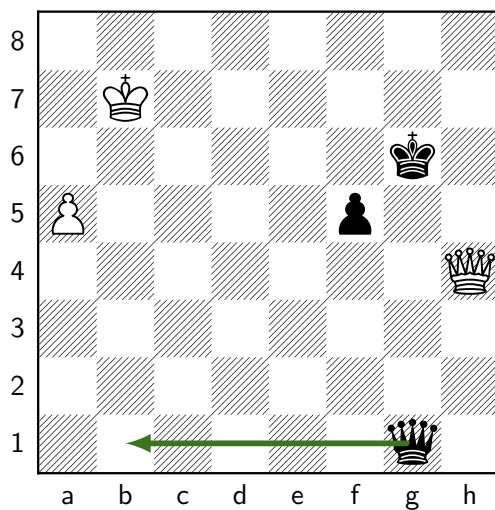


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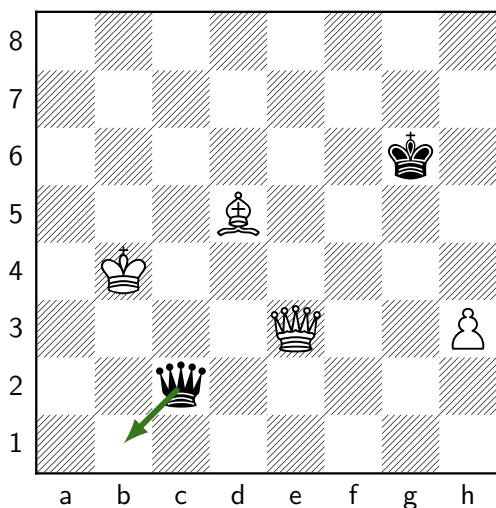
Figure 63: **Move 69, Black: Ka4.** (a) Our position; Melanie plays Ka4 (a5a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1845` (ply 133) [structural distance  $d=8$ , QAP obj=1]: the master played Kg7; under  $\mu$  our Ka4 plays the role of that game’s Kh1; cross-type roles: our nf8 as their pb4; transferred move a5a4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3607` (ply 137) [structural distance  $d=7$ , QAP obj=1]: the master played Kb8; under  $\mu$  our Ka4 plays the role of that game’s Ka2; cross-type roles: our nf8 as their bf5; transferred move a5a4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1662` (ply 131) [structural distance  $d=8$ , QAP obj=1]: the master played Kg6; under  $\mu$  our Ka4 plays the role of that game’s Kg2; cross-type roles: our nf8 as their pd4; transferred move a5a4 (**endorses this move**). Synthesis: 18 retrieved analogs transferred to Ka4 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



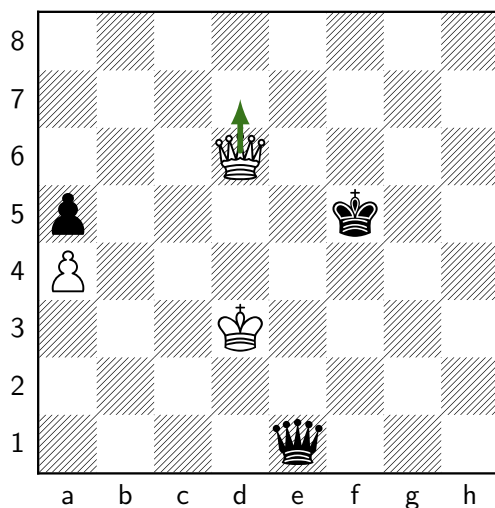
(a)



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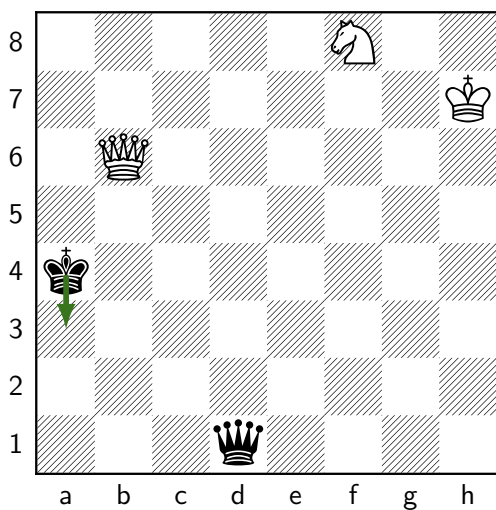


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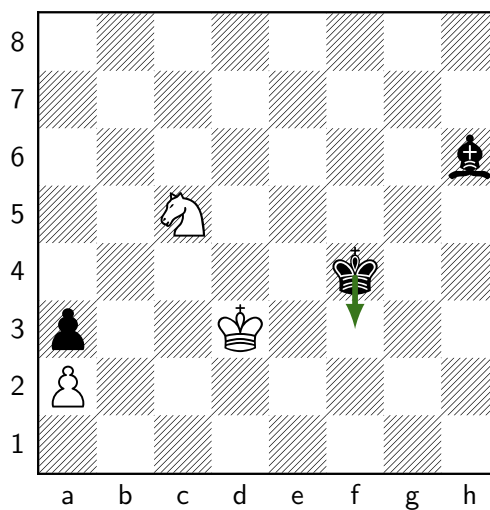


(d)

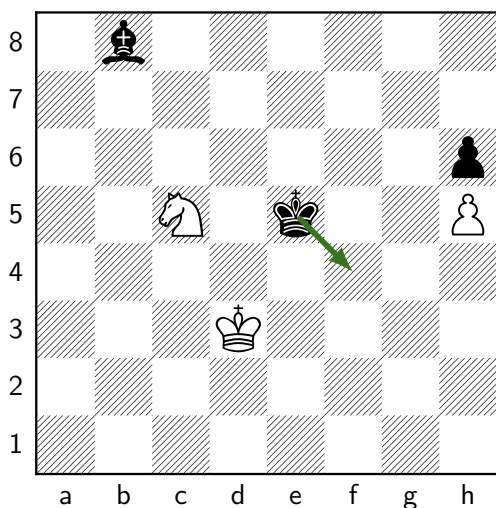
Figure 64: **Move 70, Black: Qd1**. (a) Our position; Melanie plays Qd1 (g4d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#936` (ply 139) [structural distance  $d=11$ , QAP obj=2]: the master played Qb1+; under  $\mu$  our Qg5 plays the role of that game's Qg8; cross-type roles: our nf8 as their pa5; transferred move g4d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#430` (ply 125) [structural distance  $d=10$ , QAP obj=2]: the master played Qb1+; under  $\mu$  our Qg5 plays the role of that game's Qc7; cross-type roles: our nf8 as their bd5; transferred move g4d1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1666` (ply 132) [structural distance  $d=12$ , QAP obj=2]: the master played Qd7+; under  $\mu$  our Qg5 plays the role of that game's Qd6; cross-type roles: our nf8 as their pa5; transferred move g4d1 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Qd1 (summed QAP objective 21, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



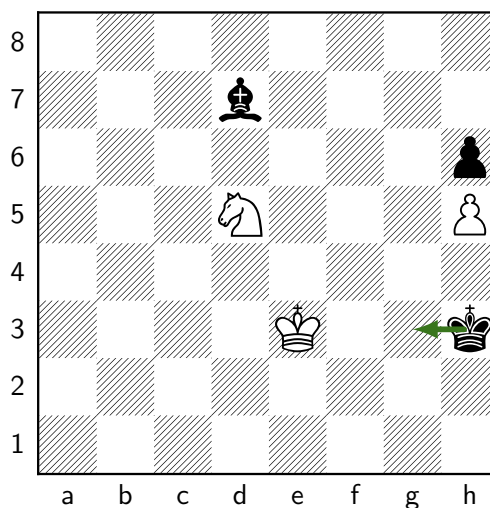
(a)



(b)

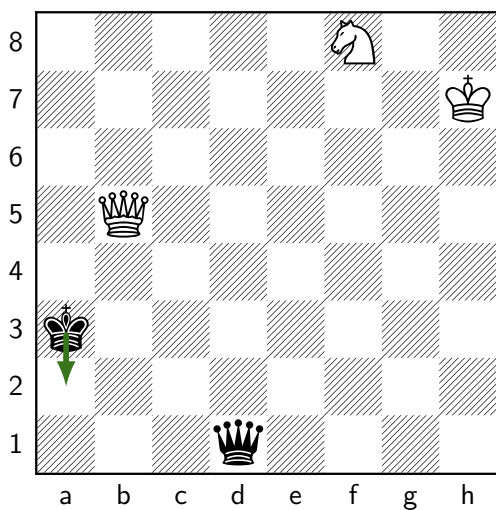


(c)

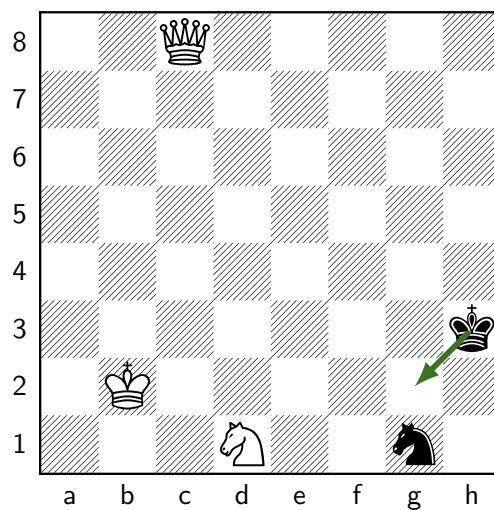


(d)

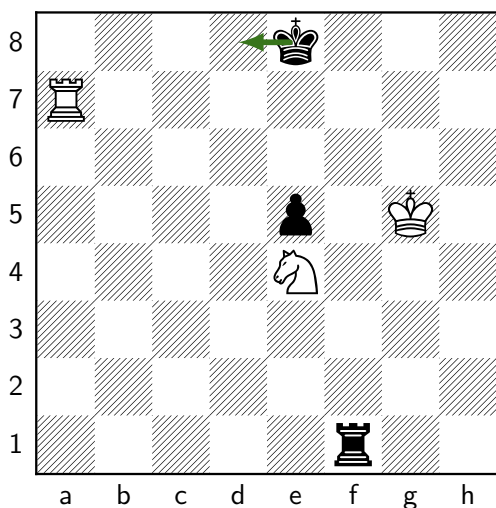
Figure 65: **Move 71, Black: Ka3.** (a) Our position; Melanie plays Ka3 (a4a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2372` (ply 151) [structural distance  $d=14$ , QAP obj=3]: the master played Kf3; under  $\mu$  our Ka5 plays the role of that game's Kf5; cross-type roles: our Qd1 as their Bh6, our qb6 as their Pa3; transferred move a4a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1155` (ply 139) [structural distance  $d=13$ , QAP obj=3]: the master played Kf4; under  $\mu$  our Ka5 plays the role of that game's Ke4; cross-type roles: our Qd1 as their Bb8, our qb6 as their Ph6; transferred move a4a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2179` (ply 133) [structural distance  $d=14$ , QAP obj=3]: the master played Kg3; under  $\mu$  our Ka5 plays the role of that game's Kh6; cross-type roles: our Qd1 as their Bd7, our qb6 as their Ph6; transferred move a4a3 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Ka3 (summed QAP objective 28, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



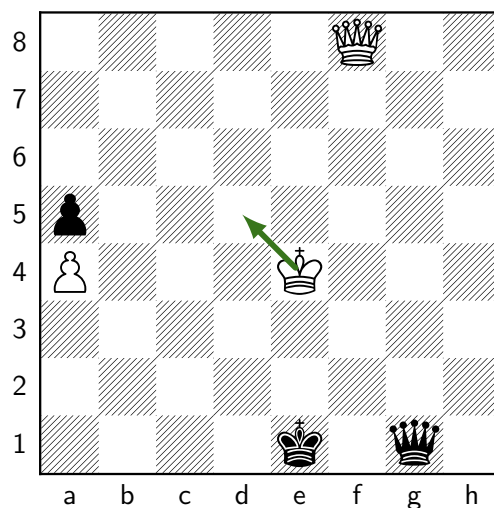
(a)



(b)

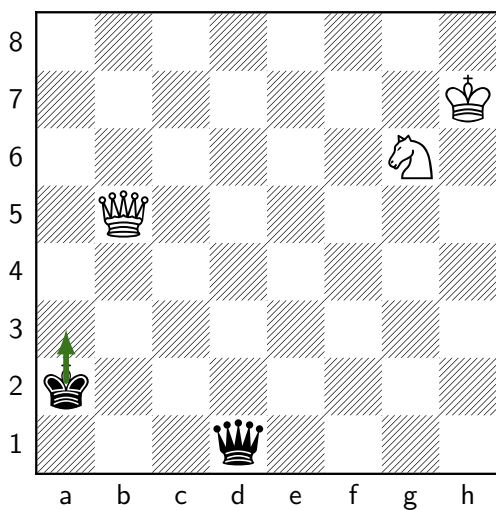


(c)

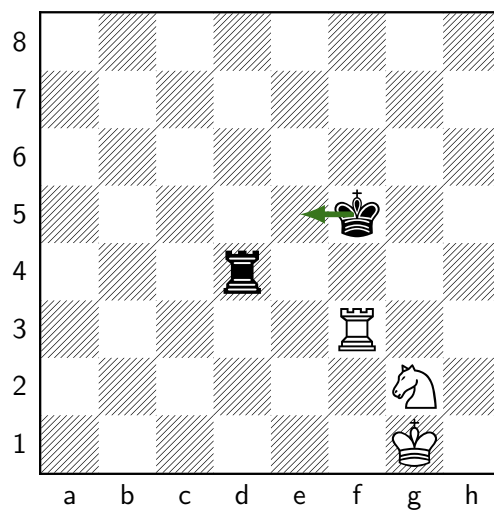


(d)

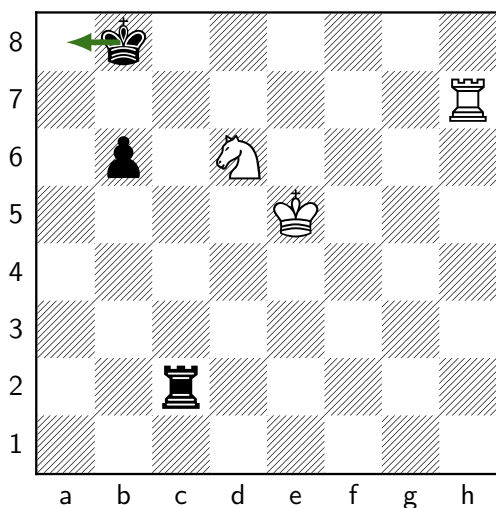
Figure 66: **Move 72, Black: Ka2.** (a) Our position; Melanie plays Ka2 (a3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#310` (ply 157) [structural distance  $d=11$ , QAP obj=3]: the master played Kg2; under  $\mu$  our Ka6 plays the role of that game's Kh6; cross-type roles: our Qd1 as their Ng1; transferred move a3a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#396` (ply 145) [structural distance  $d=11$ , QAP obj=2]: the master played Kd8; under  $\mu$  our Ka6 plays the role of that game's Ke1; cross-type roles: our qb5 as their ra7, our Qd1 as their Rf1; transferred move a3a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1512` (ply 148) [structural distance  $d=11$ , QAP obj=2]: the master played Kd5; under  $\mu$  our Ka6 plays the role of that game's Ke4; cross-type roles: our nf8 as their qg1, our qb5 as their Pa4; transferred move a3a2 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Ka2 (summed QAP objective 39, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



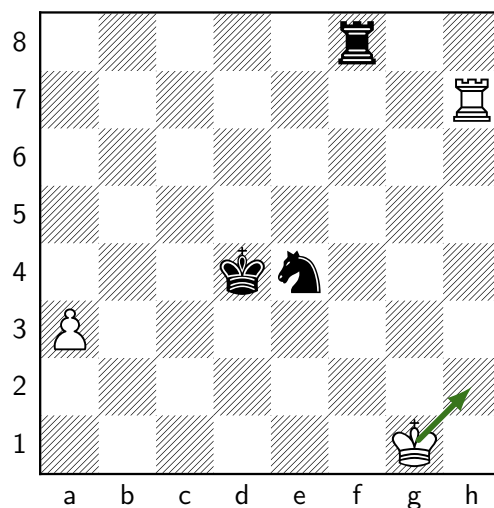
(a)



(b)

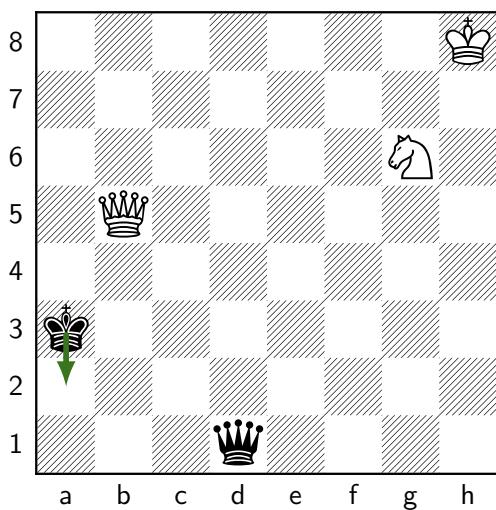


(c)

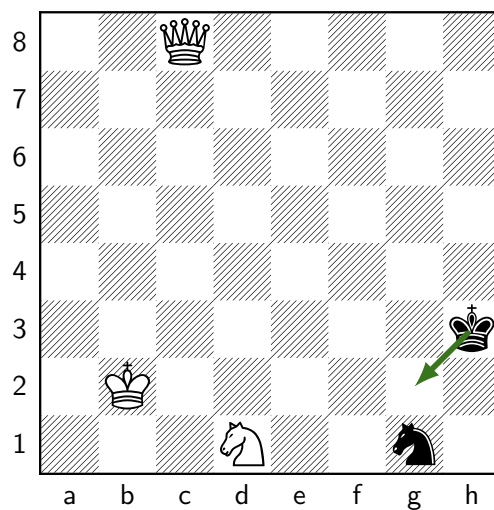


(d)

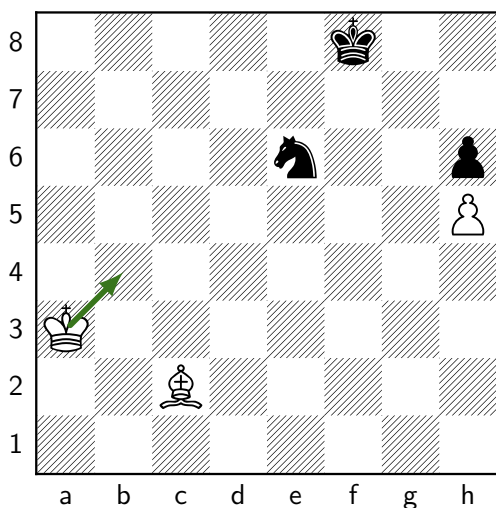
Figure 67: **Move 73, Black: Ka3**. (a) Our position; Melanie plays Ka3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#931` (ply 143) [structural distance  $d=10$ , QAP obj=3]: the master played Ke5; under  $\mu$  our Ka7 plays the role of that game’s Kf4; cross-type roles: our Qd1 as their Rd4, our qb5 as their rf3; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1061` (ply 147) [structural distance  $d=10$ , QAP obj=3]: the master played Ka8; under  $\mu$  our Ka7 plays the role of that game’s Kb1; cross-type roles: our qb5 as their rh7, our Qd1 as their Rc2; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1426` (ply 142) [structural distance  $d=8$ , QAP obj=2]: the master played Kh2; under  $\mu$  our Ka7 plays the role of that game’s Kg1; cross-type roles: our Qd1 as their Rh7, our qb5 as their rf8; transferred move a2a3 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Ka3 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



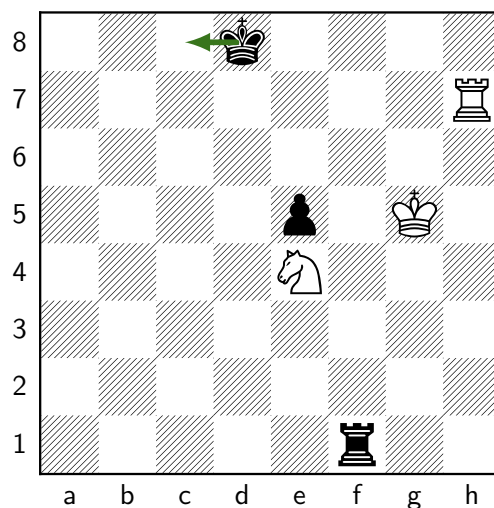
(a)



(b)

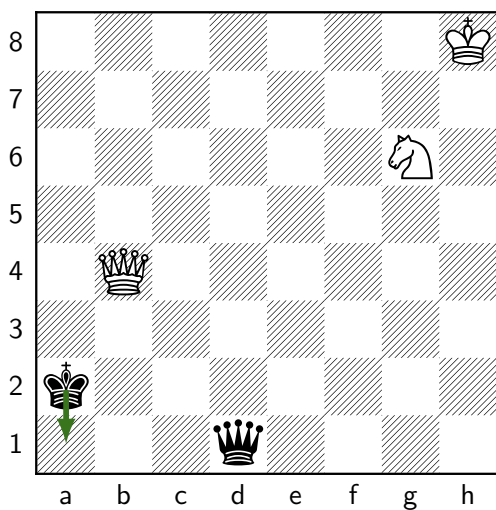


(c)

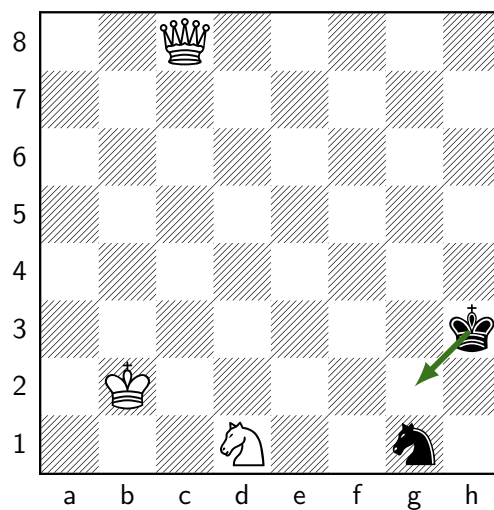


(d)

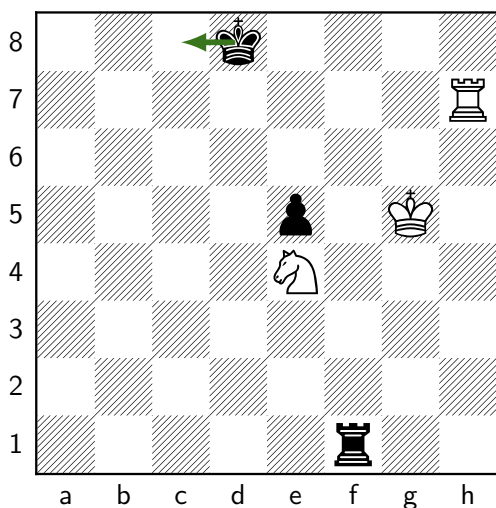
Figure 68: **Move 74, Black: Ka2**. (a) Our position; Melanie plays Ka2 (a3a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#310` (ply 157) [structural distance  $d=11$ , QAP obj=3]: the master played Kg2; under  $\mu$  our Ka6 plays the role of that game’s Kh6; cross-type roles: our Qd1 as their Ng1; transferred move a3a2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#235` (ply 140) [structural distance  $d=10$ , QAP obj=2]: the master played Kb4; under  $\mu$  our Ka6 plays the role of that game’s Ka3; cross-type roles: our Qd1 as their Ph5, our qb5 as their ph6; transferred move a3a2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#396` (ply 147) [structural distance  $d=11$ , QAP obj=2]: the master played Kc8; under  $\mu$  our Ka6 plays the role of that game’s Kd1; cross-type roles: our qb5 as their rh7, our Qd1 as their Pe5; transferred move a3a2 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Ka2 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



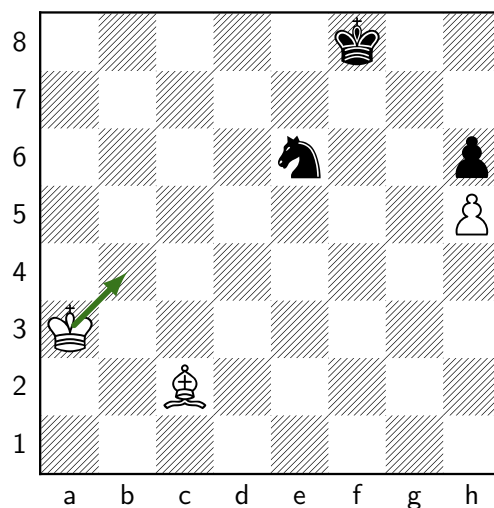
(a)



(b)

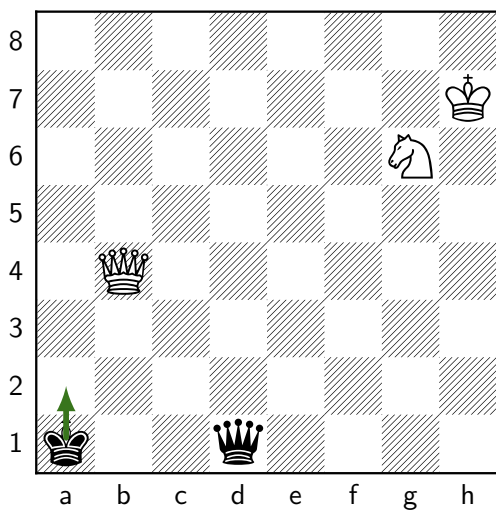


(c)

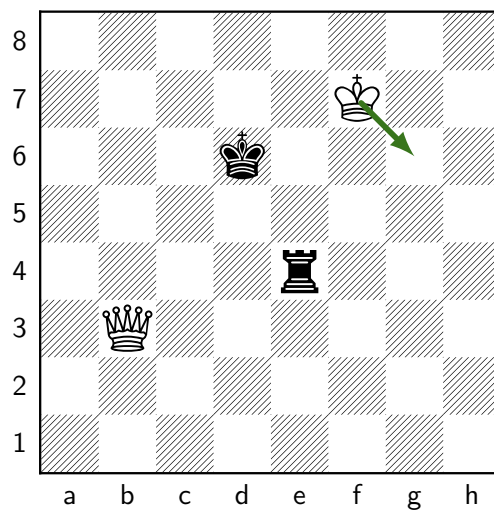


(d)

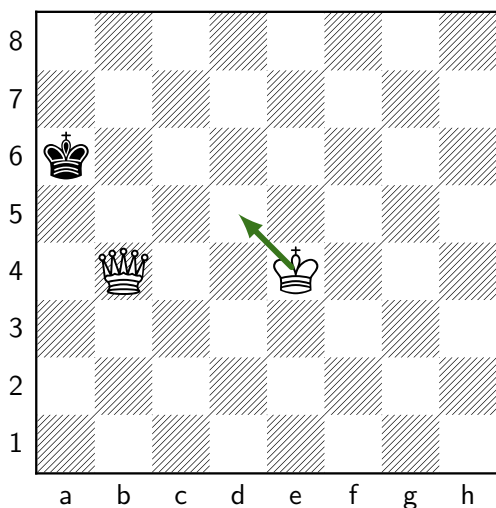
Figure 69: **Move 75, Black: Ka1**. (a) Our position; Melanie plays Ka1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#310` (ply 157) [structural distance  $d=11$ , QAP obj=3]: the master played Kg2; under  $\mu$  our Ka7 plays the role of that game’s Kh6; cross-type roles: our Qd1 as their Ng1; transferred move a2a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#396` (ply 147) [structural distance  $d=11$ , QAP obj=2]: the master played Kc8; under  $\mu$  our Ka7 plays the role of that game’s Kd1; cross-type roles: our qb4 as their rh7, our Qd1 as their Pe5; transferred move a2a1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#235` (ply 140) [structural distance  $d=10$ , QAP obj=2]: the master played Kb4; under  $\mu$  our Ka7 plays the role of that game’s Ka3; cross-type roles: our Qd1 as their Ph5, our qb4 as their ph6; transferred move a2a1 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Ka1 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



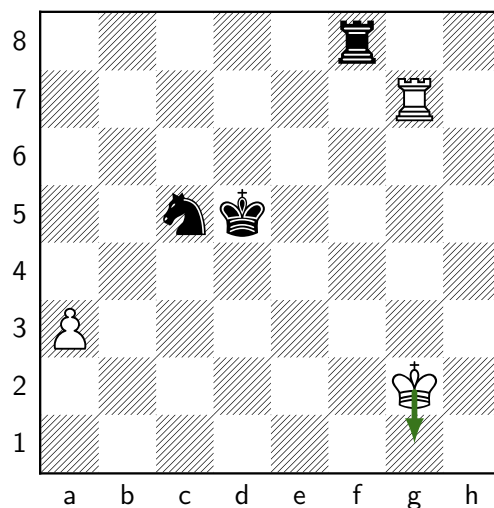
(a)



(b)

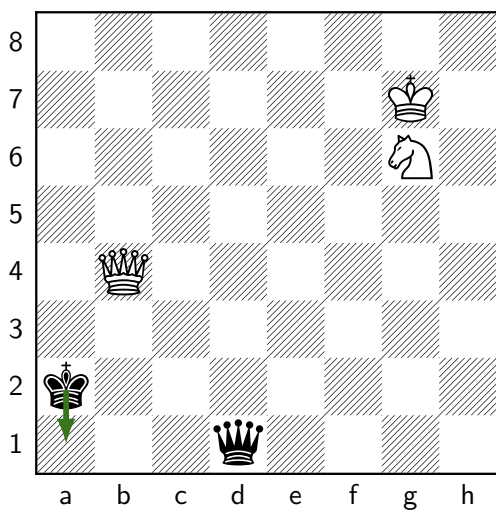


(c)

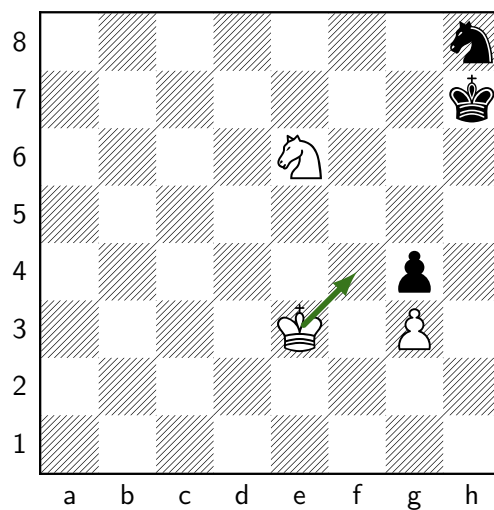


(d)

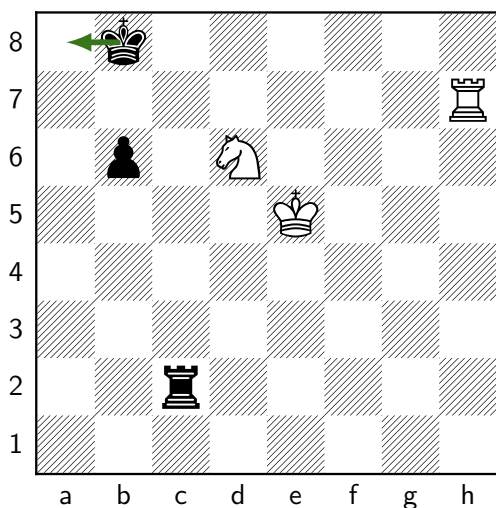
Figure 70: **Move 76, Black: Ka2.** (a) Our position; Melanie plays Ka2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#270` (ply 150) [structural distance  $d=10$ , QAP obj=2]: the master played Kg6; under  $\mu$  our Ka8 plays the role of that game’s Kf7; cross-type roles: our qb4 as their re4; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#187` (ply 136) [structural distance  $d=12$ , QAP obj=2]: the master played Kd5; under  $\mu$  our Ka8 plays the role of that game’s Ke4; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1426` (ply 136) [structural distance  $d=13$ , QAP obj=2]: the master played Kg1; under  $\mu$  our Ka8 plays the role of that game’s Kg2; cross-type roles: our ng6 as their Pa3, our Qd1 as their Rg7; transferred move a1a2 (**endorses this move**). Synthesis: 19 retrieved analogs transferred to Ka2 (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



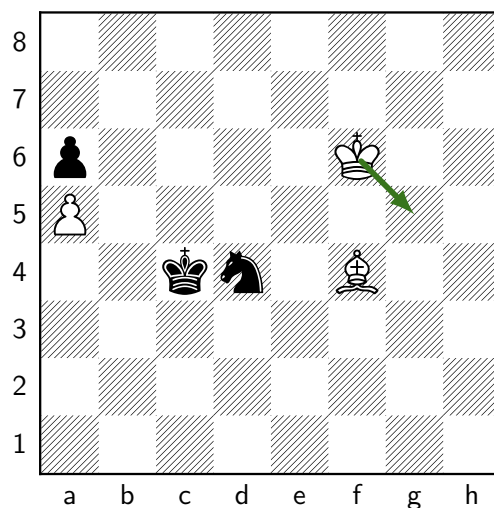
(a)



(b)

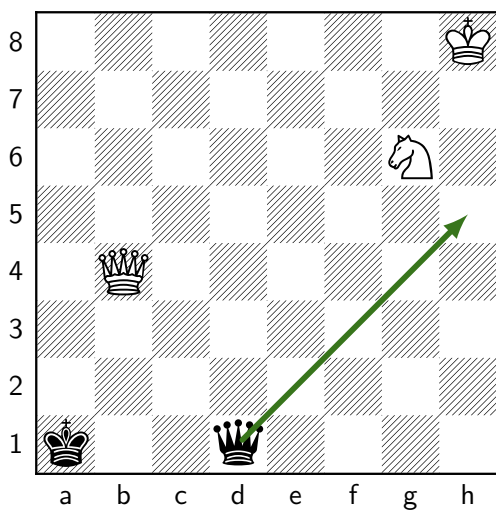


(c)

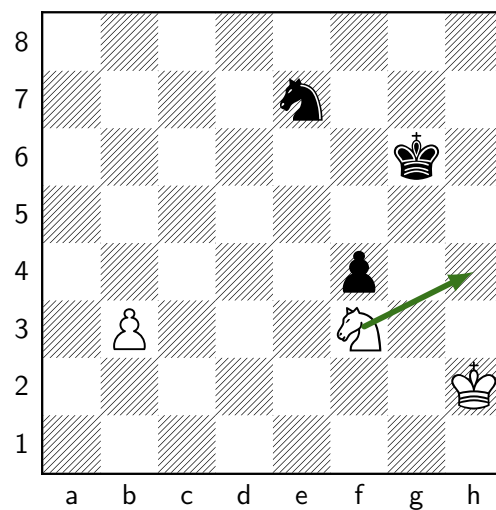


(d)

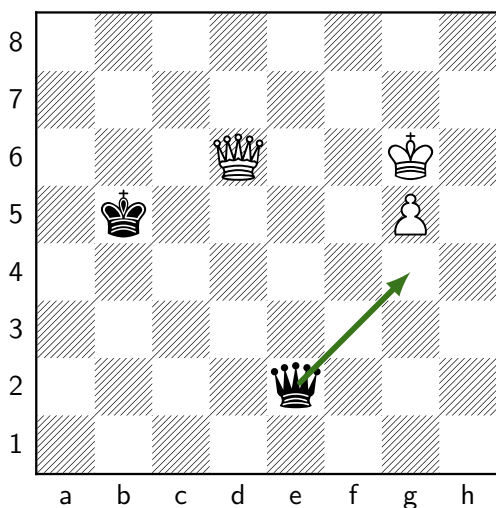
Figure 71: **Move 77, Black: Ka1.** (a) Our position; Melanie plays Ka1 (a2a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#19` (ply 146) [structural distance  $d=9$ , QAP obj=2]: the master played Kf4; under  $\mu$  our Ka7 plays the role of that game’s Ke3; cross-type roles: our Qd1 as their Pg3, our qb4 as their pg4; transferred move a2a1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1061` (ply 147) [structural distance  $d=10$ , QAP obj=2]: the master played Ka8; under  $\mu$  our Ka7 plays the role of that game’s Kb1; cross-type roles: our qb4 as their rh7, our Qd1 as their Pb6; transferred move a2a1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1997` (ply 144) [structural distance  $d=9$ , QAP obj=2]: the master played Kg5; under  $\mu$  our Ka7 plays the role of that game’s Kf6; cross-type roles: our qb4 as their Pa5, our Qd1 as their pa6; transferred move a2a1 (**endorses this move**). Synthesis: 23 retrieved analogs transferred to Ka1 (summed QAP objective 36, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



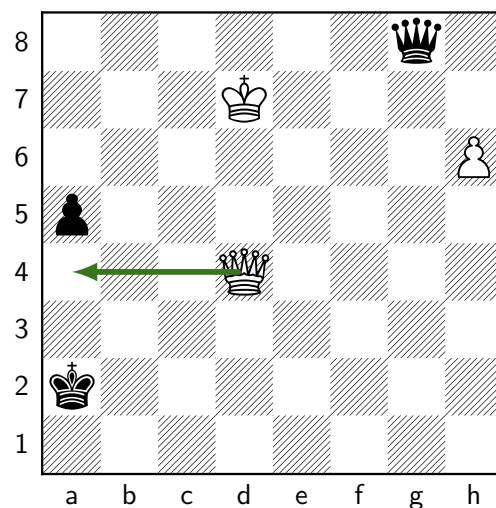
(a)



(b)

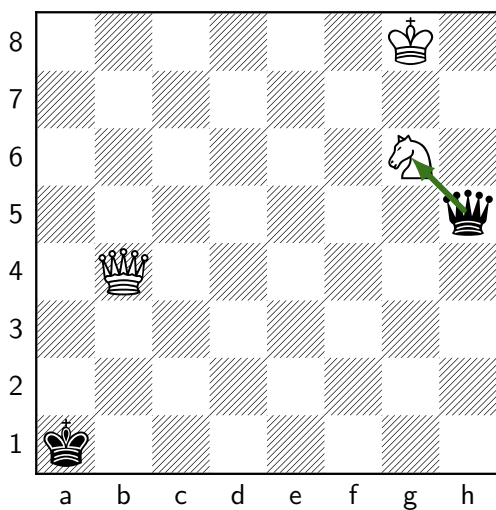


(c)

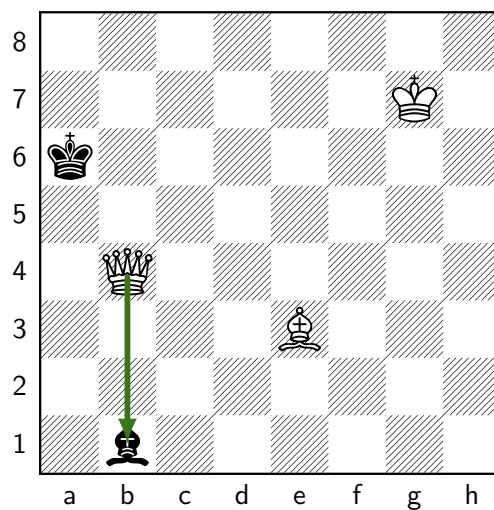


(d)

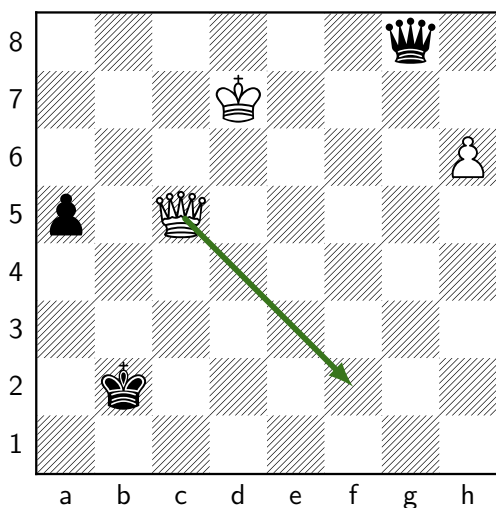
Figure 72: **Move 78, Black: Qh5+.** (a) Our position; Melanie plays Qh5+ (d1h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2077` (ply 152) [structural distance  $d=13$ , QAP obj=4]: the master played Nh4+; under  $\mu$  our Qd8 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our qb4 as their pf4; transferred move d1h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1434` (ply 167) [structural distance  $d=12$ , QAP obj=2]: the master played Qg4; under  $\mu$  our Qd8 plays the role of that game’s Qe7; cross-type roles: our ng6 as their pg5; transferred move d1h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#127` (ply 168) [structural distance  $d=13$ , QAP obj=2]: the master played Qa4+; under  $\mu$  our Qd8 plays the role of that game’s Qd4; cross-type roles: our qb4 as their pa5, our ng6 as their qg8; transferred move d1h5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Qh5+ (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



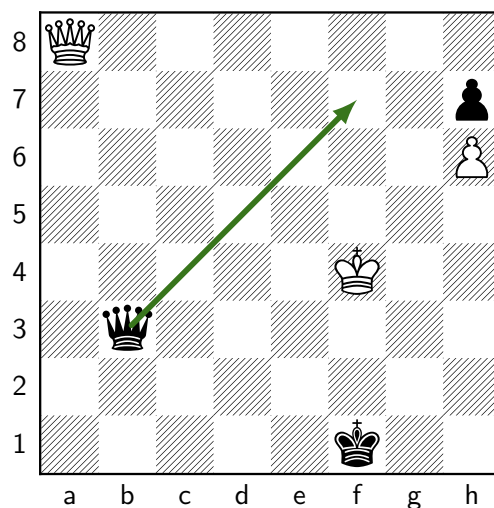
(a)



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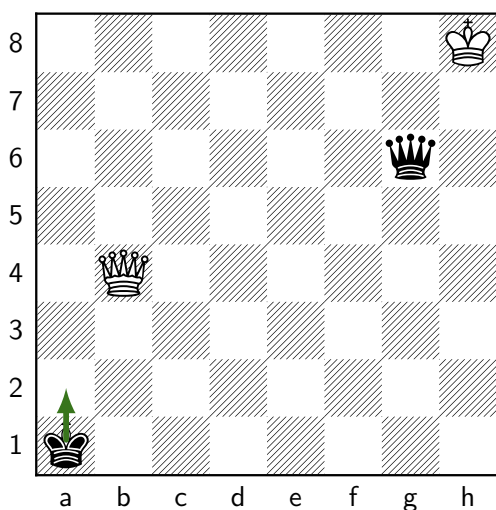


(c)

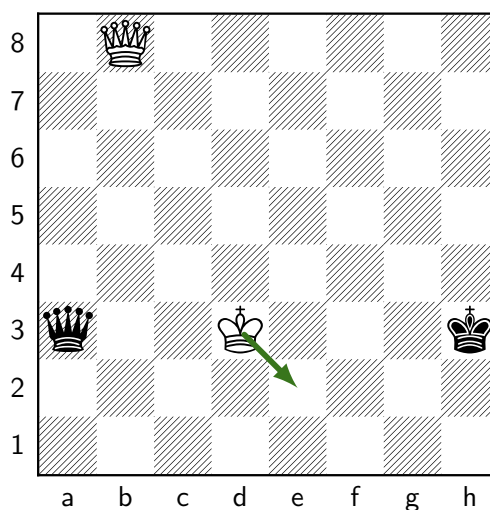


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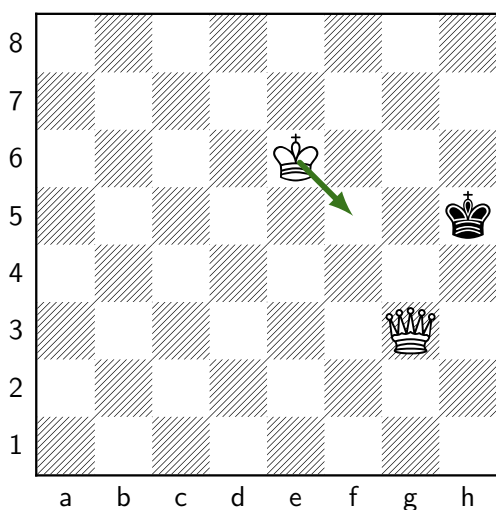
Figure 73: **Move 79, Black: Qxg6+.** (a) Our position; Melanie plays Qxg6+ (h5g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#38` (ply 148) [structural distance  $d=9$ , QAP obj=6]: the master played Qxb1; under  $\mu$  our Qh4 plays the role of that game's Qb4; cross-type roles: our ng6 as their bb1, our qb4 as their Be3; transferred move h5g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#127` (ply 158) [structural distance  $d=8$ , QAP obj=1]: the master played Qf2+; under  $\mu$  our Qh4 plays the role of that game's Qc5; cross-type roles: our ng6 as their pa5; transferred move h5g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2313` (ply 155) [structural distance  $d=9$ , QAP obj=1]: the master played Qf7+; under  $\mu$  our Qh4 plays the role of that game's Qb6; cross-type roles: our ng6 as their Ph7; transferred move h5g6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxg6+ (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



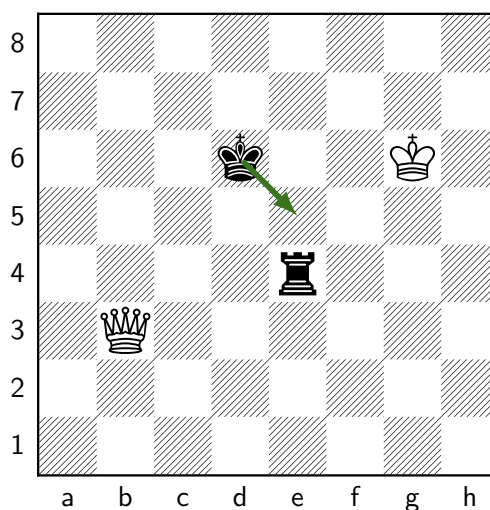
(a)



(b)

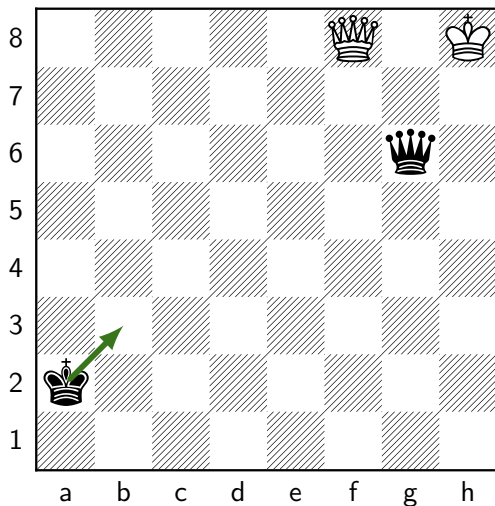


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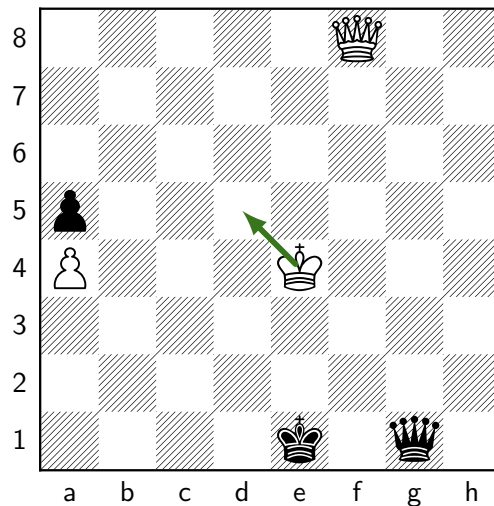


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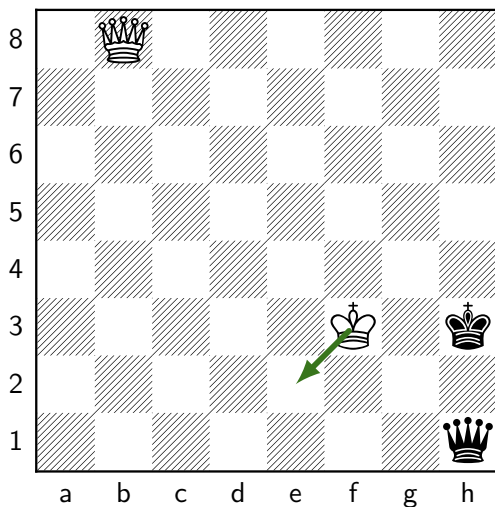
Figure 74: **Move 80, Black: Ka2.** (a) Our position; Melanie plays Ka2 (a1a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1624` (ply 154) [structural distance  $d=4$ , QAP obj=1]: the master played Ke2; under  $\mu$  our *Ka8* plays the role of that game’s *Kd3*; transferred move a1a2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1390` (ply 172) [structural distance  $d=4$ , QAP obj=0]: the master played Kf5; under  $\mu$  our *Ka8* plays the role of that game’s *Ke6*; transferred move a1a2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#270` (ply 151) [structural distance  $d=5$ , QAP obj=0]: the master played Ke5; under  $\mu$  our *Ka8* plays the role of that game’s *Kd3*; cross-type roles: our *Qg6* as their *Re4*; transferred move a1a2 (**endorses this move**). Synthesis: 24 retrieved analogs transferred to Ka2 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



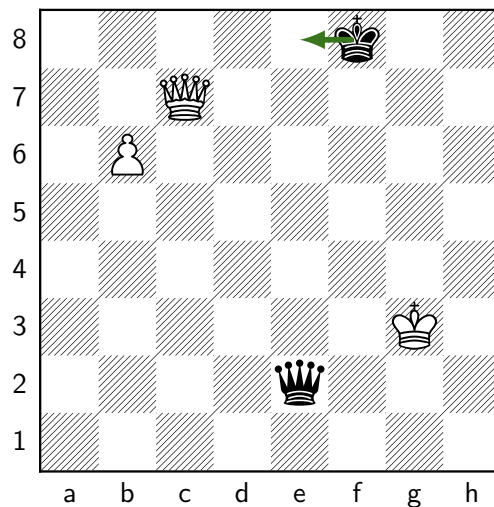
(a)



(b)

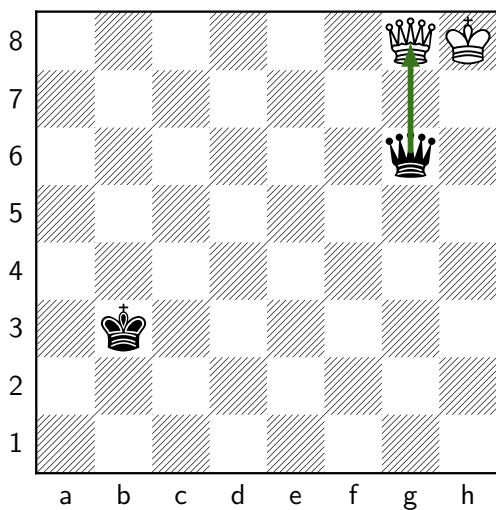


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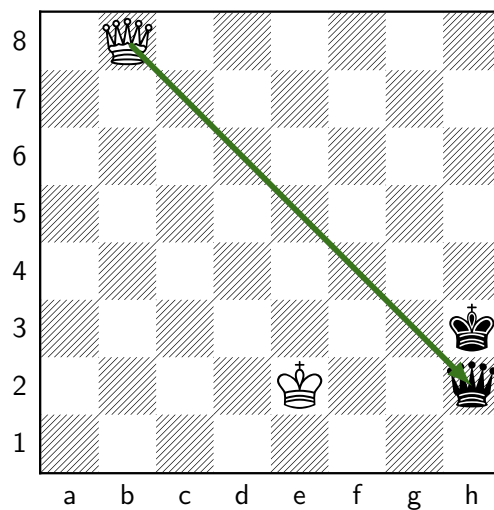


(d)

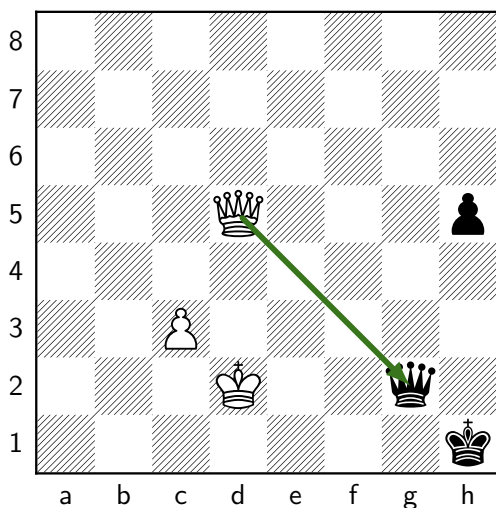
Figure 75: **Move 81, Black: Kb3.** (a) Our position; Melanie plays Kb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1512` (ply 148) [structural distance  $d=8$ , QAP obj=2]: the master played Kd5; under  $\mu$  our *Ka7* plays the role of that game’s *Ke4*; transferred move a2b3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1624` (ply 162) [structural distance  $d=4$ , QAP obj=2]: the master played Ke2; under  $\mu$  our *Ka7* plays the role of that game’s *Kf3*; transferred move a2b3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2865` (ply 145) [structural distance  $d=9$ , QAP obj=1]: the master played Ke8; under  $\mu$  our *Ka7* plays the role of that game’s *Kf1*; transferred move a2b3 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to Kb3 (summed QAP objective 14, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



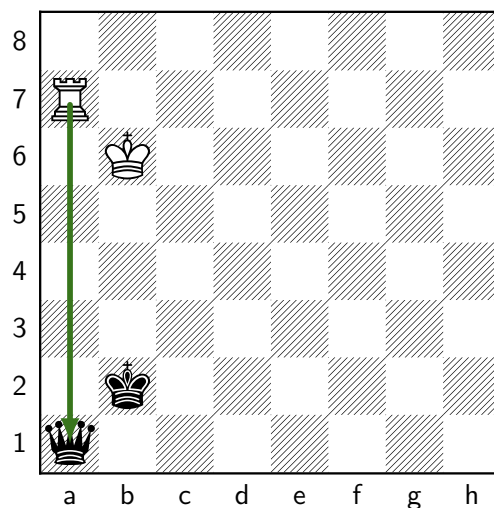
(a)



(b)



(c)



(d)

Figure 76: **Move 82, Black: Qxg8+.** (a) Our position; Melanie plays Qxg8+ (g6g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1624` (ply 168) [structural distance  $d=1$ , QAP obj=47]: the master played Qxh2+; under  $\mu$  our  $Qg3$  plays the role of that game’s  $Qb8$ ; transferred move g6g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#156` (ply 164) [structural distance  $d=20$ , QAP obj=44]: the master played Qxg2+; under  $\mu$  our  $Qg3$  plays the role of that game’s  $Qd5$ ; transferred move g6g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1462` (ply 154) [structural distance  $d=16$ , QAP obj=4]: the master played Rxa1; under  $\mu$  our  $Qg3$  plays the role of that game’s  $Ra7$  (a **cross-type** correspondence); transferred move g6g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxg8+ (summed QAP objective 95, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.